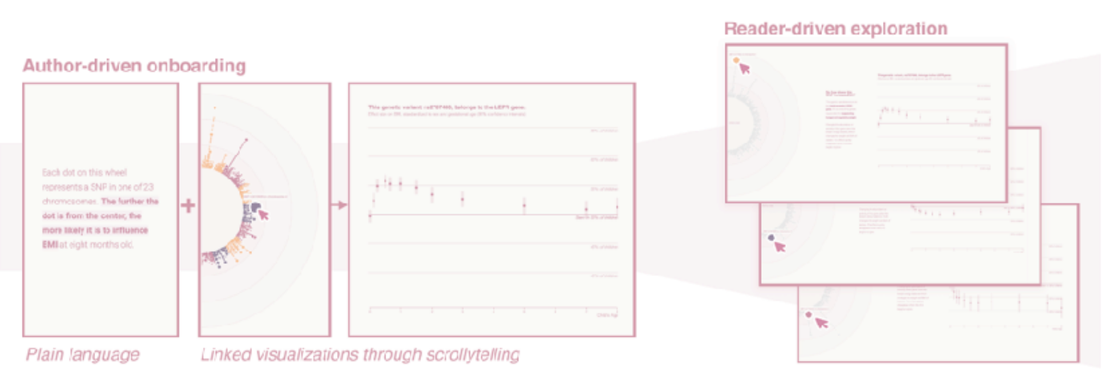
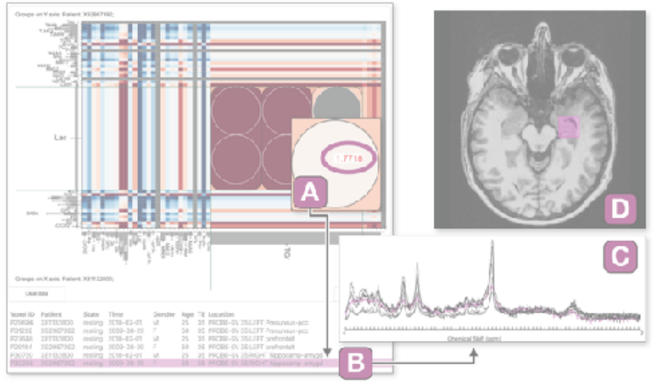
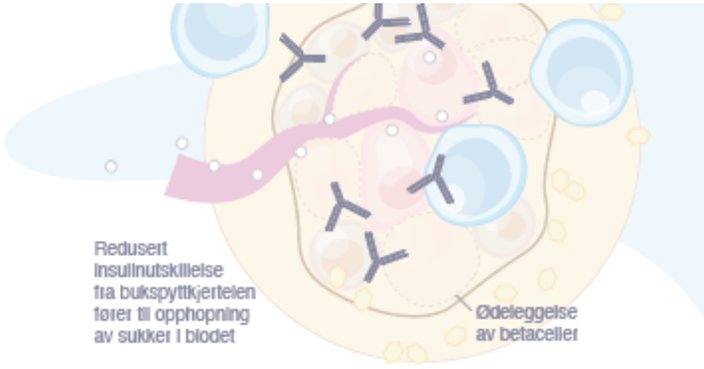
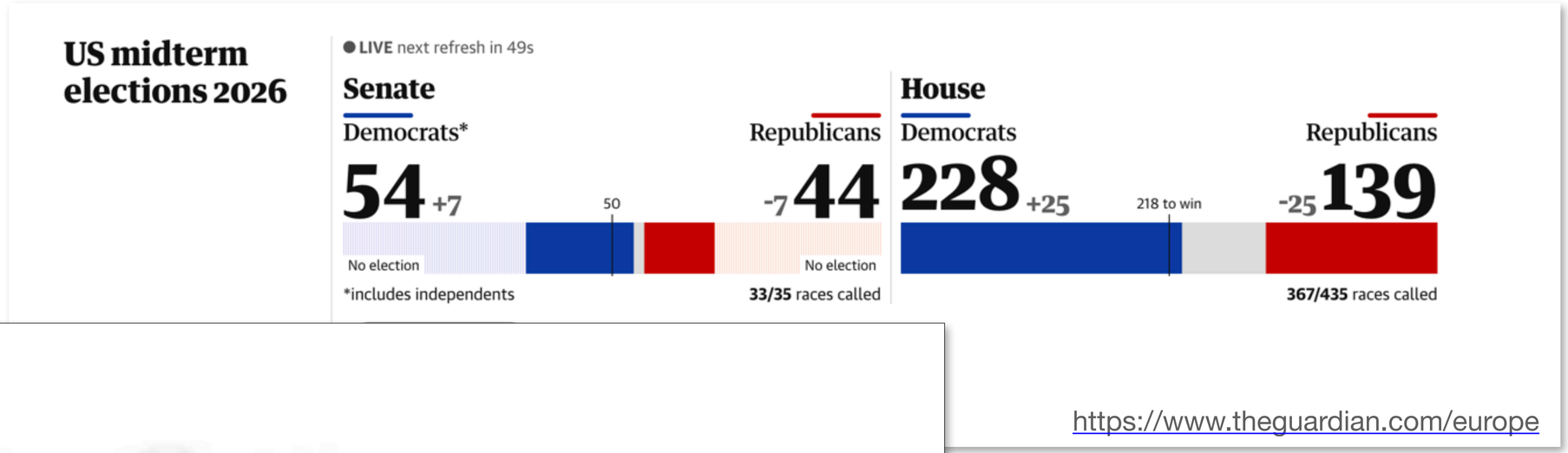






Communicating your science with **transparency** and **care**

Laura Garrison, Associate Professor
Visualization Group, Dept of Informatics, UiB
PRESIMAL Autumn Research School 2026



https://www.theguardian.com/europe

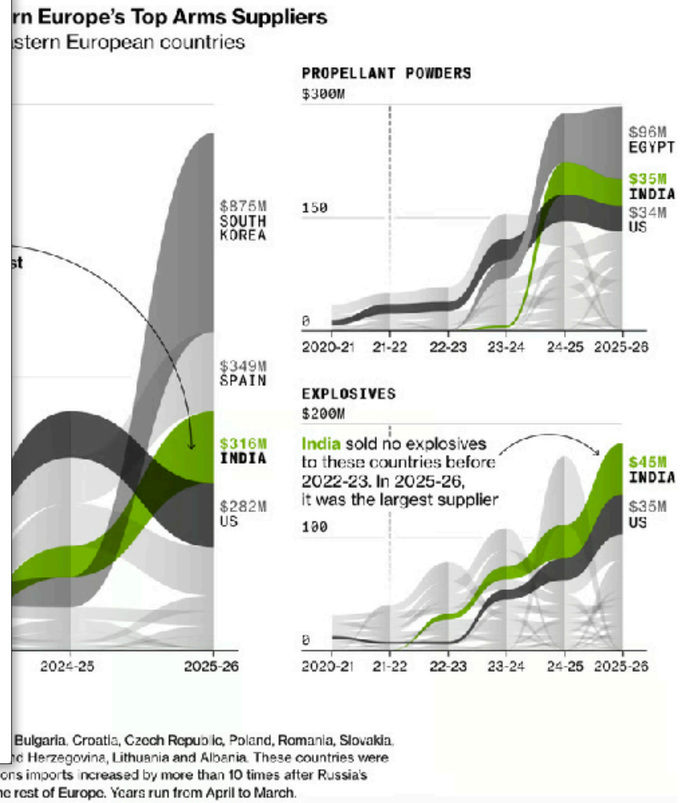
An Inside Look at Covid's Lasting Damage to the Lungs

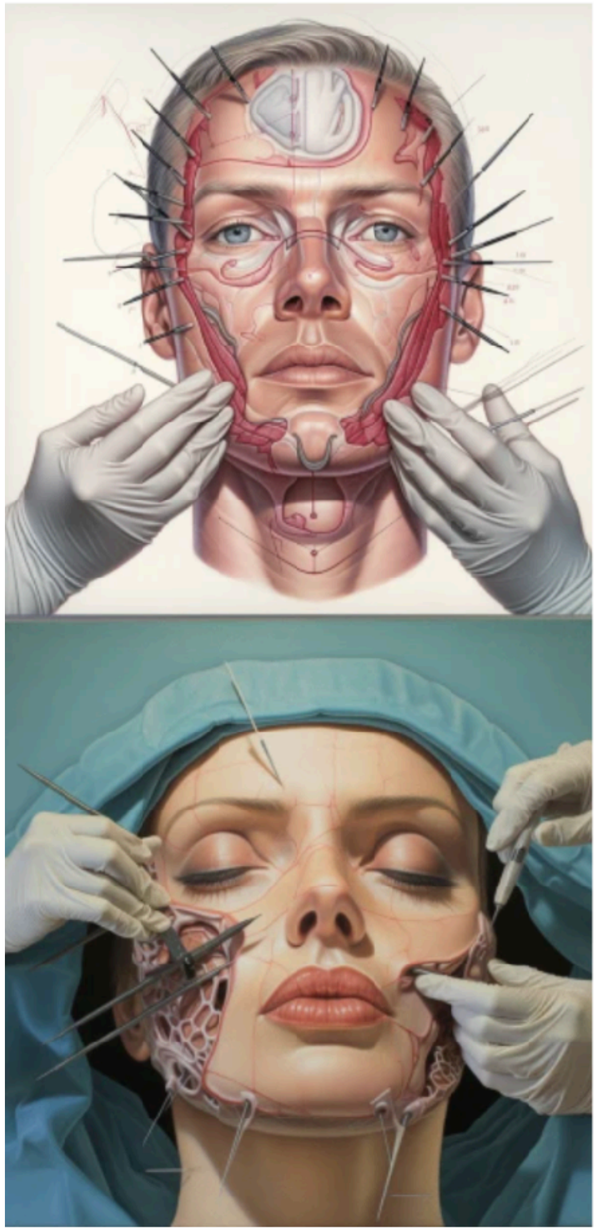
By Jeremy White, Pam Belluck, Noah Bassetti-Blum, Eleanor Lutz and Hang Do Thi Duc

May 31, 2023

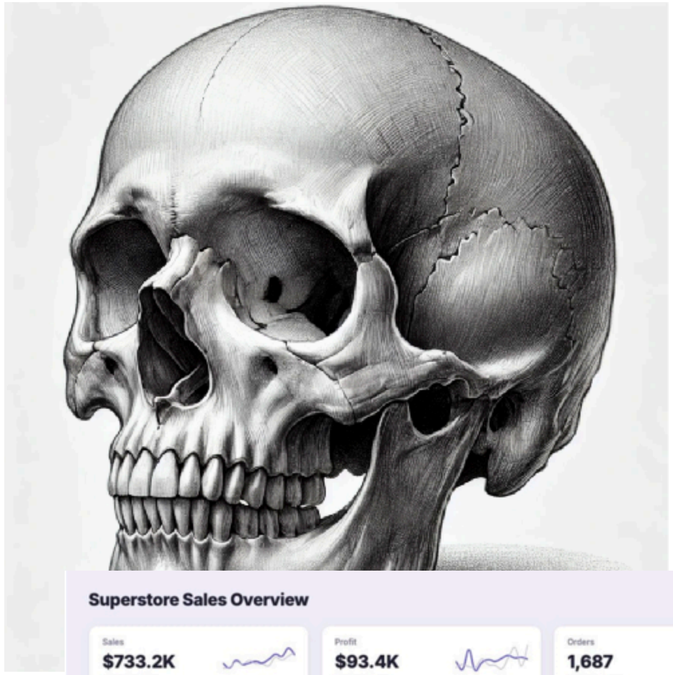
Share full article 114

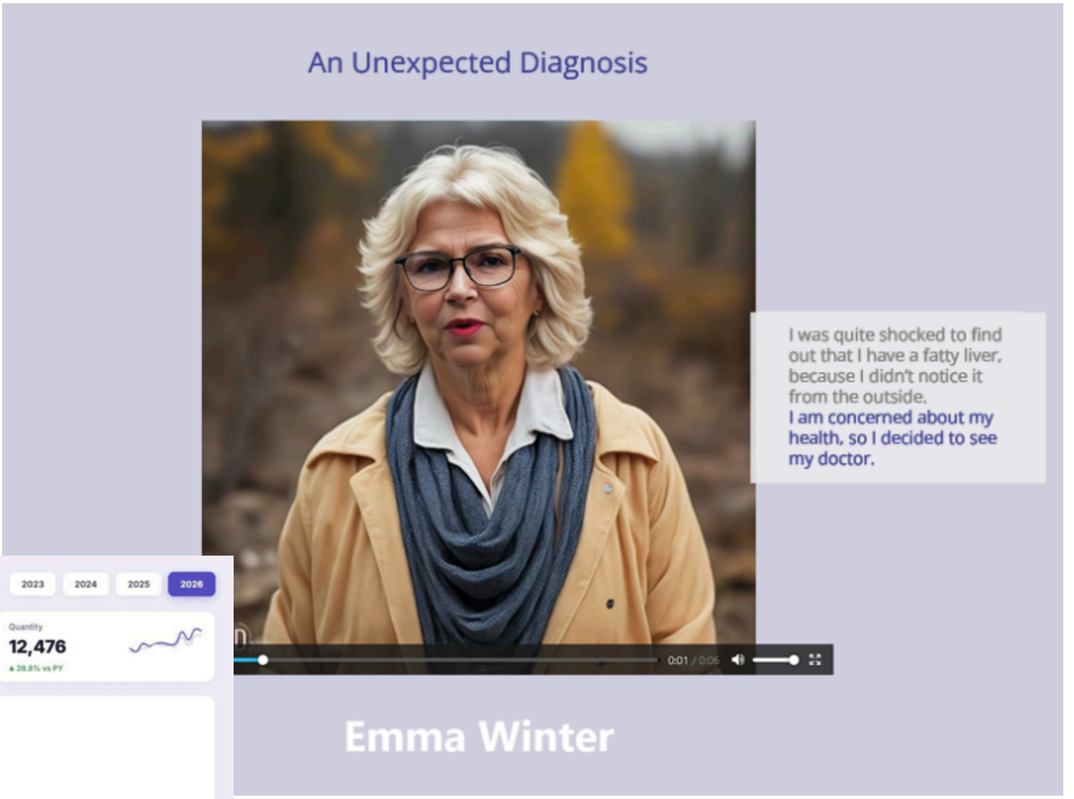
The New York Times: https://www.nytimes.com/interactive/2023/05/31/health/covid-lung-damage.html





Ziman et al. 2025

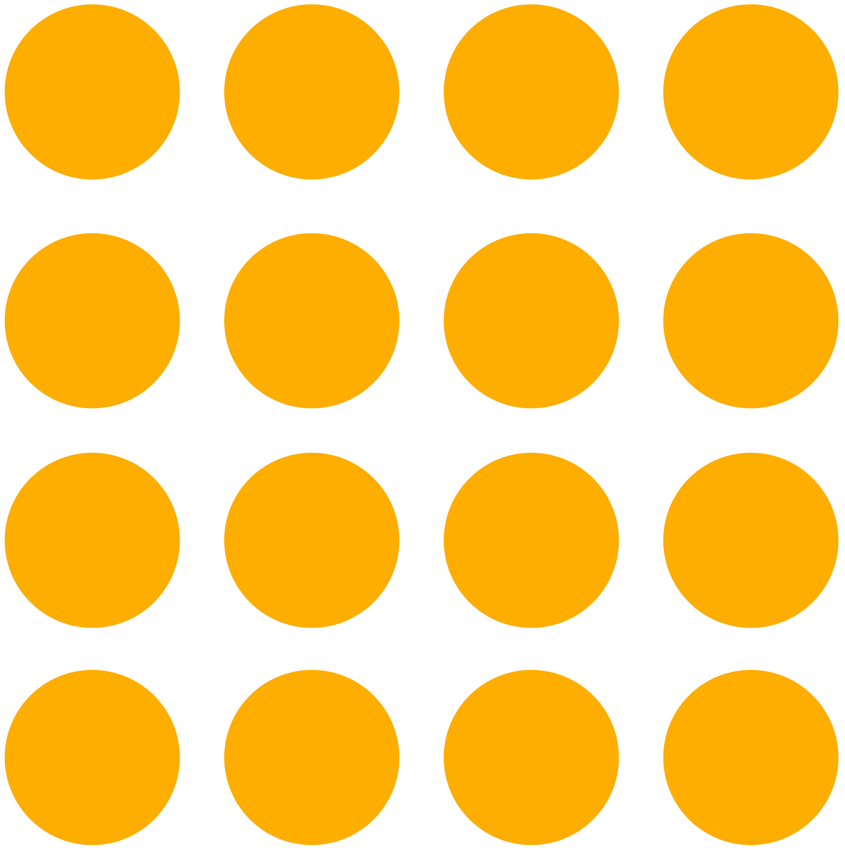




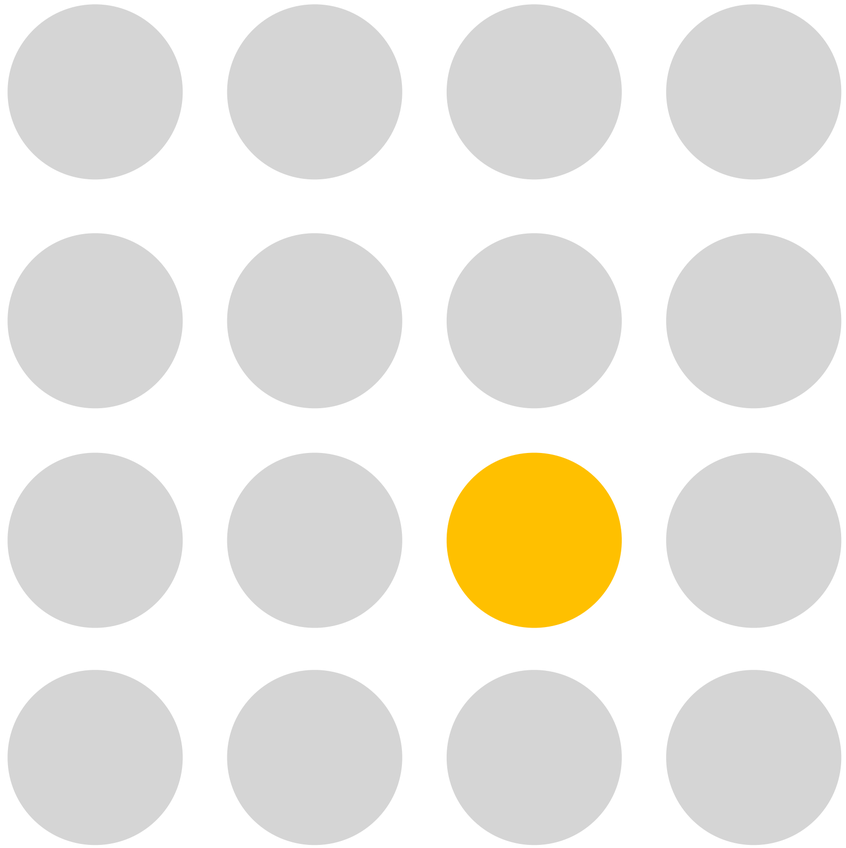
Budich et al. 2023

Claude remake of a sales dashboard (PolicyViz blog)

Visuals are powerful.

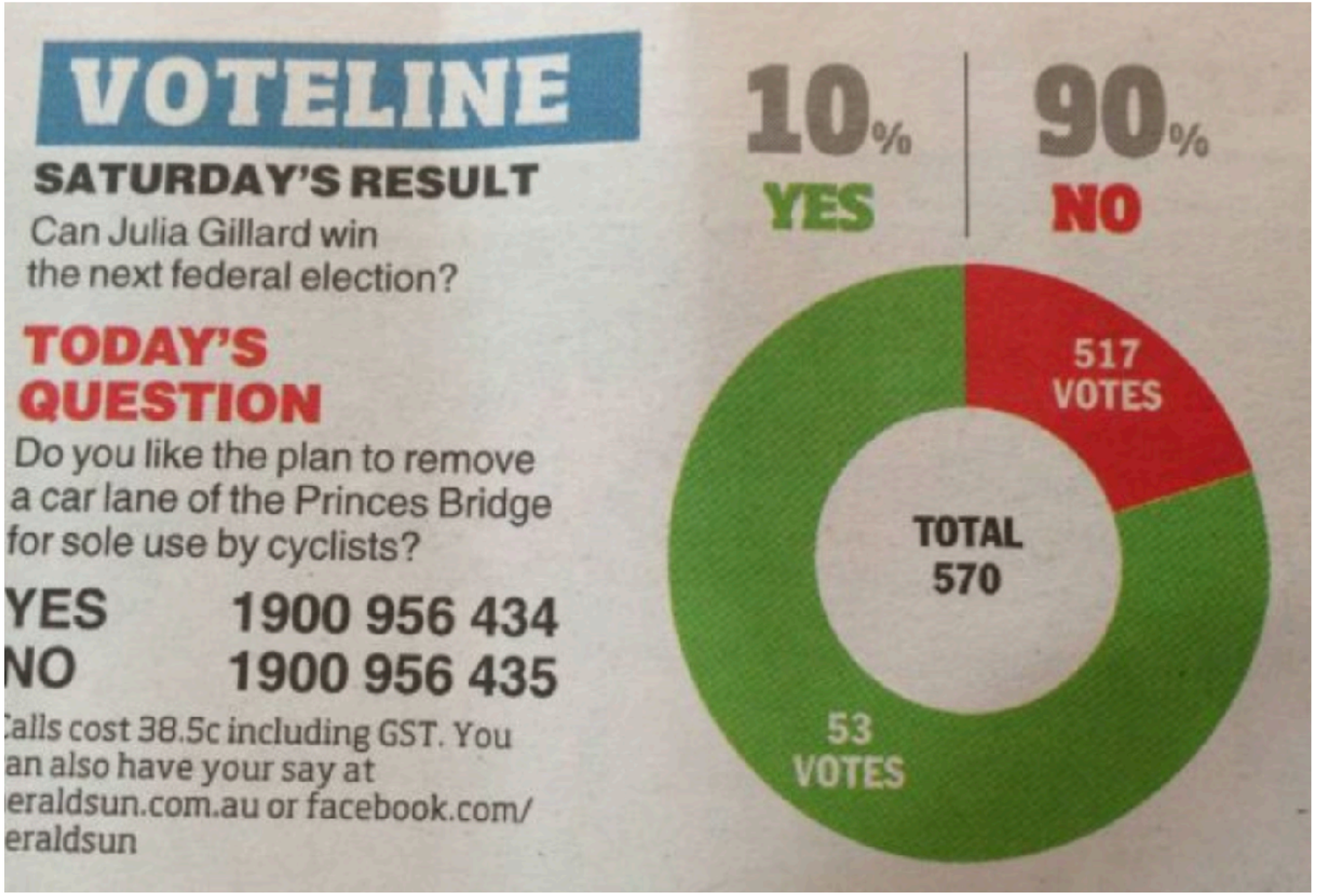


See **patterns….**



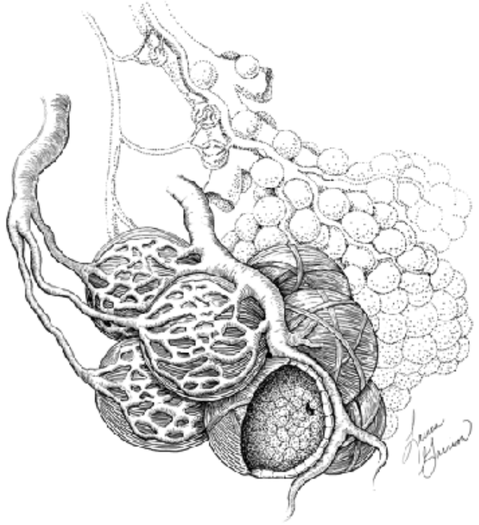
…and **exceptions**.

...but can be misused.

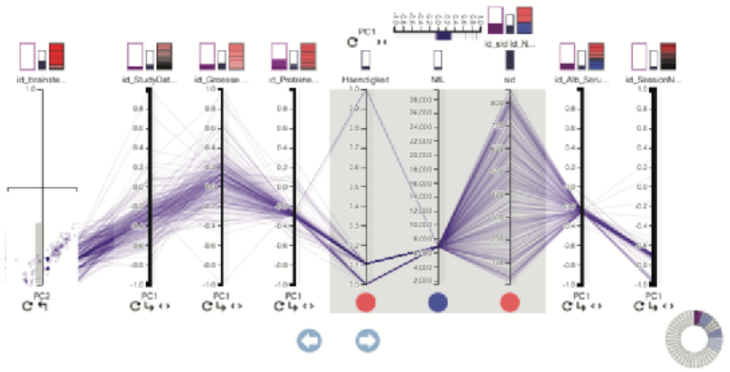


https://viz.wtf/

An **informative visual** is more than a pretty picture or a well-defined prompt.



MS **Biomedical visualization**
Univ. Illinois-Chicago

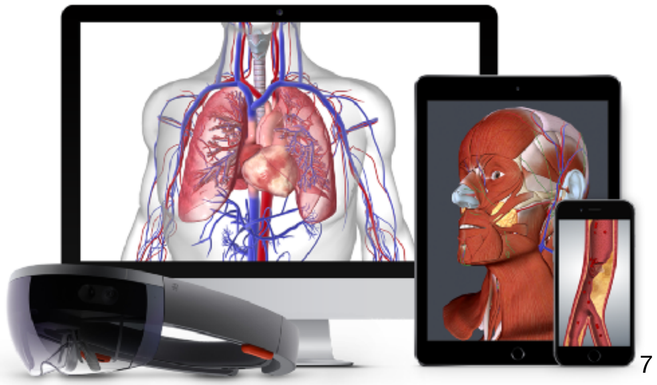


PhD **Visualization**
Univ. Bergen

VisGROUP UNIVERSITY OF BERGEN

BA **Anatomy & Physiology**
Northern Michigan Univ.

US-based health tech startups



bouvet

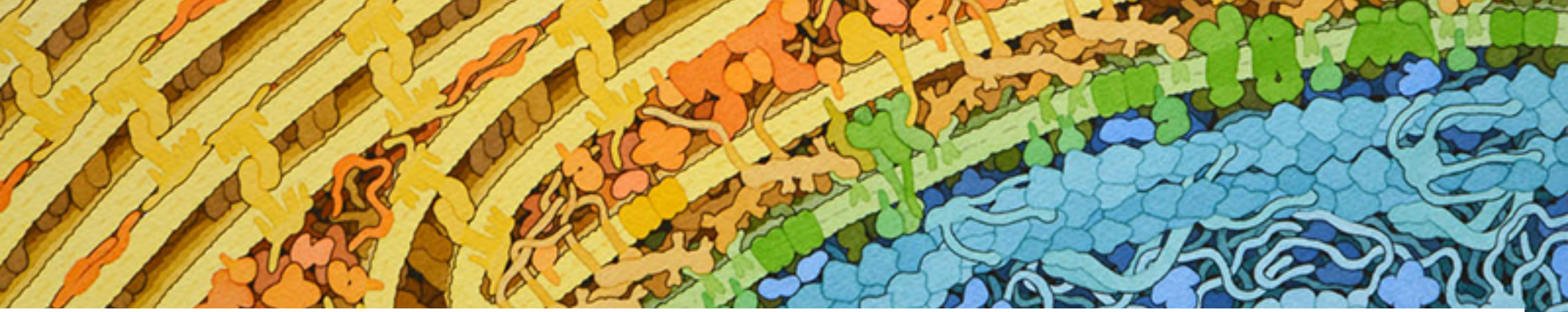
MMIV

MMIV

Center for Digital Narrative

VisGROUP UNIVERSITY OF BERGEN

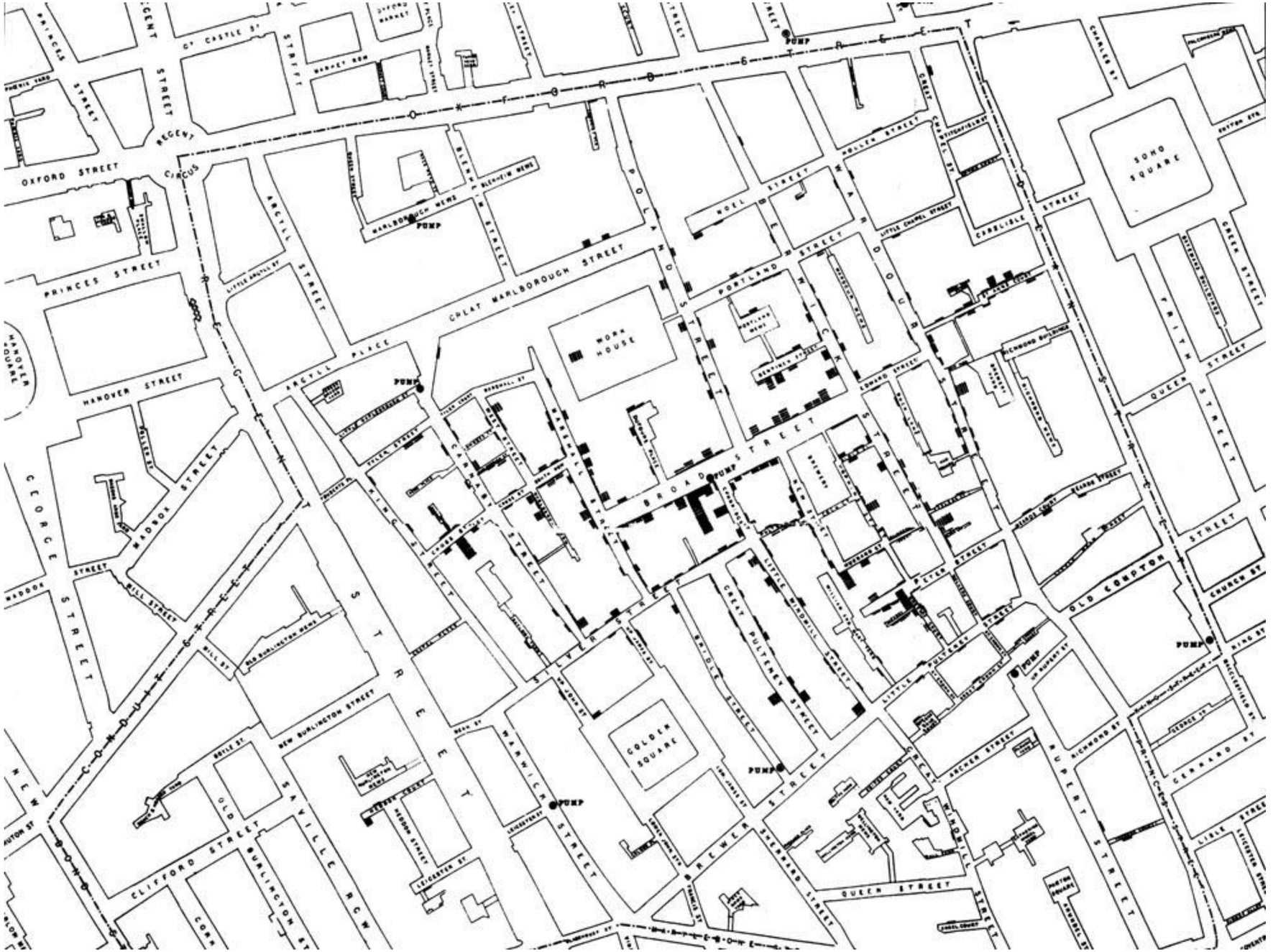
Associate Professor in **Visualization**
Univ. Bergen



Visualization for Communication

…the creation and use of visuals to express ideas, communicate meaning, and to invite exploration of information.

Myelin, David Goodsell (PDB-101, 2020)



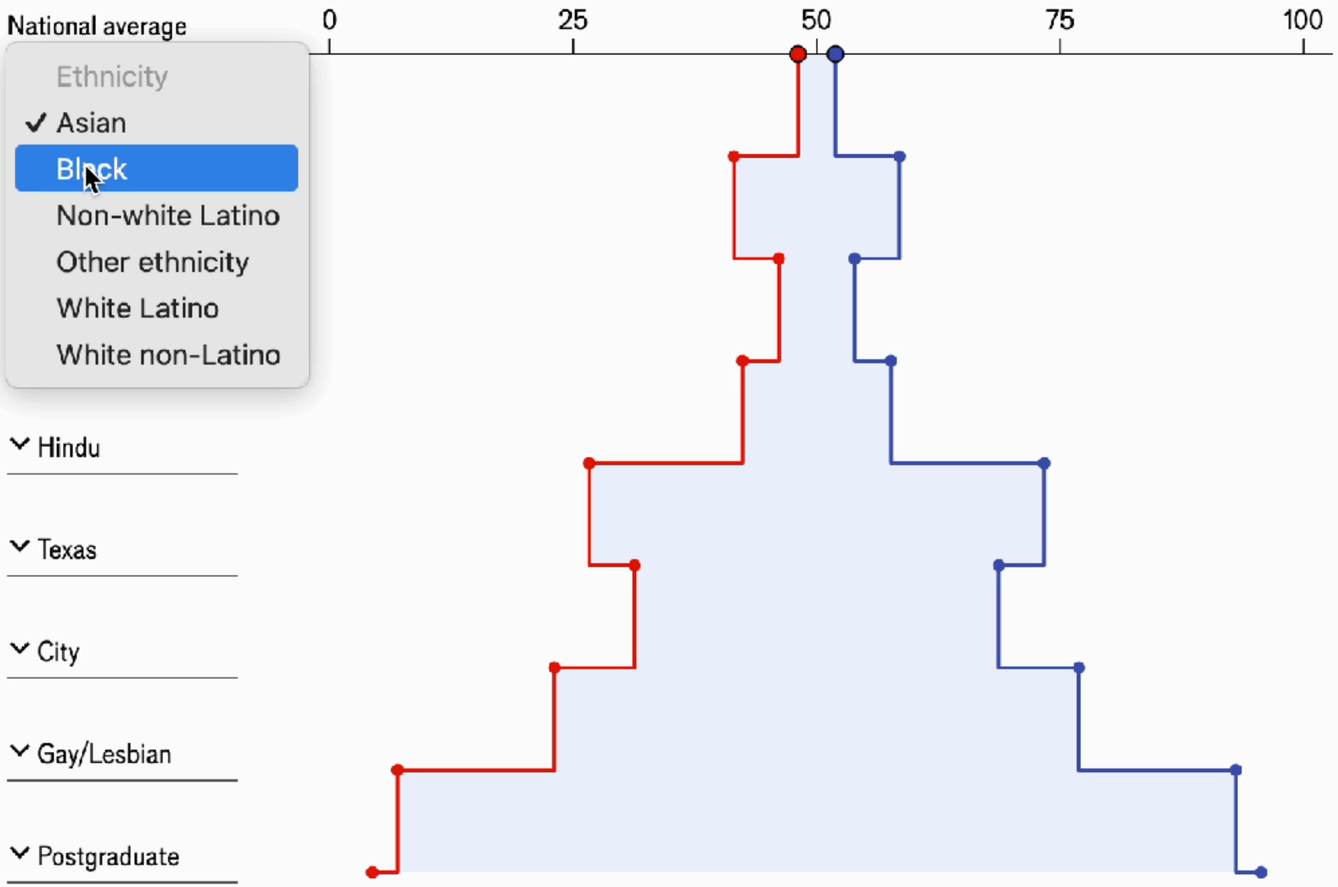
Cholera Outbreak Map

by John Snow (1854)

Build a voter
The Economist

Build an American voter

Chance of voting for candidate*, %

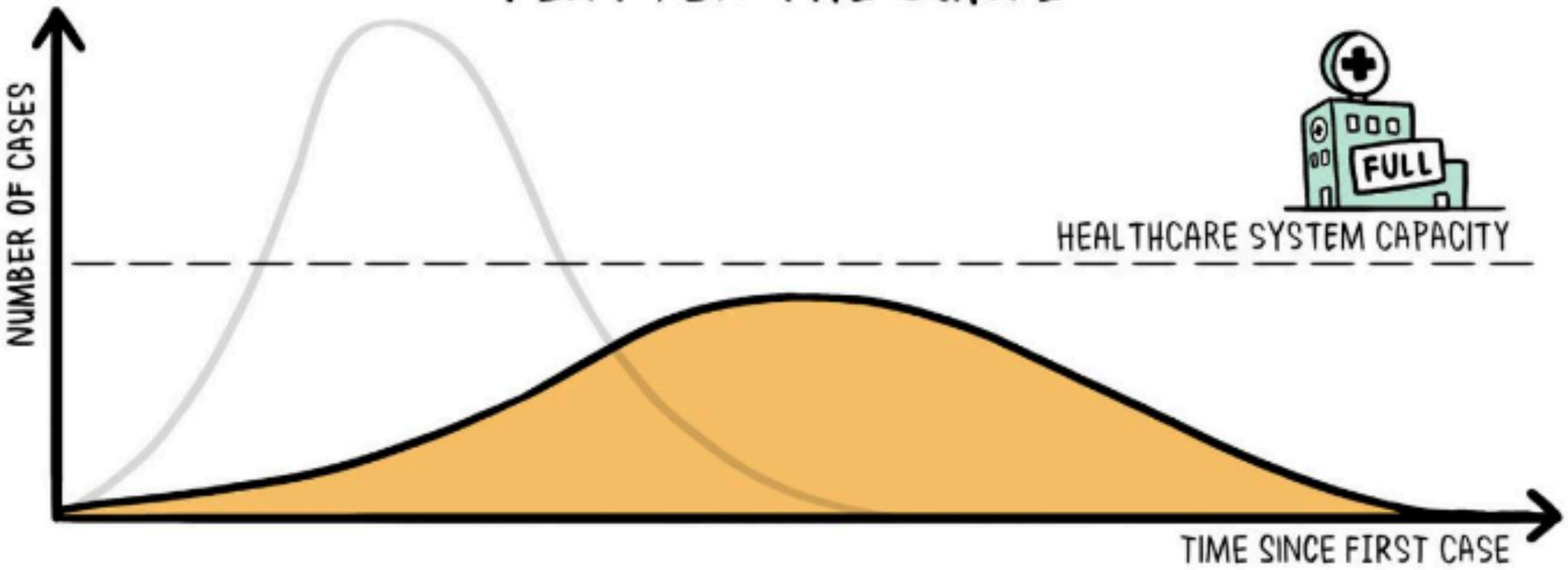


	Trump	Harris
2024 prediction	4%	96%
2020 estimate	5%	95%

How this group compares to others

Gay Asian men aged 35-44 from a city in Texas, who are Hindu and

FLATTEN THE CURVE



@SIOUXSIEW @XTOTL @THESPINOFFTV

'ADAPTED FROM @DREWAHARRIS, THOMAS SPLETTSTÖßER (@SPLETTE) AND THE CDC'

CC-BY-SA

https://issuu.com/smcnz/docs/drawing_science_guide_web (p. 3)

Important in peer communication!

Higher impact correlates with more diagrams, more plots [1]

***Reflect**: How do you review papers? What makes you spend more time with a paper?*

[1] Viziometrics: Analyzing Visual Information in the Scientific Literature, P. Lee, J. D. West, and B. Howe, IEEE Transactions on Big Data 2018.

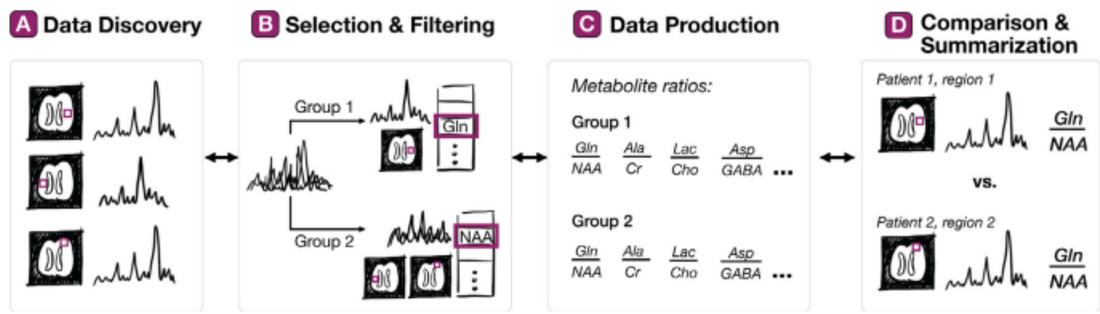


Fig. 2. Typical task flow for MRS data analysis. Users begin with data discovery (A) to review spatial voxel position, associated spectral graphs, and relevant acquisition parameters. (B) continues with data selection and filtering, where spectral voxels of interest are selected and divided into groups. Data production (C) calculates all possible ratios of selected metabolites, e.g., Glutamine (Gln) to N-acetylaspartate (NAA). In (D) ratios are compared and summarized between, e.g., Gln/NAA for different patients or different brain regions. Each of theses steps may be revisited.

differences between healthy and hypoxic newborns. This can be achieved by evaluating ratios of the same metabolites between both groups, e.g., NAA healthy vs. NAA hypoxic. Futhermore, researchers would like to understand the metabolic profile of hypoxic newborns on a spatial and individual level. For instance, the basal ganglia region of the brain is known to be sensitive to oxygen deprivation, so it is clinically relevant to compare this region to a less sensitive region. Within a given region of interest researchers then wish to compare individuals to identify clinically relevant outliers in order to answer questions such as *"How does Lactate/Choline compare for Patient X versus Y?"* Moreover, oxygen-deprived newborns who survive often experience developmental disabilities later in life. Longitudinal MRS studies allow researchers to understand how the metabolic profiles of affected individuals change over time relative to healthy individuals. In a different scenario, researchers studying schizophrenia are interested in comparing the metabolic profiles of individuals when their brains are *active* relative to their *resting* brain state. Different metabolites present in different concentrations in these states, and identification of these differences may help progress understanding of this disorder.

Following comparison of interesting metabolite ratios, researchers often wish to refine their hypotheses and revisit metabolic input data. This task sequence then repeats, following an iterative analytical approach to hypothesis exploration and verification in MRS data.

4.2. Design requirements

Following the identification of tasks important for our collaborators in MRS analysis, we developed the design requirements for our application. First, on a technical and infrastructure level, our colleagues often switch between hospital workstations while accessing sensitive patient data. Thus, for practical utility it is critical to provide a tool that enables a *machine-independent workflow* (**R1**)

Based on the types of questions outlined in **T4**, users must be able to *compare metabolite ratios of interest* (**R5**). This should be accomplished for any permutation through four key attributes: *spatial region, individual, time point, and brain activity state*. Additionally, appropriate mechanisms to compare ratios over time as well as between spatial regions and individuals are critical for longitudinal or single-run studies. Furthermore, for functional MRS studies it is important to support comparison of metabolite ratios in an active relative to a resting brain state.

5. SpectraMosaic workflow and interface

We provide an overview of the SpectraMosaic interface in Fig. 3. Following an offline processing step, data are loaded into the web tool (Fig. 3(A)). Data of interest for analysis can be explored, selected, and added (Fig. 3(B)–(D)) to a spectral ratio heatmap for deeper inquiry and hypothesis verification (Fig. 3(E)). A legend provides information on the encodings used in the tool (Fig. 3(F)). A table below the heatmap summarizes salient acquisition information (Fig. 3(G)).

Data processing and loading. We first perform an offline processing step that automates spectral processing and quantification from Tarquin and MATLAB [33]. We utilize MATLAB to process the structural imaging files, which includes patient data de-identification (**R2**). The resulting output contains a structural image to localize the voxel sample, the spectral graph, quantified metabolites, and associated metadata; these data remain semantically linked in the visual tool. We use a custom data format because the DICOM standard is not universally or consistently adopted for MRS data.

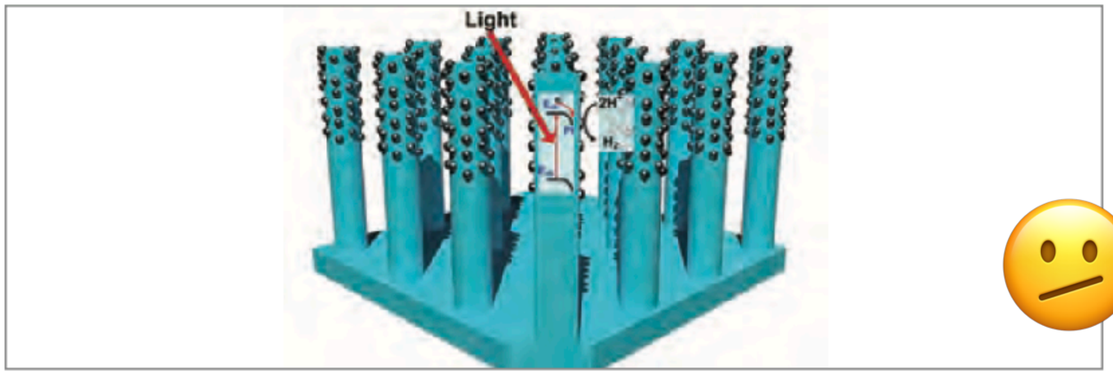
Visual inspection of voxel positioning and spectral graphs. Following data loading (Fig. 3(A)), the spatial voxel overview panel (Fig. 3(B)) is used to review the spectral graph, associated anatomical image, and included metadata for each acquisition. This panel consists of a set of images for each patient. In each structural im-

Important in peer communication!

High-quality visualizations are perceived as…

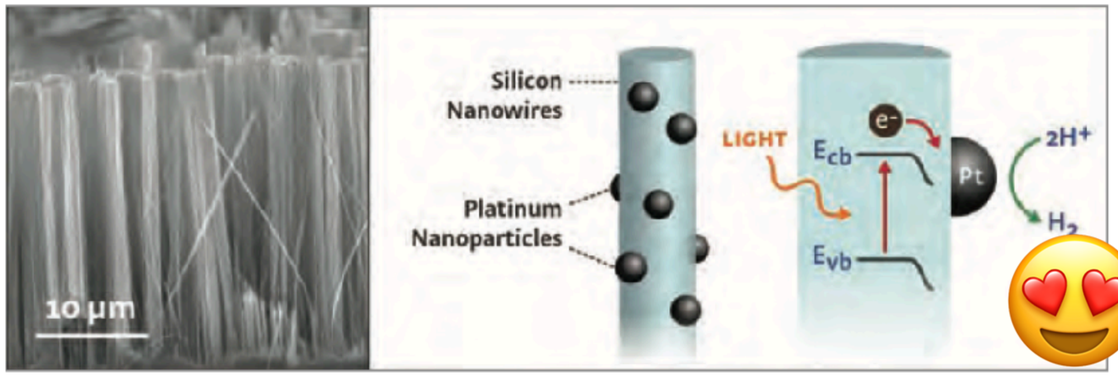
- More **interesting**
- **Clearly written**
- More **scientifically rigorous**

Before redesign:



😐

After redesign:

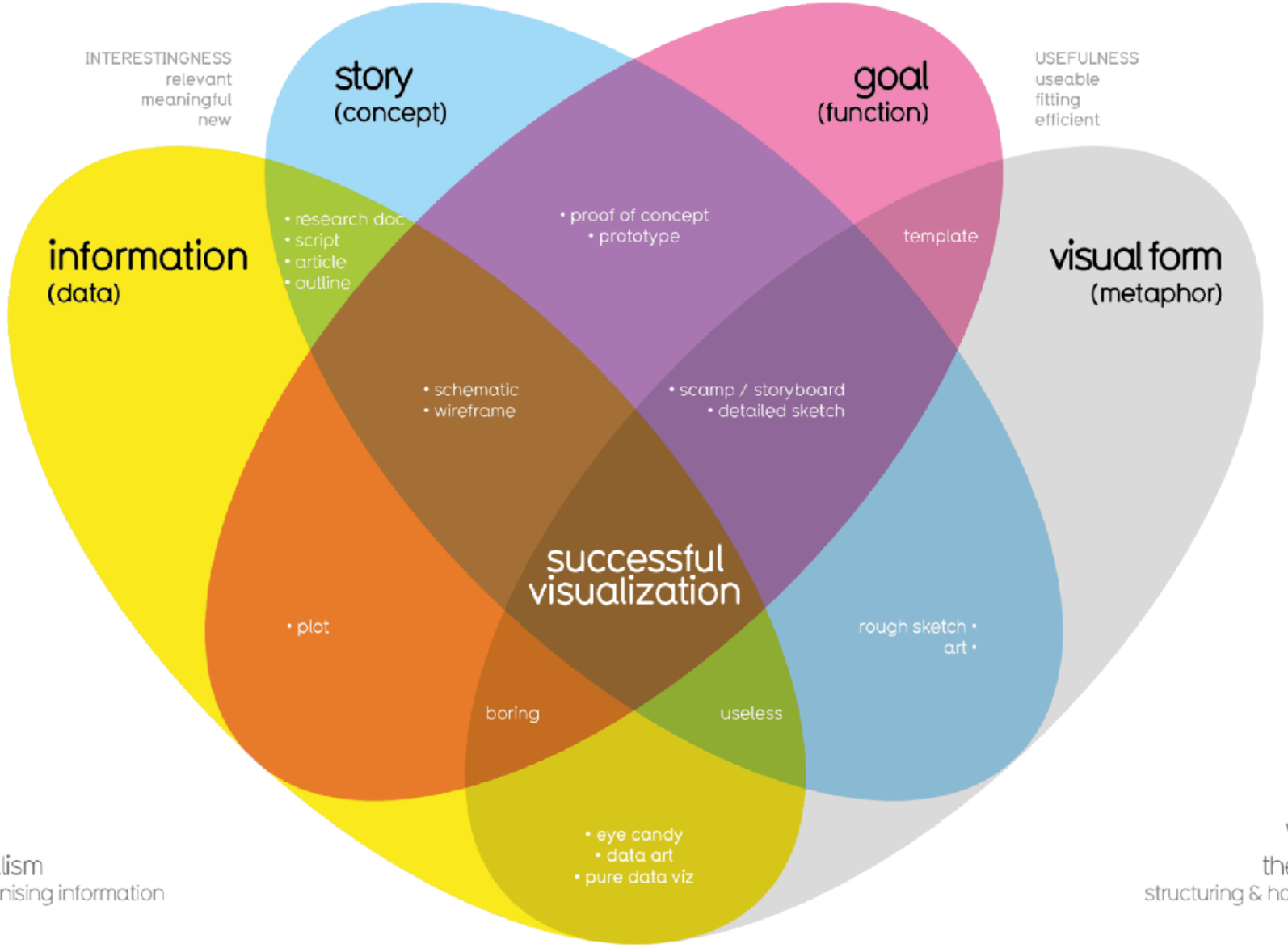


😍

Proving the Value of Visual Design in Scientific Communication, K. Cheng et al., Information Design Journal 2017 *(ref + image source)*

What Makes a Good Visualization?

explicit (implicit)



information
the art of journalism
structuring & harmonising information

visualization
the art of design
structuring & harmonising visuals

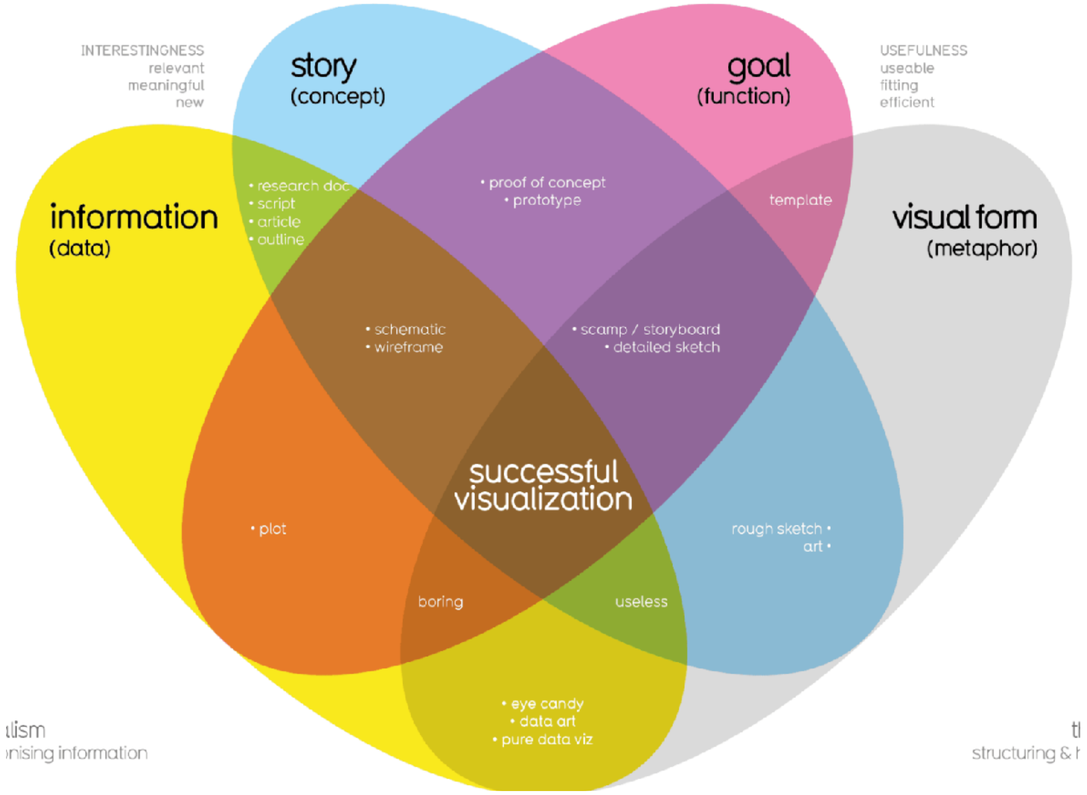
David McCandless
InformationisBeautiful.net

taken from new book
Knowledge is Beautiful

find out more
bit.ly/KIB_Books

⚒ Fundamentals of a “good” visualization

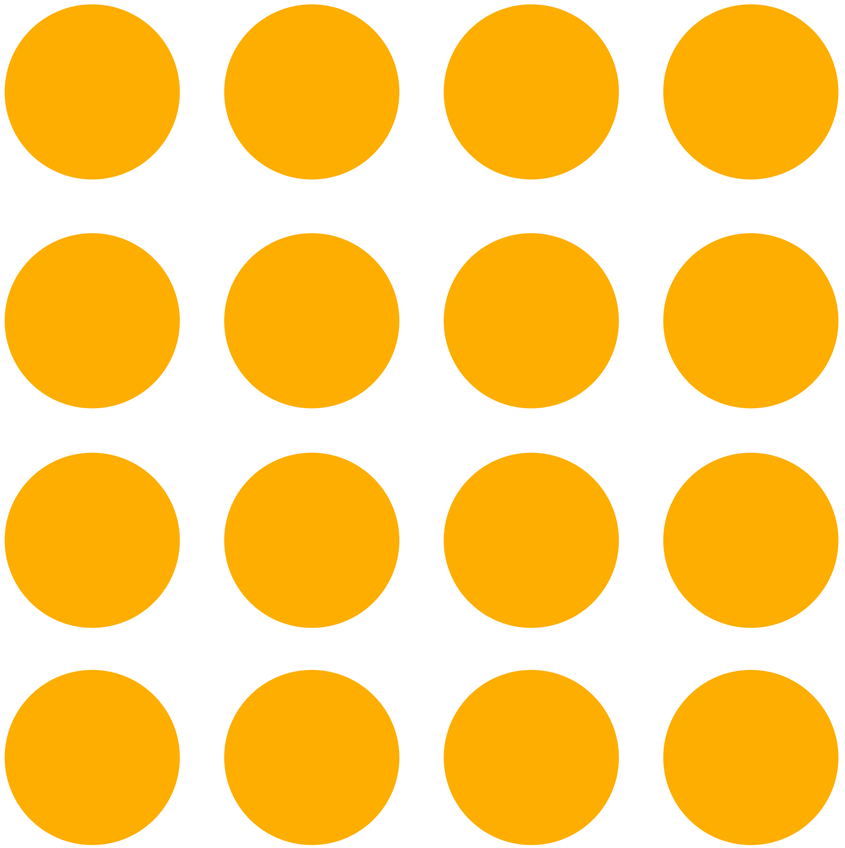
1. Show **information** integrity
2. Main **goal** to create something useful and fitting to your audience
3. Make an interesting **story**
4. Don’t skip (or overdo) **aesthetics**



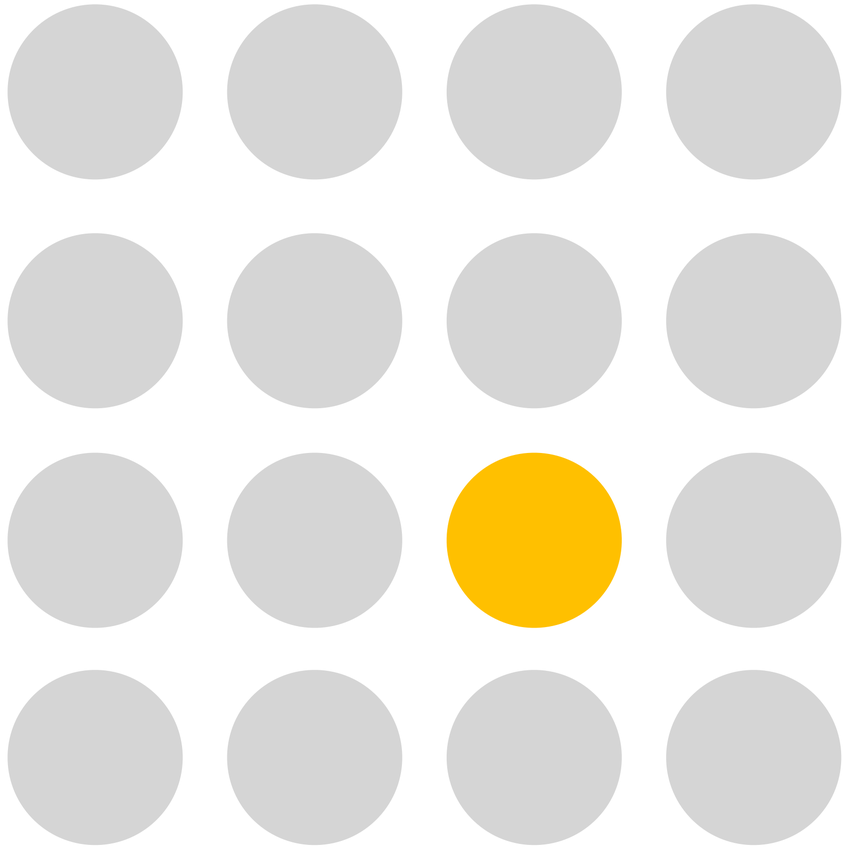
Goal

Make a functional visualization for your audience.

Visuals are powerful.

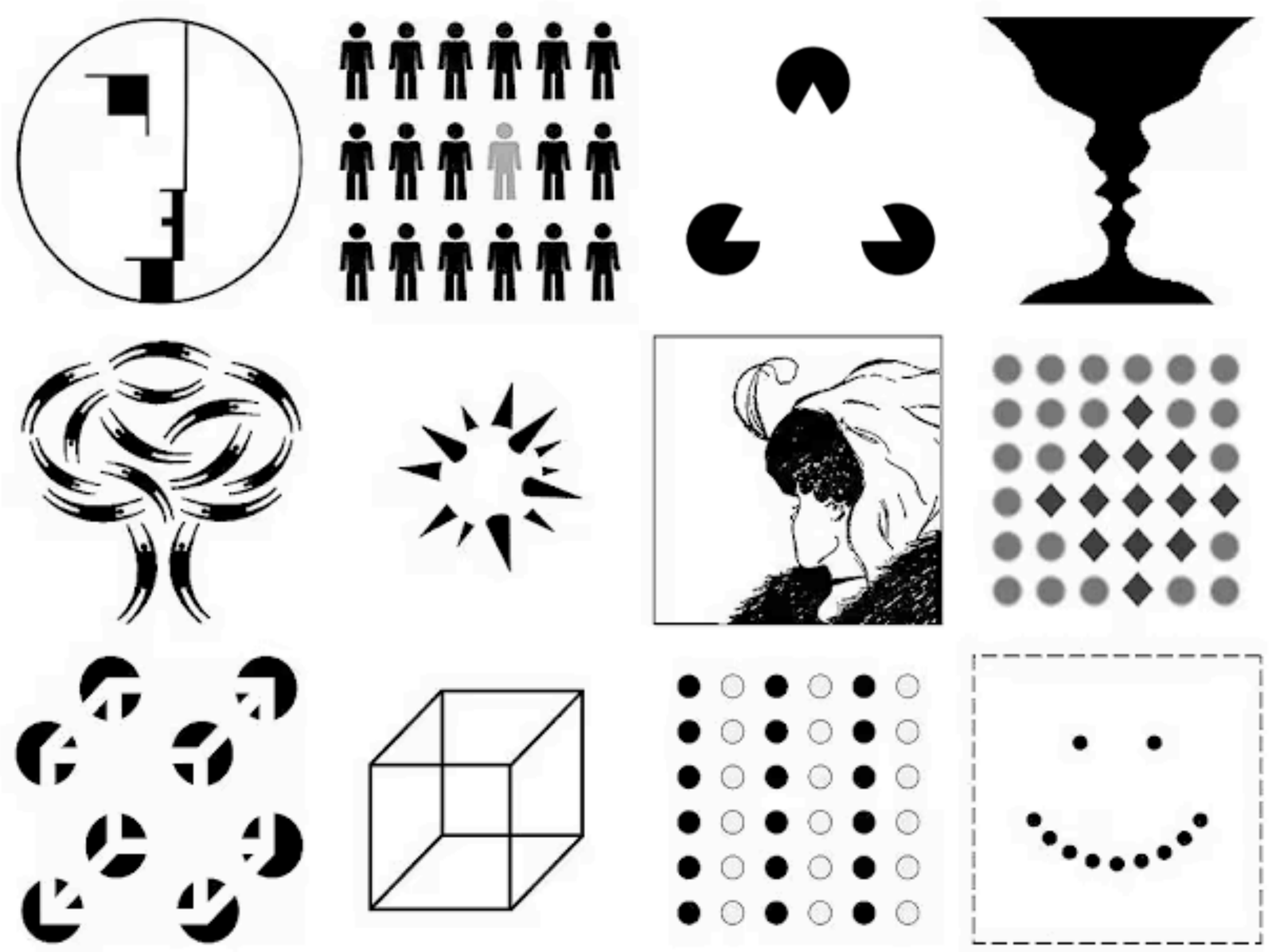


See **patterns….**

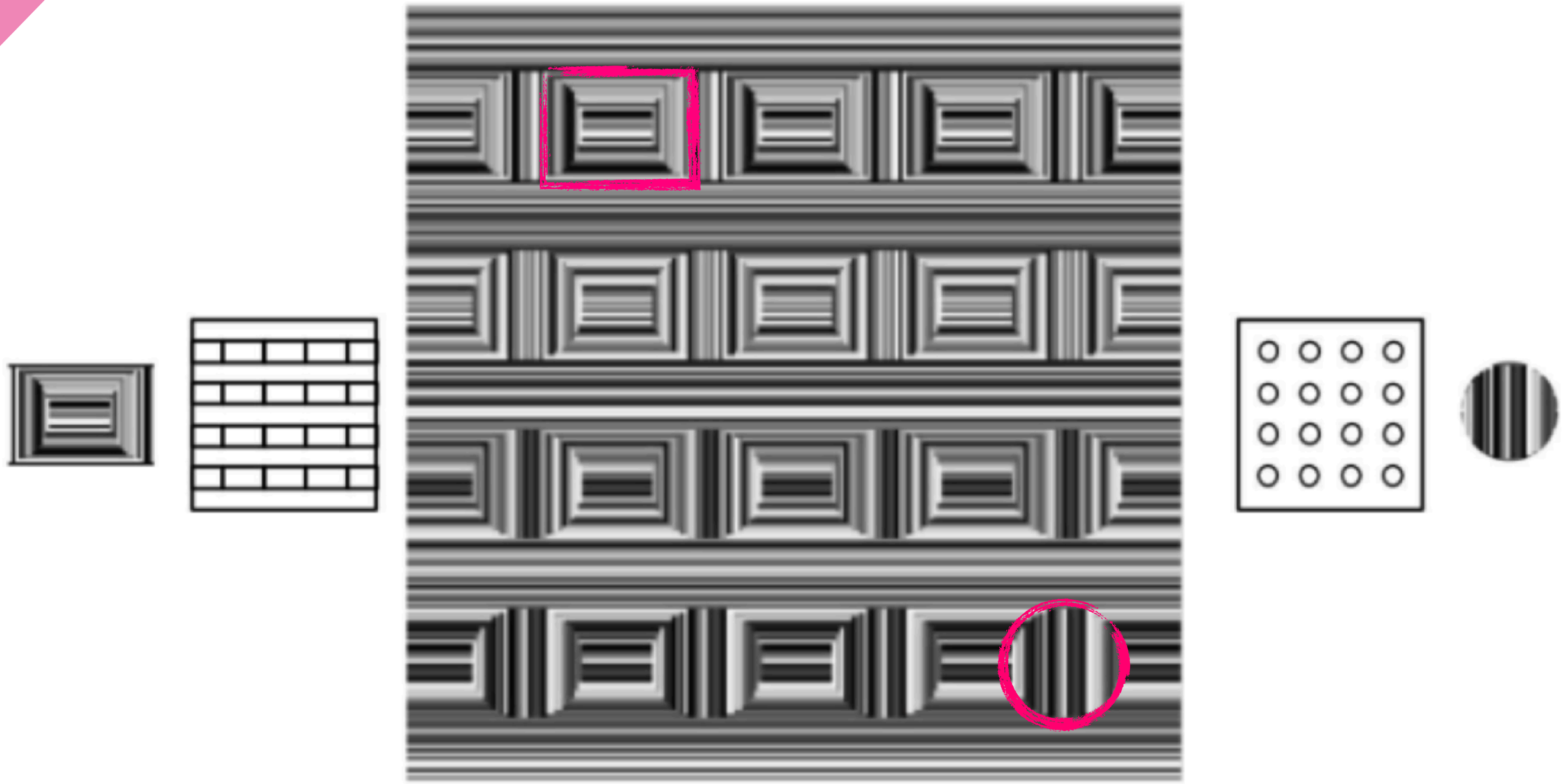


…and **exceptions**.

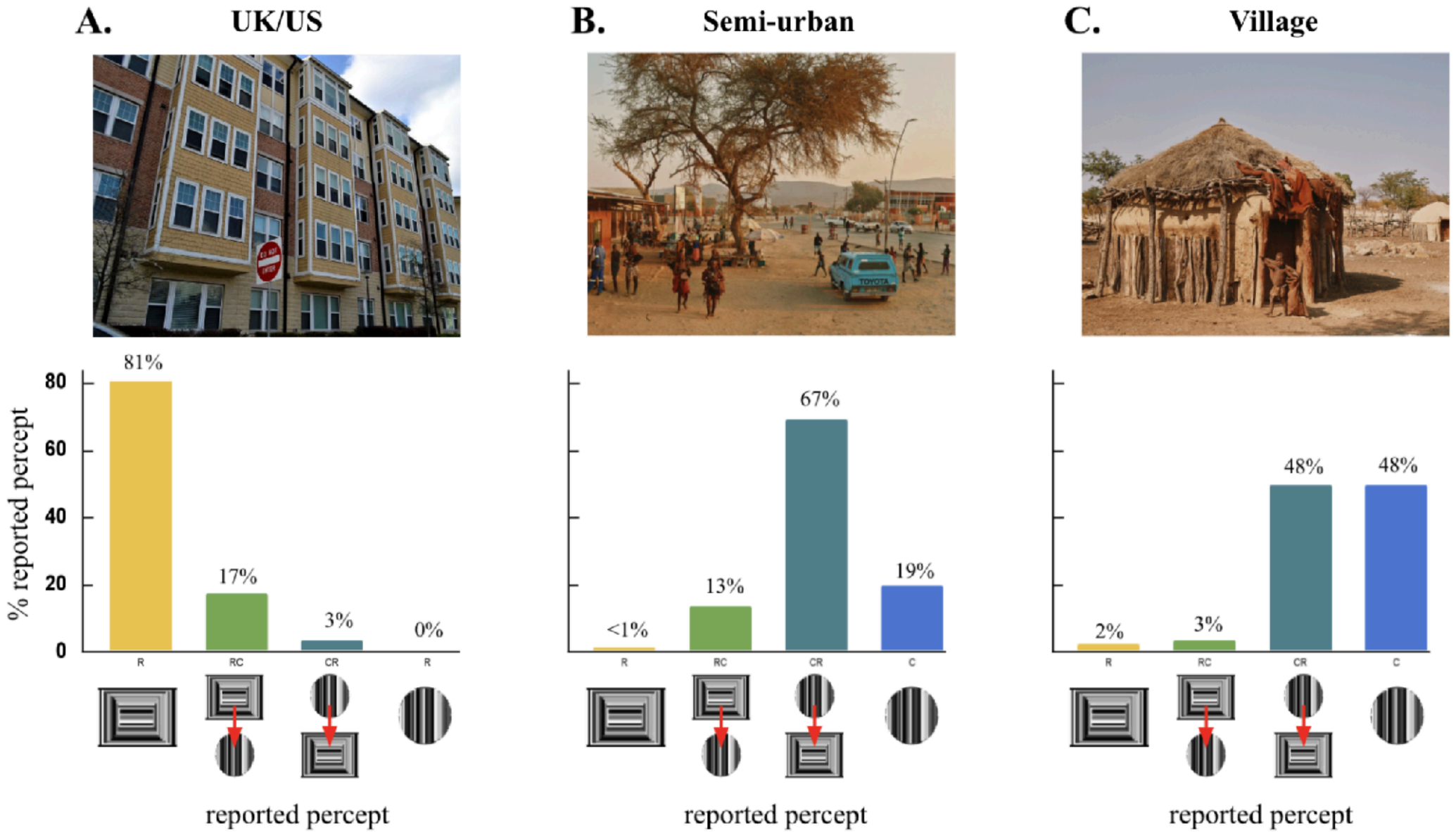
Are gestalt principles really universal?



Gestalt psychology. Image source: https://www.interaction-design.org/literature/topics/gestalt-principles



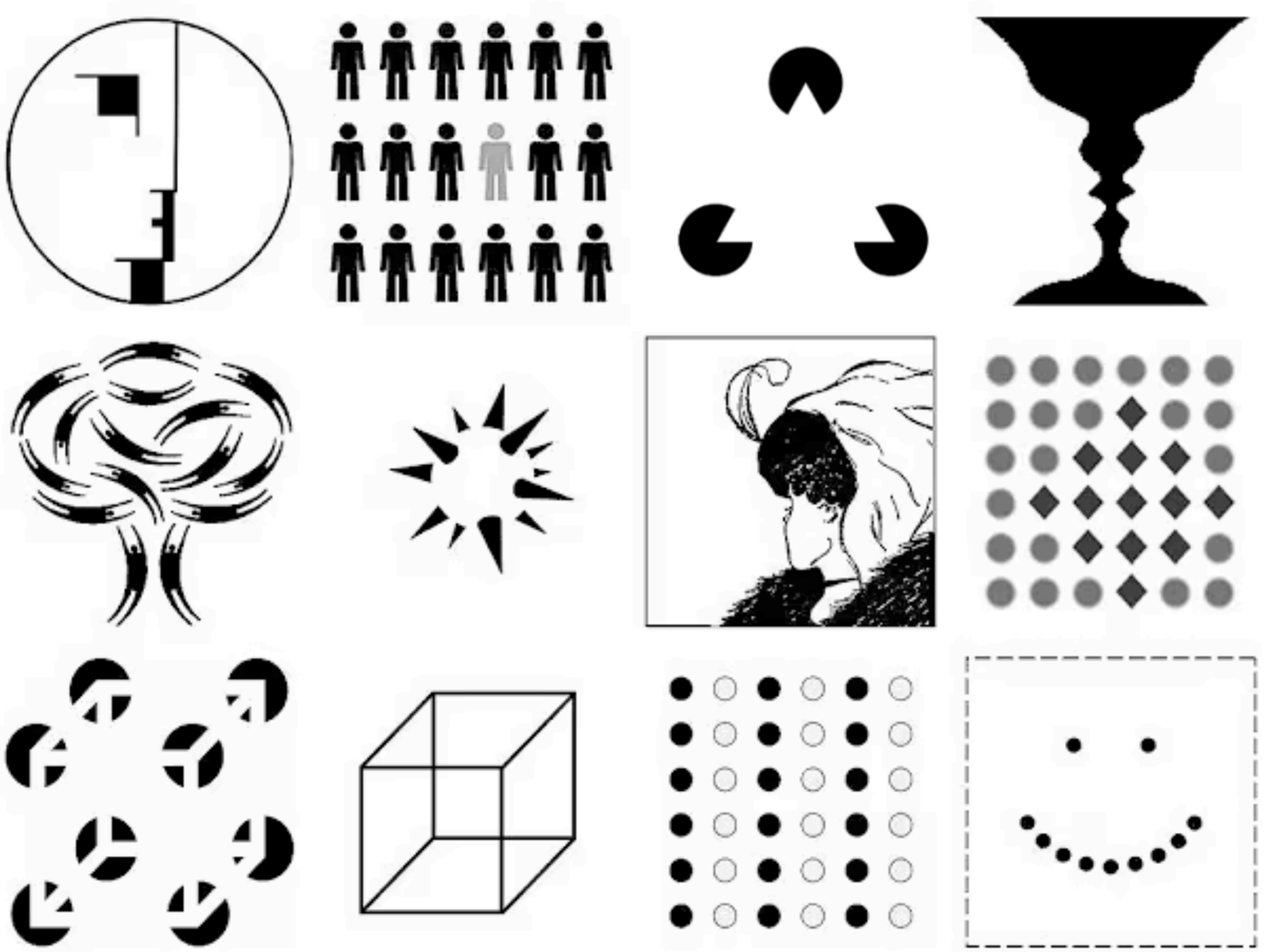
Kroupin et al., Visual illusions reveal wide range of cross-cultural differences in visual perception, *psyArxiv* 2025



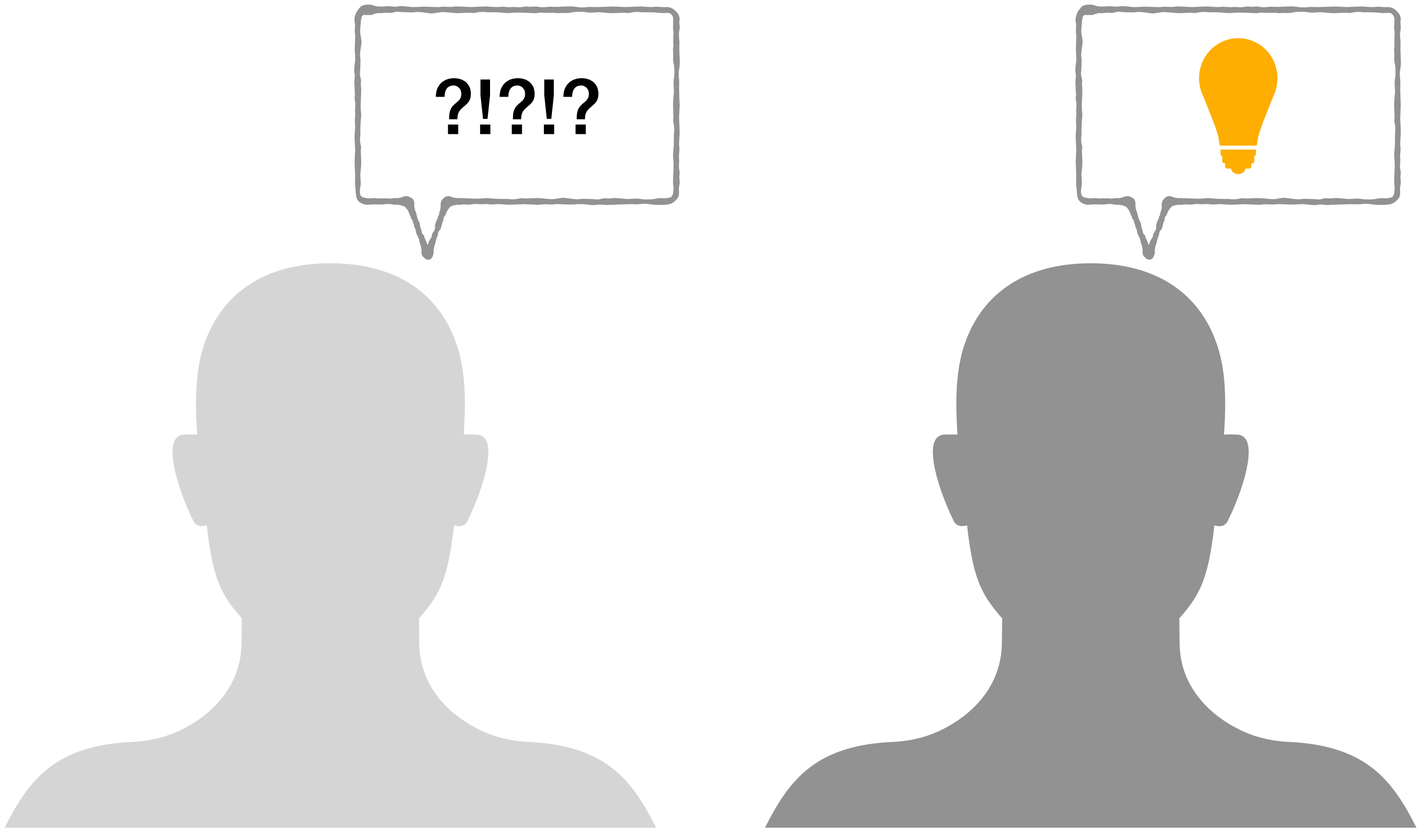
Kroupin et al., Visual illusions reveal wide range of cross-cultural differences in visual perception, *psyArxiv* 2025

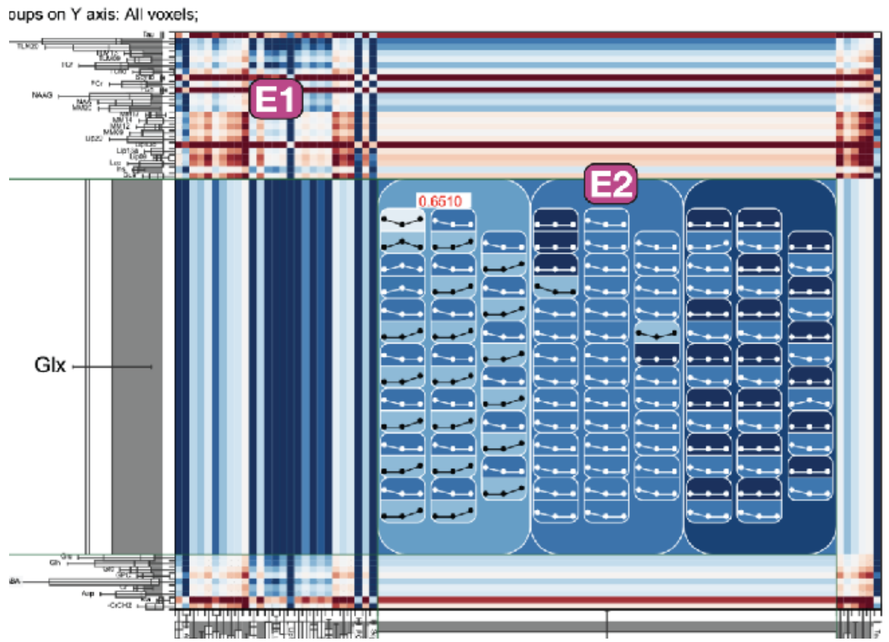
~~Are gestalt principles really universal?~~

Gestalt principles are learned and context-dependent



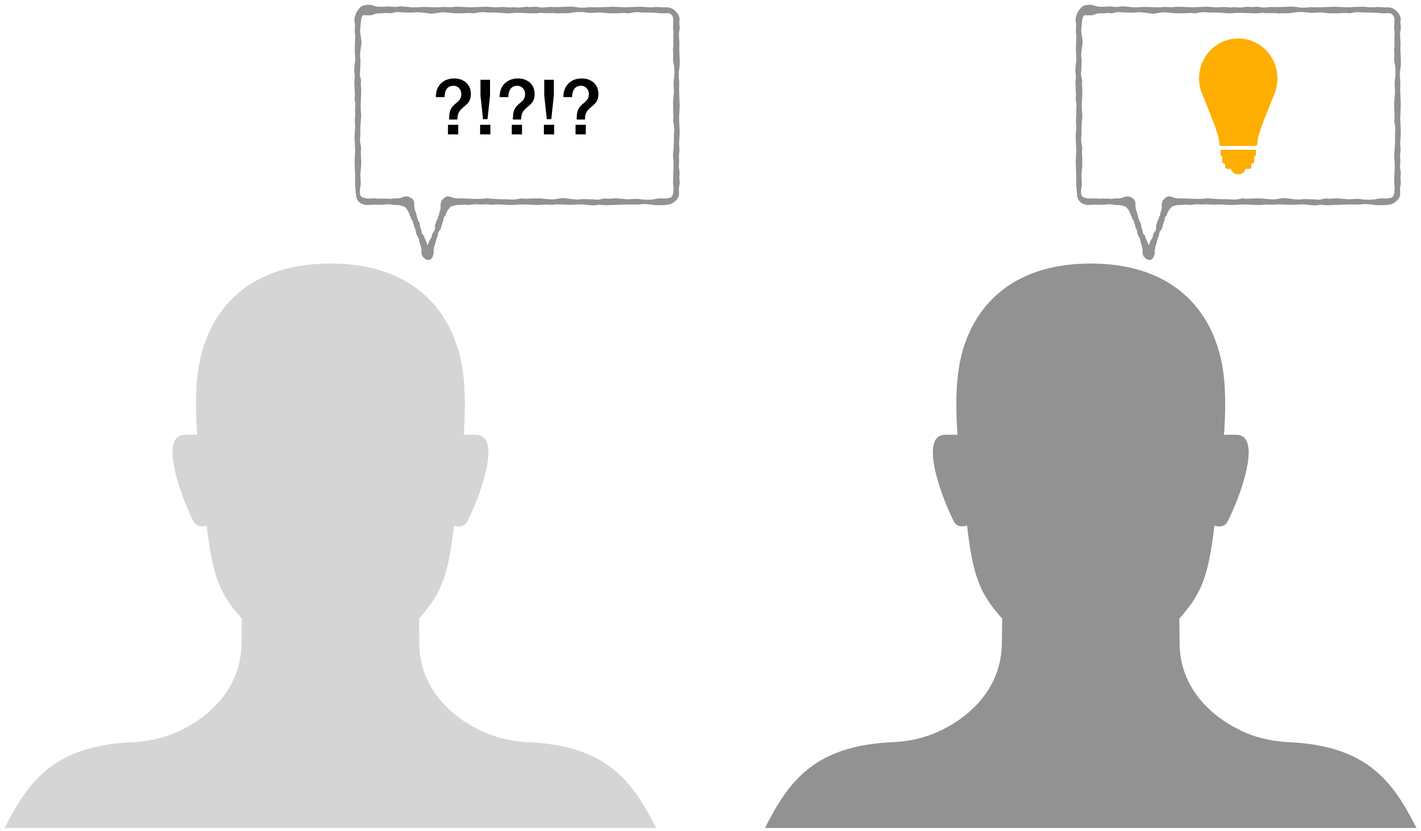
Intended audience / user / reader

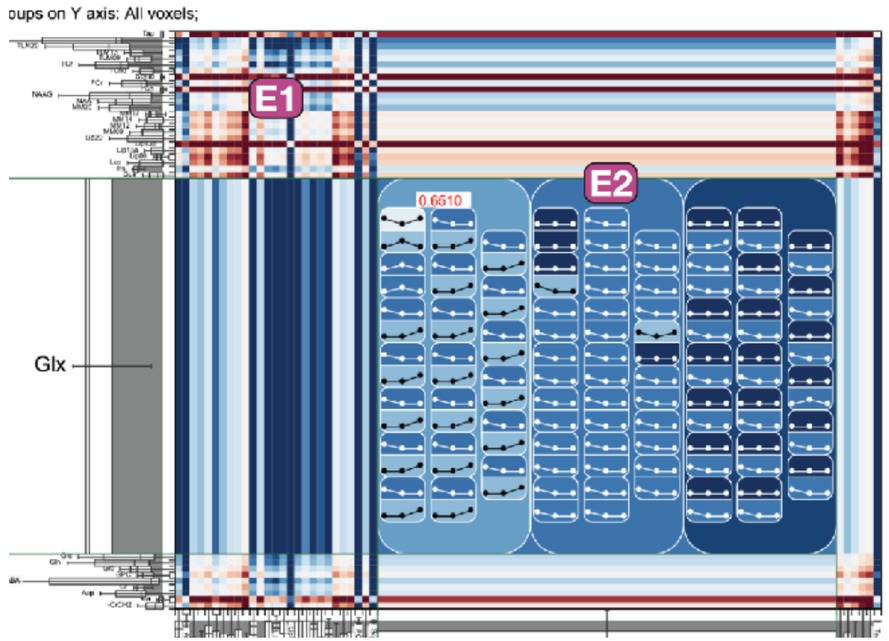




L. Neeley et al., “Linking Scholarship and Practice: Narrative and Identity in Science,” Frontiers in Communication Vol. 5. 2020.
M. C. Green and T. C. Brock, “The Role of Transportation in the Persuasiveness of Public Narratives,” Journal of Personality and Social Psychology, Vol. 79, 2000.
A. C. Graesser et al., “Advanced Outlines, Familiarity, and Text Genre on Retention of Prose,” The Journal of Experimental Education, Vol. 48, 1980

Intended audience / user / reader

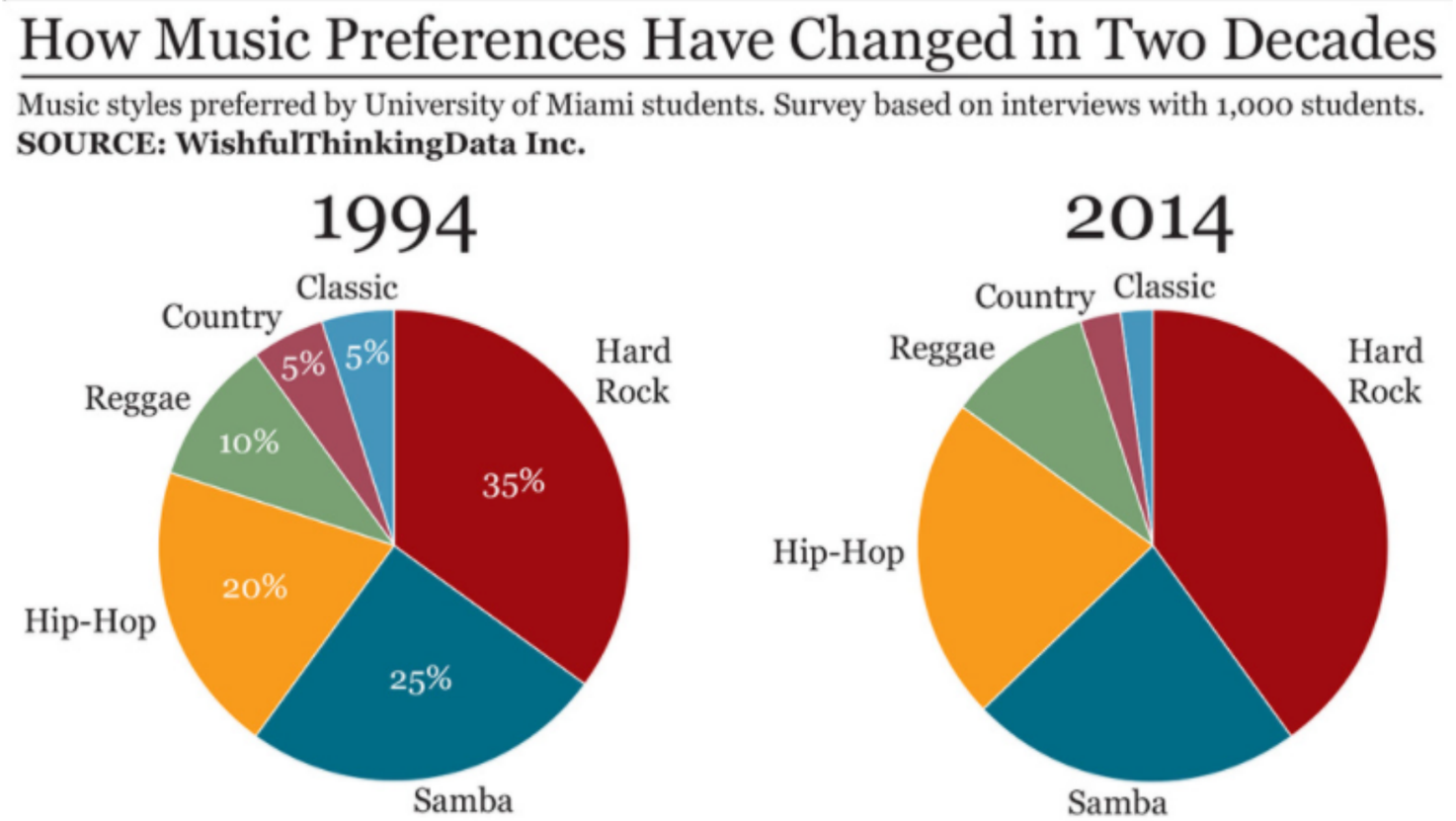




Relate/ see **oneself** in the situation or phenomenon

Goal helps restrict your visual design space

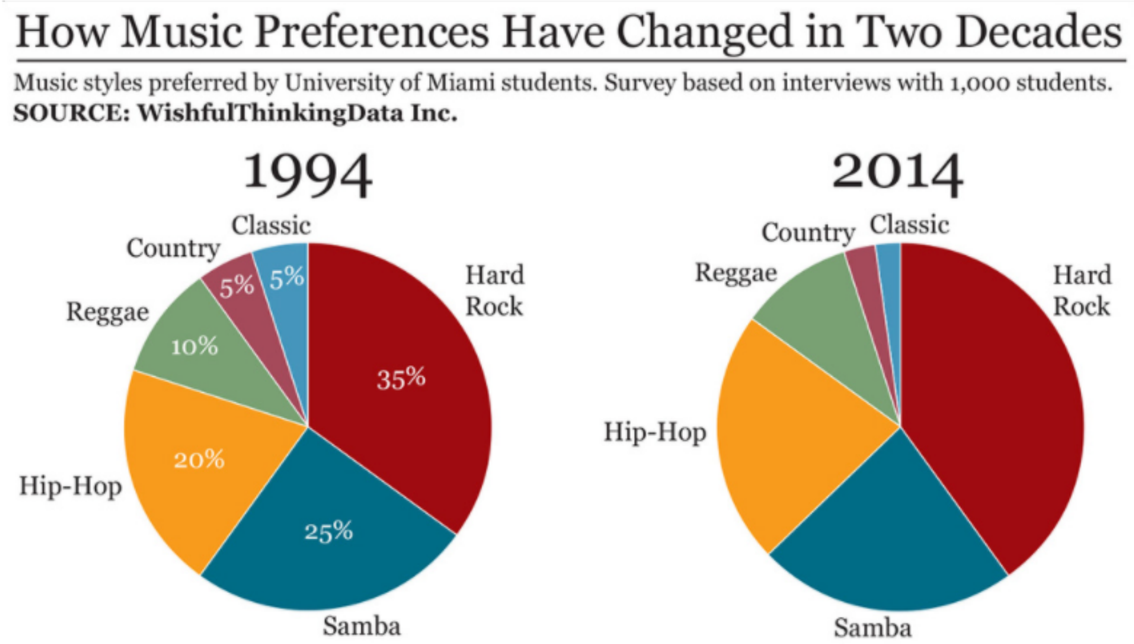
Compare?

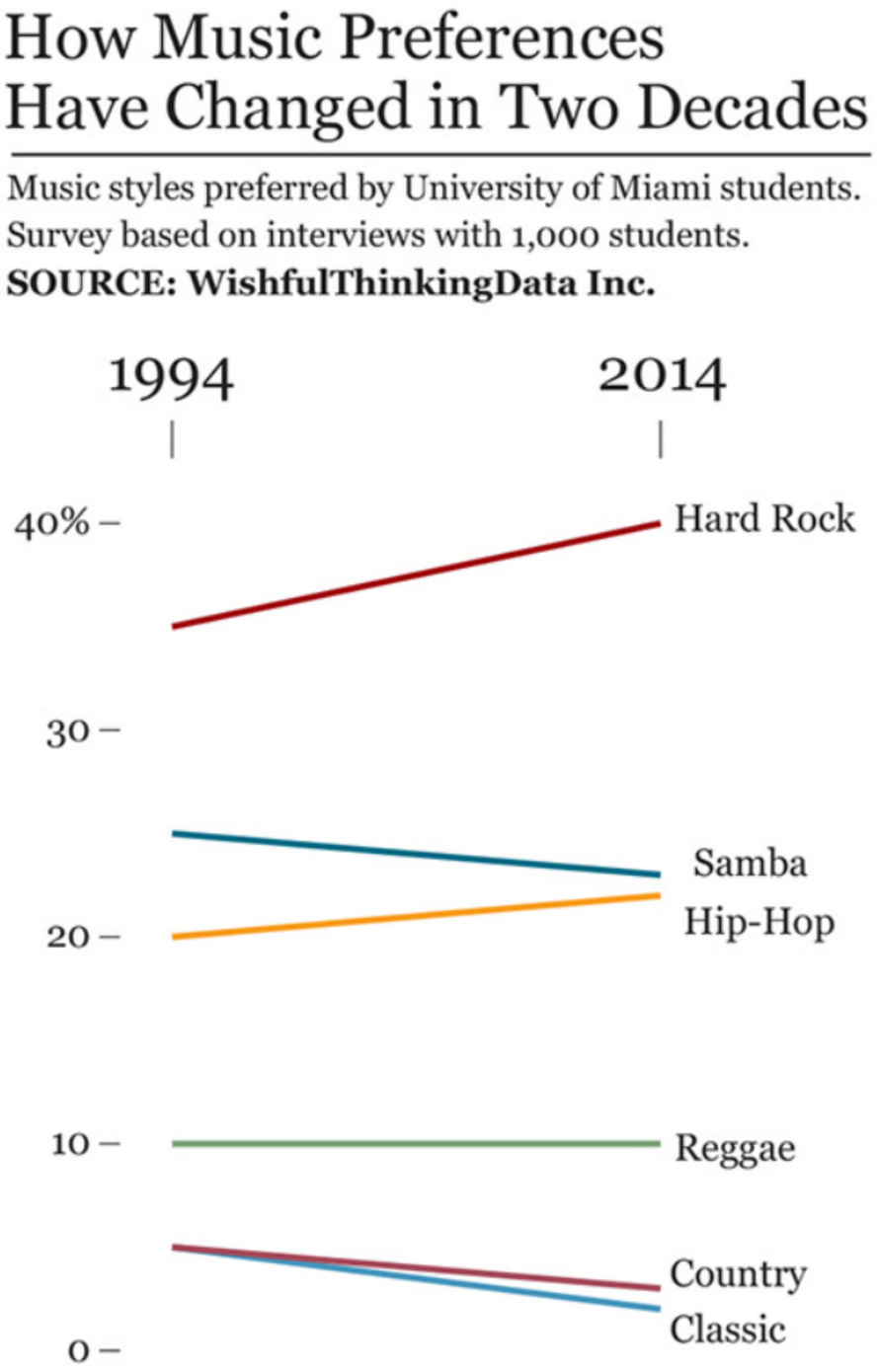


Alberto Cairo, *The Truthful Art*

Goal helps restrict your visual design space

Compare?





Alberto Cairo, *The Truthful Art*

Chart Suggestions—A Thought-Starter

www.ExtremePresentation.com
© 2009 A. Abela — a.v.abela@gmail.com

What would you like to show?

Comparison

- Among Items
  - Two Variables per Item
    - Variable Width Column Chart
  - One Variable per Item
    - Many Categories
      - Table or Table with Embedded Charts
    - Few Categories
      - Many Items
        - Bar Chart
      - Few Items
        - Column Chart
- Over Time
  - Many Periods
    - Cyclical Data
      - Circular Area Chart
    - Non-Cyclical Data
      - Line Chart
  - Few Periods
    - Single or Few Categories
      - Column Chart
    - Many Categories
      - Line Chart

Relationship

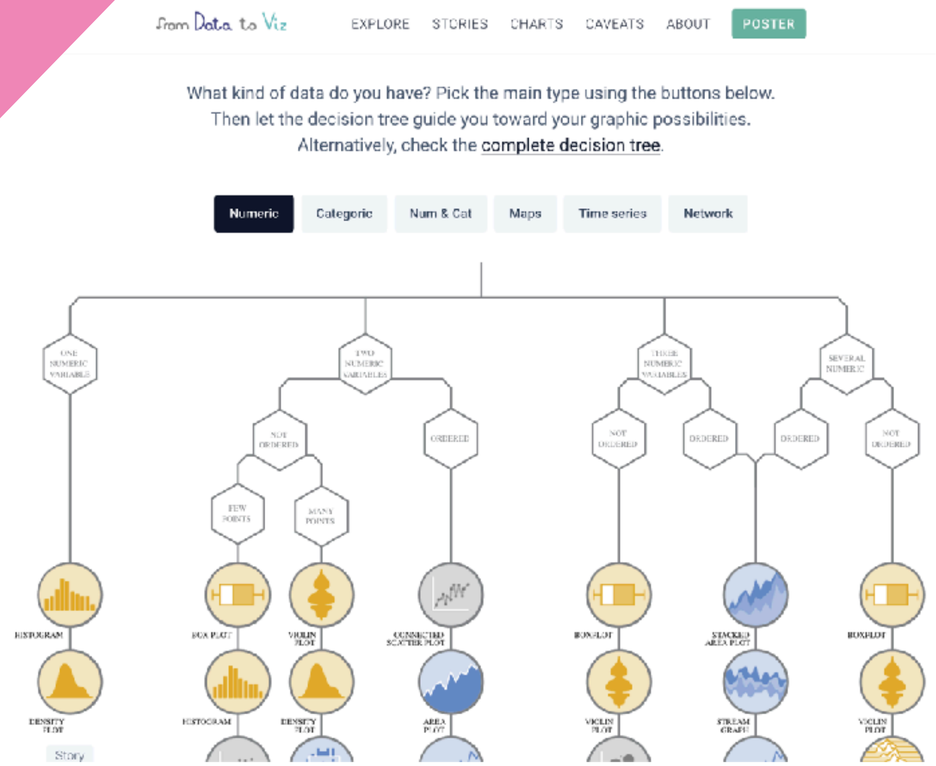
- Two Variables
  - Scatter Chart
  - Bubble Chart
- Three Variables

Distribution

- Single Variable
  - Few Data Points
    - Column Histogram
  - Many Data Points
    - Line Histogram
- Two Variables
  - Scatter Chart
- Three Variables
  - 3D Area Chart

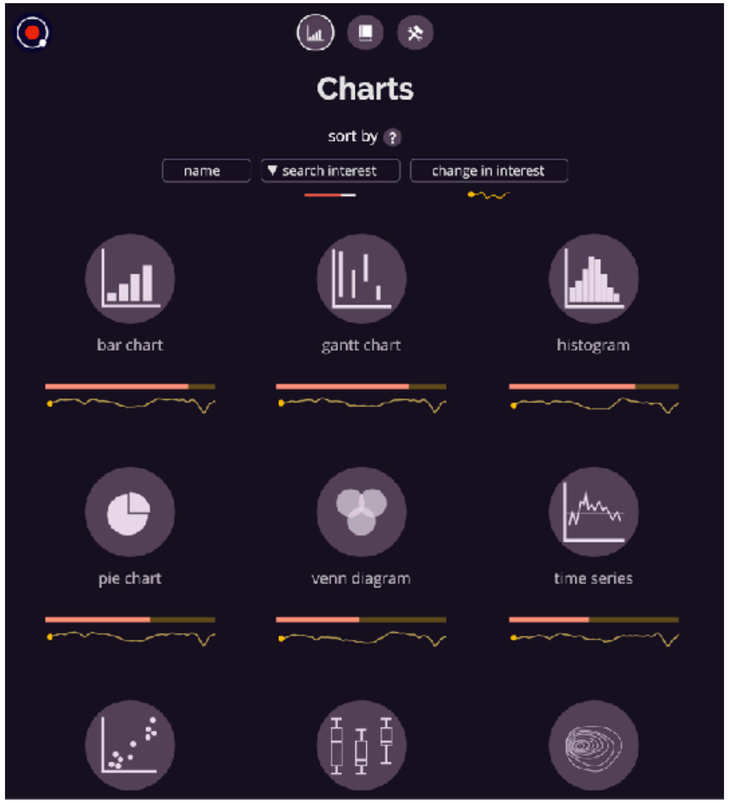
Composition

- Changing Over Time
  - Few Periods
    - Only Relative Differences Matter
      - Stacked 100% Column Chart
    - Relative and Absolute Differences Matter
      - Stacked Column Chart
  - Many Periods
    - Only Relative Differences Matter
      - Stacked 100% Area Chart
    - Relative and Absolute Differences Matter
      - Stacked Area Chart
- Static
  - Simple Share of Total
    - Pie Chart
  - Accumulation or Subtraction to Total
    - Waterfall Chart
  - Components of Components
    - Stacked 100% Column Chart with Subcomponents



data-to-viz.com

datavizproject.com



visualizationuniverse.adioma.com/charts

Visual Vocabulary

Designing with data

There are so many ways to visualise data – how do we know which one to pick? Click on the coloured categories below to decide which data relationship is most important in your story, then look at the different types of chart within the category to form some initial ideas about what might work best. This list is not meant to be exhaustive, nor a wizard, but is a useful starting point for making informative and meaningful data visualisations

Inspired by the Graphic Continuum by Jon Schwabish and Severino Ribecca

Deviation Correlation Change v Time Ranking Distribution Part to whole Magnitude Spatial Flow

Part to whole

Show how a single entity can be broken down into its component elements. If the reader's interest is solely in the size of the components, consider a magnitude-type chart instead

Examples of use

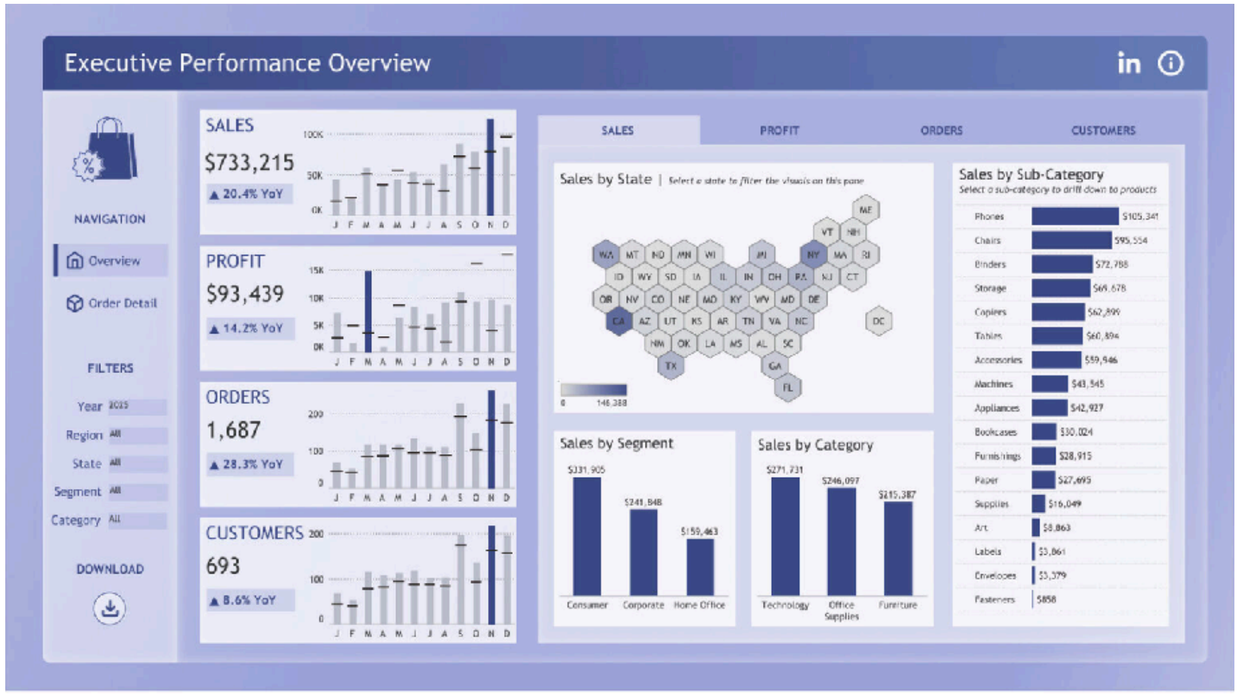
https://ft-interactive.github.io/visual-vocabulary/

Information

Understand your data, be **accurate**, and be **honest** in your depiction.

Take time to understand your data

If you use a genAI tool, check the outputs thoroughly.

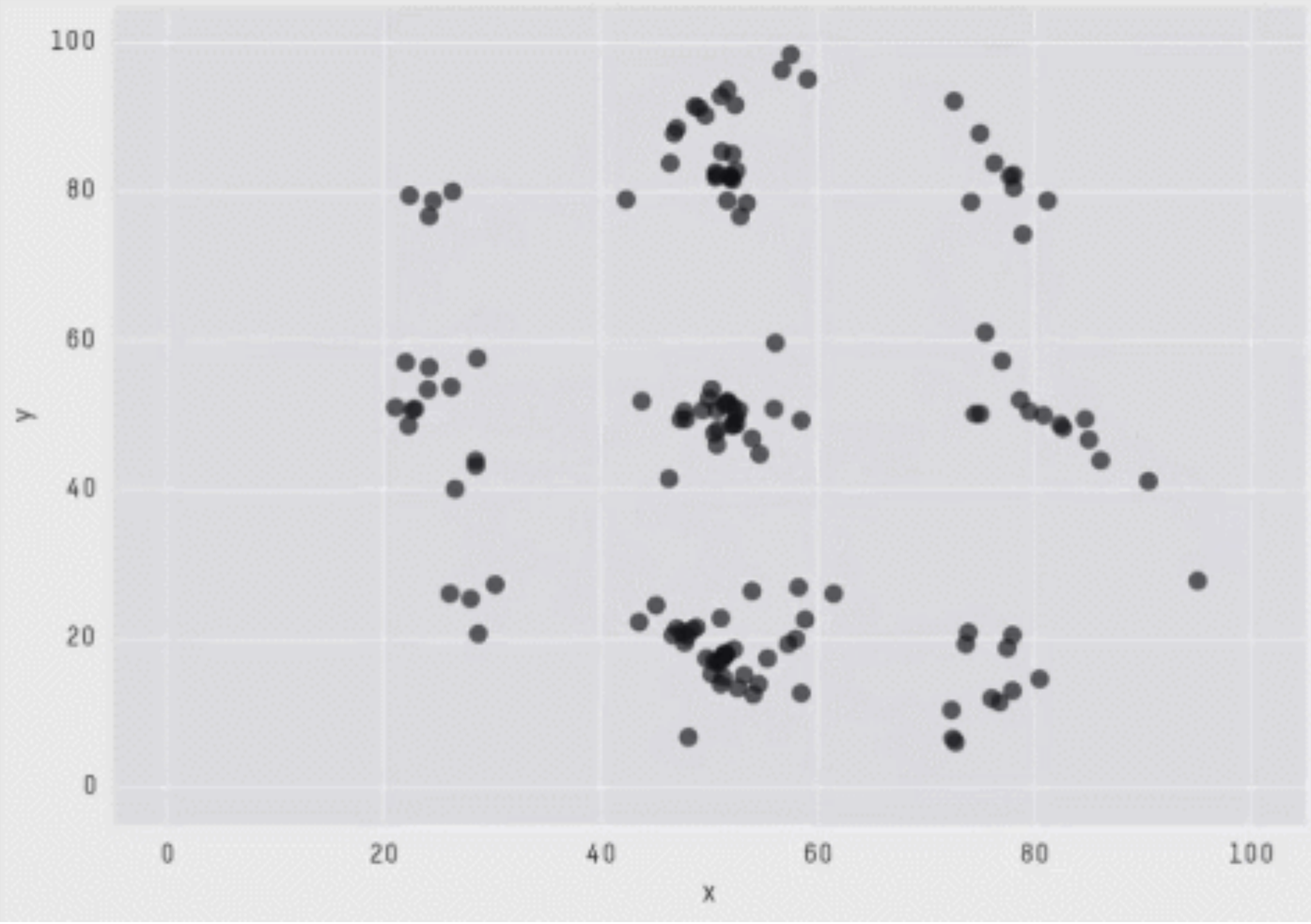


Superstore Executive Performance Dashboard

Claude remake

https://policyviz.com/2026/09/22/i-replicated-two-tableau-dashboards-with-claude-heres-what-broke-and-what-didnt/

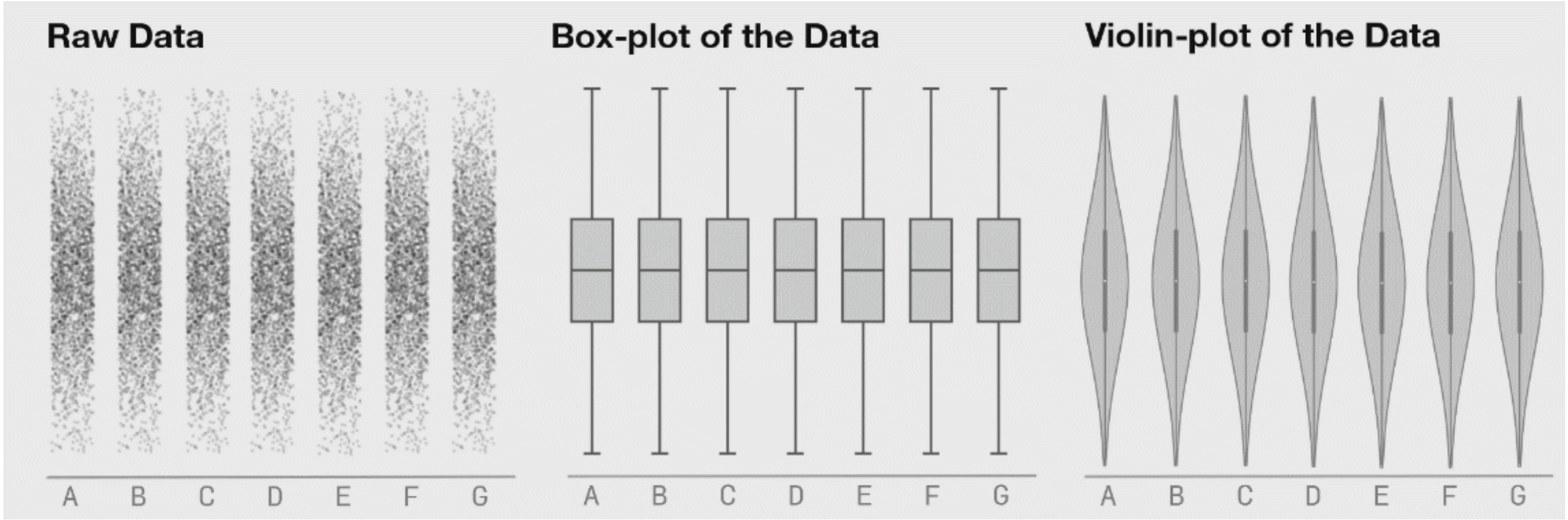
Show the data (beyond summaries)



X Mean: 54.2627564
Y Mean: 47.8334283
X SD : 16.7697622
Y SD : 26.9313112
Corr. : -0.0603570

Same stats, different graphs: generating datasets with varied appearance and identical statistics through simulated annealing. Matejka, J. and Fitzmaurice, G., Proc CHI 2017

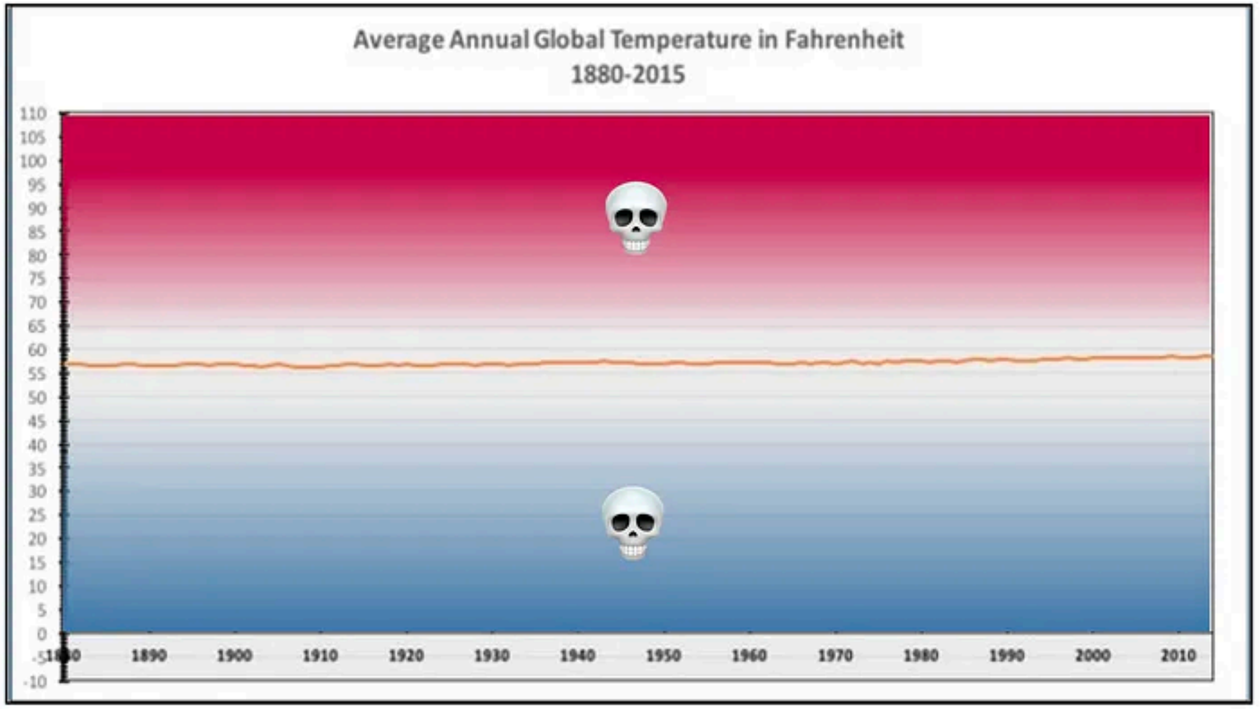
Show the data (beyond summaries)



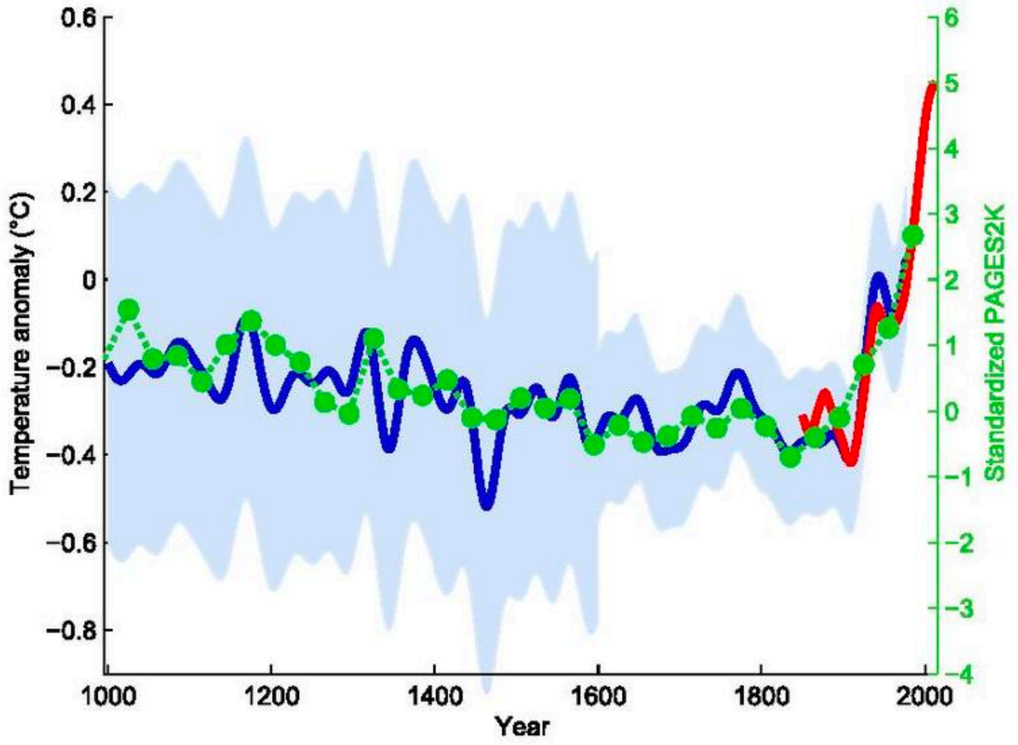
Same stats, different graphs: generating datasets with varied appearance and identical statistics through simulated annealing. Matejka, J. and Fitzmaurice, G., Proc CHI 2017

Be honest in how you show your data.

Which is more truthful to the phenomenon?



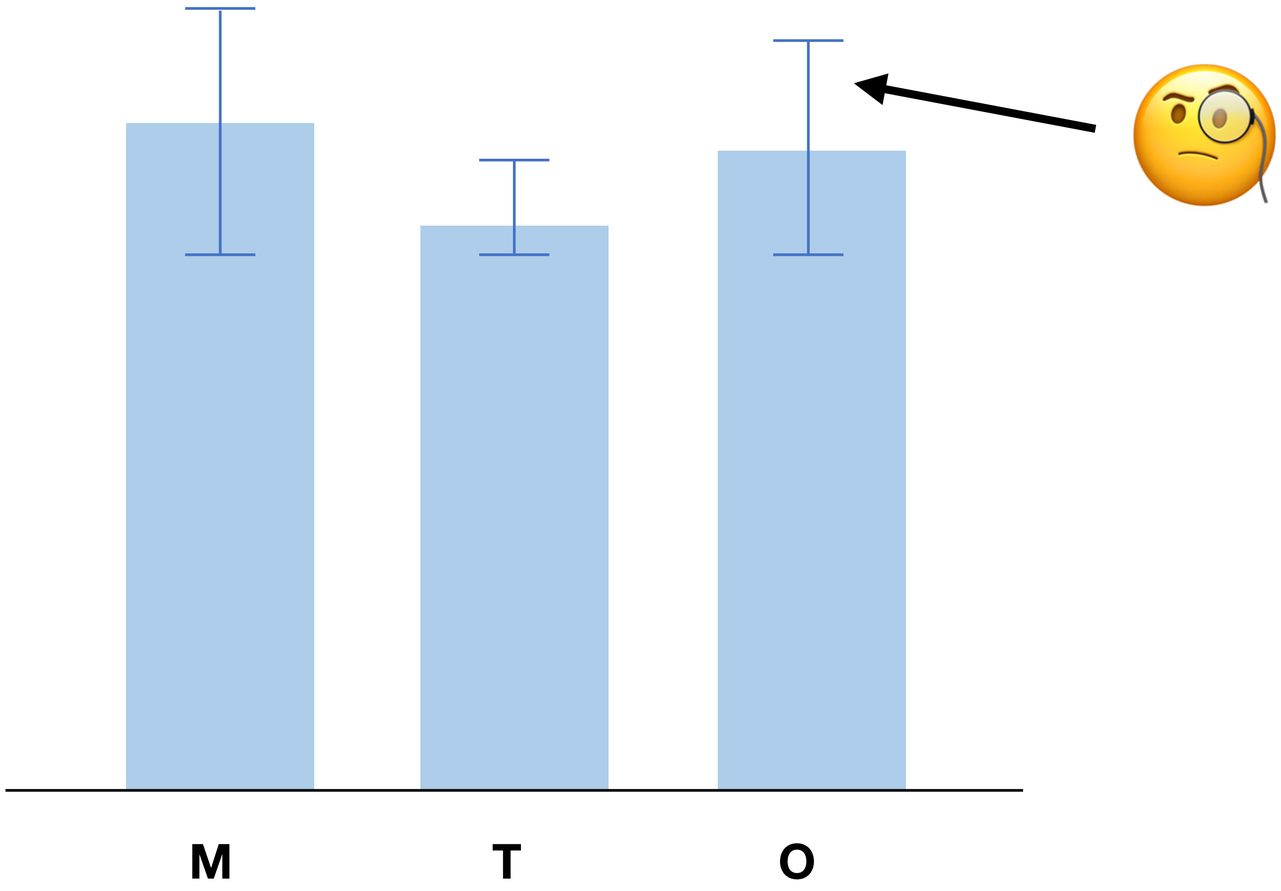
No truncation of temperature range
No uncertainty



Truncated to show temp. anomalies
Shows uncertainty

https://mcorrell.medium.com/an-inconvenient-graph-addc218579e9

Communicate uncertainty

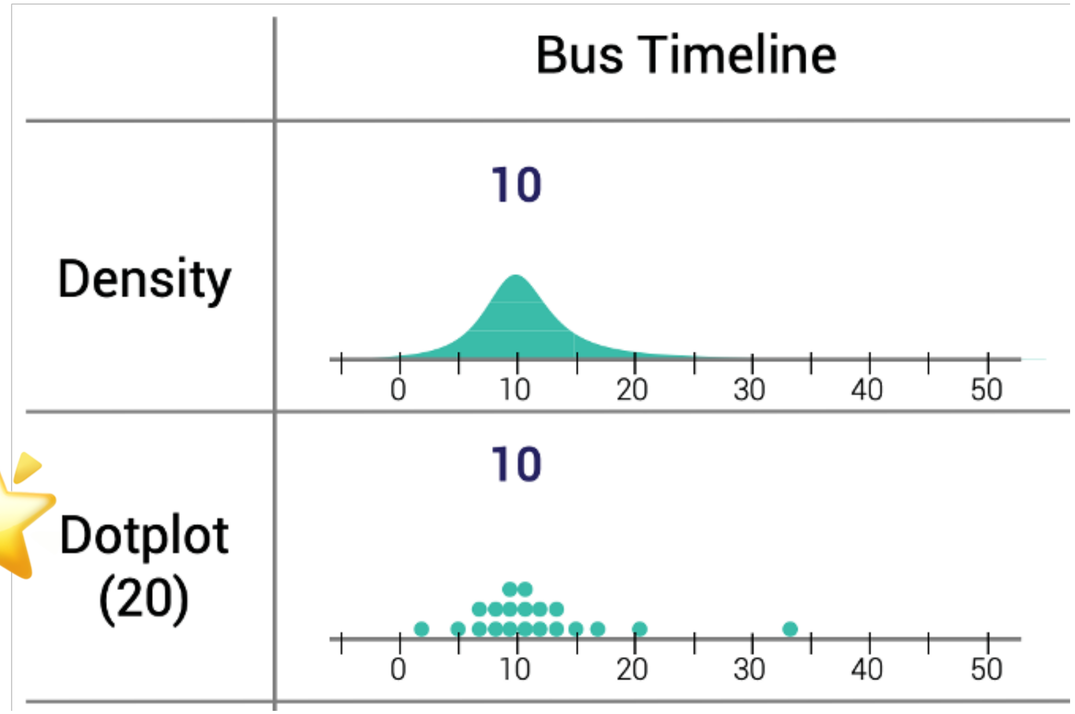


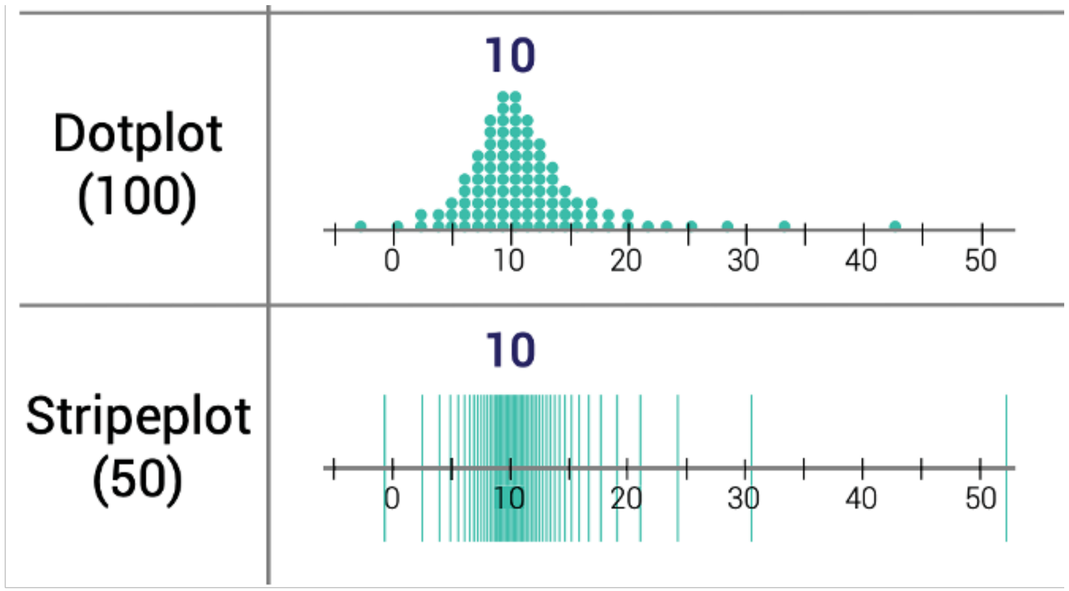
CIs not always seen as parameter estimates.

Instead mistaken as discrete range of data.

Joslyn & LeClerc, 2012

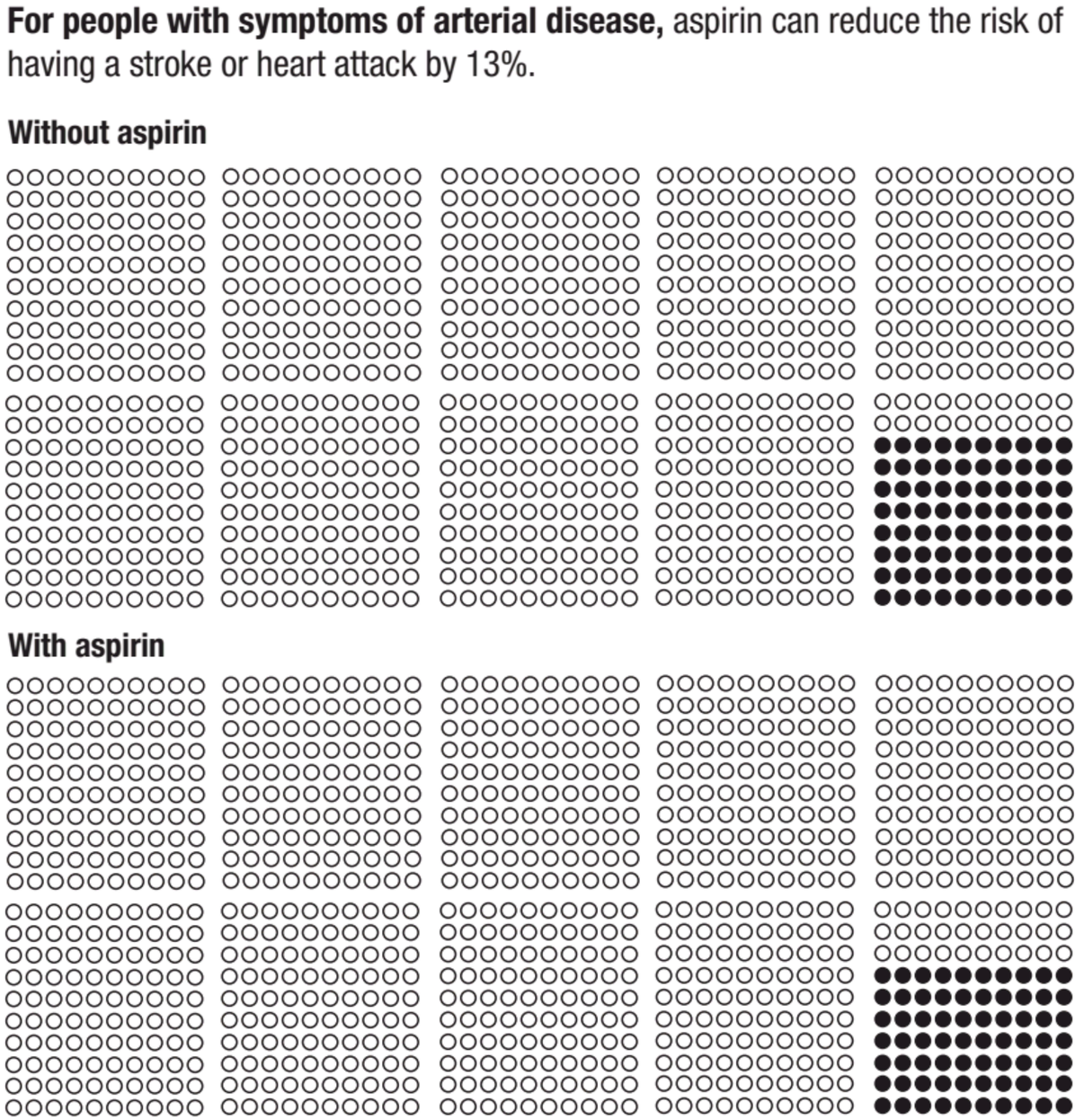
Dotplots and stripeplots are more intuitive alternatives to CIs.





M. Kay et al. "When (ish) is my bus? user-centered visualizations of uncertainty in everyday, mobile predictive systems." CHI 2016.

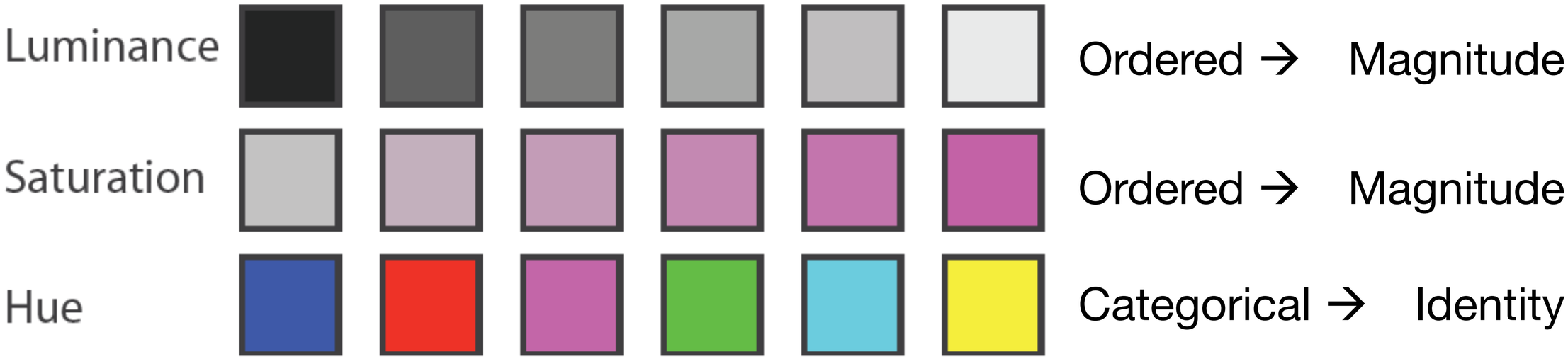
Icon arrays to communicate (absolute) risk



The Science of Visual Data Communication: What Works, Franconeri et al. 2021

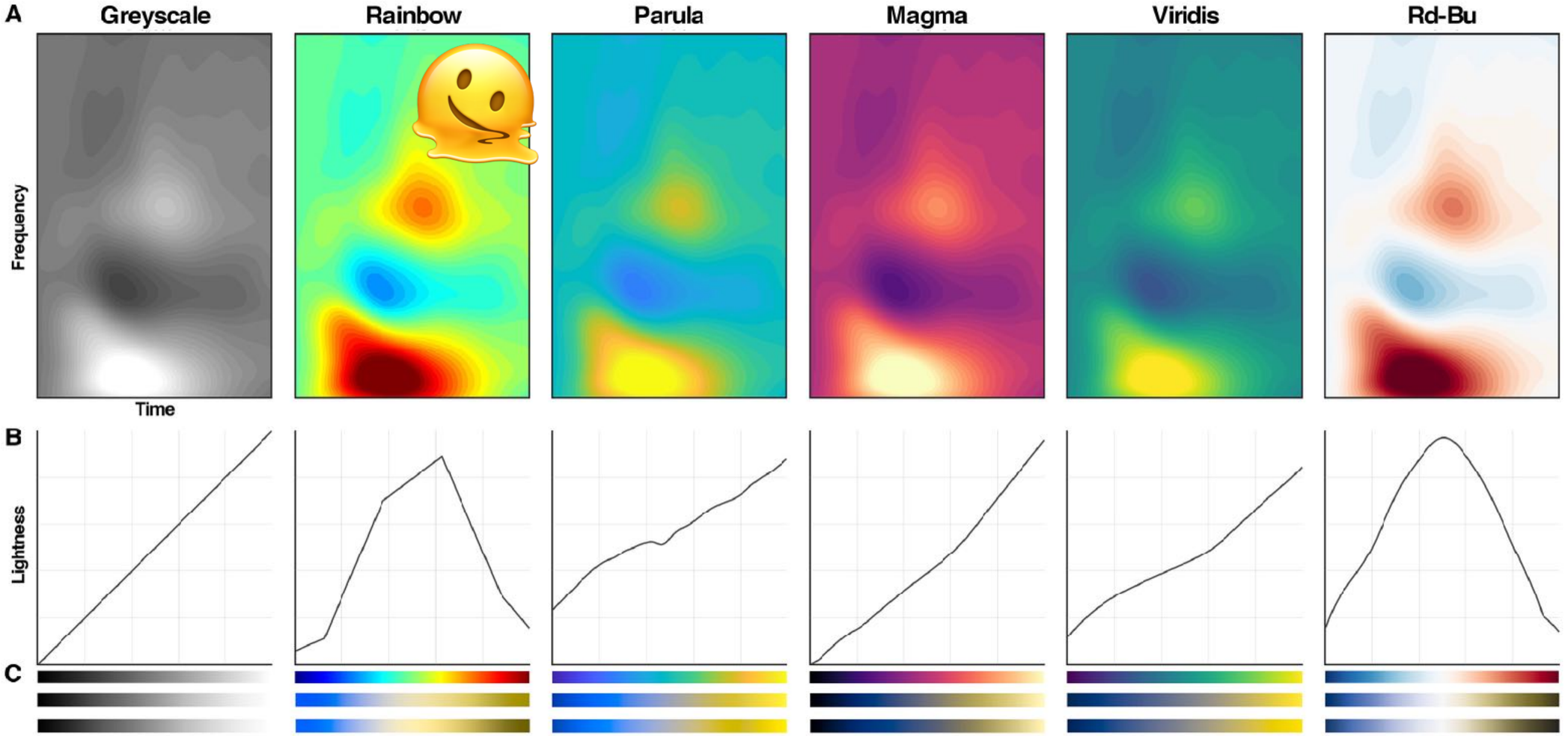
Be honest with your visuals

Color is easily misused



Color is easily misused.

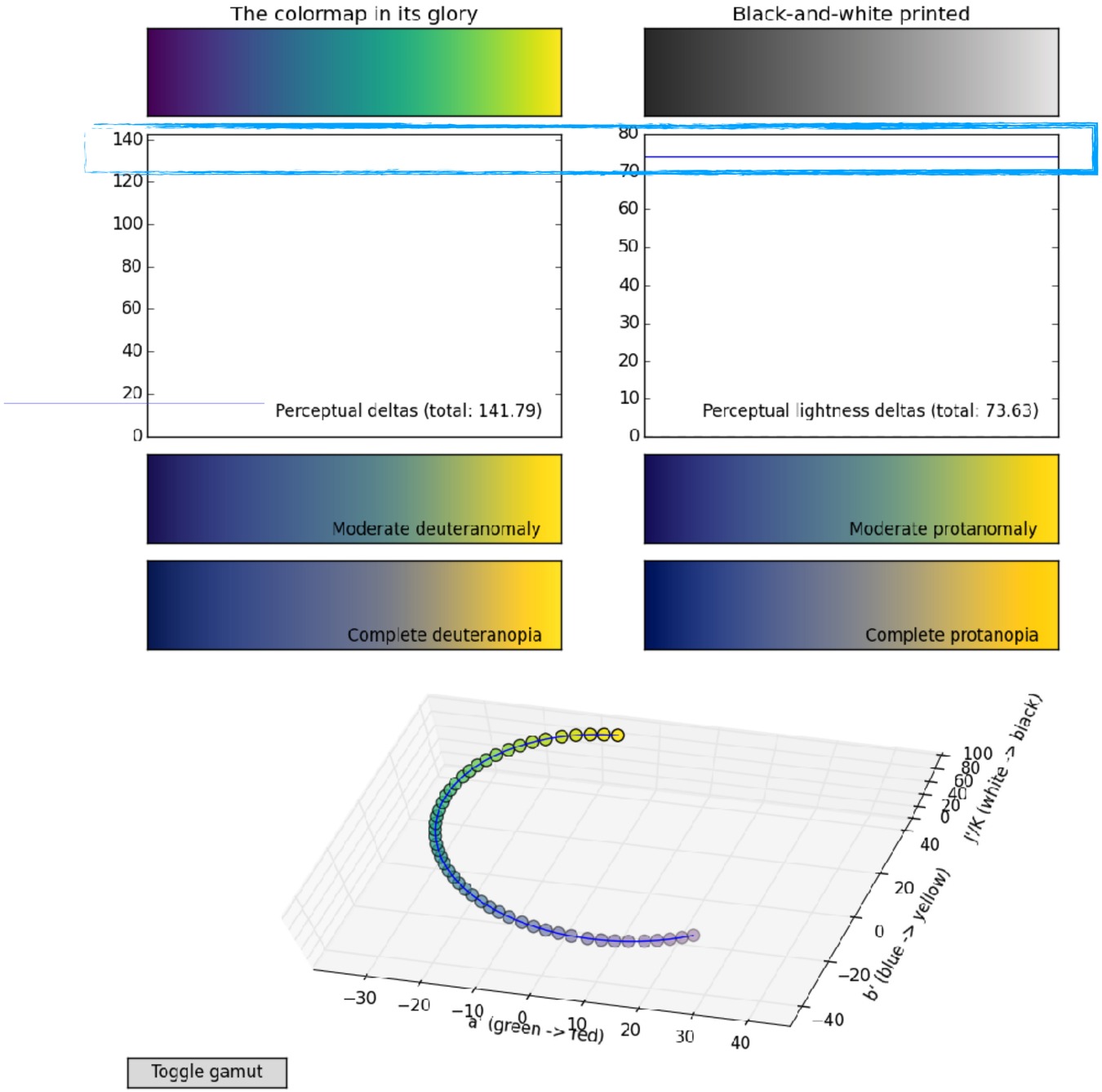
Look for maps with monotonically (de/in)creasing luminance



Over the rainbow: Guidelines for meaningful use of colour maps in neurophysiology, Cooper et al. 2021

Color is easily misused.

Viridis is a perceptually uniform alternative.



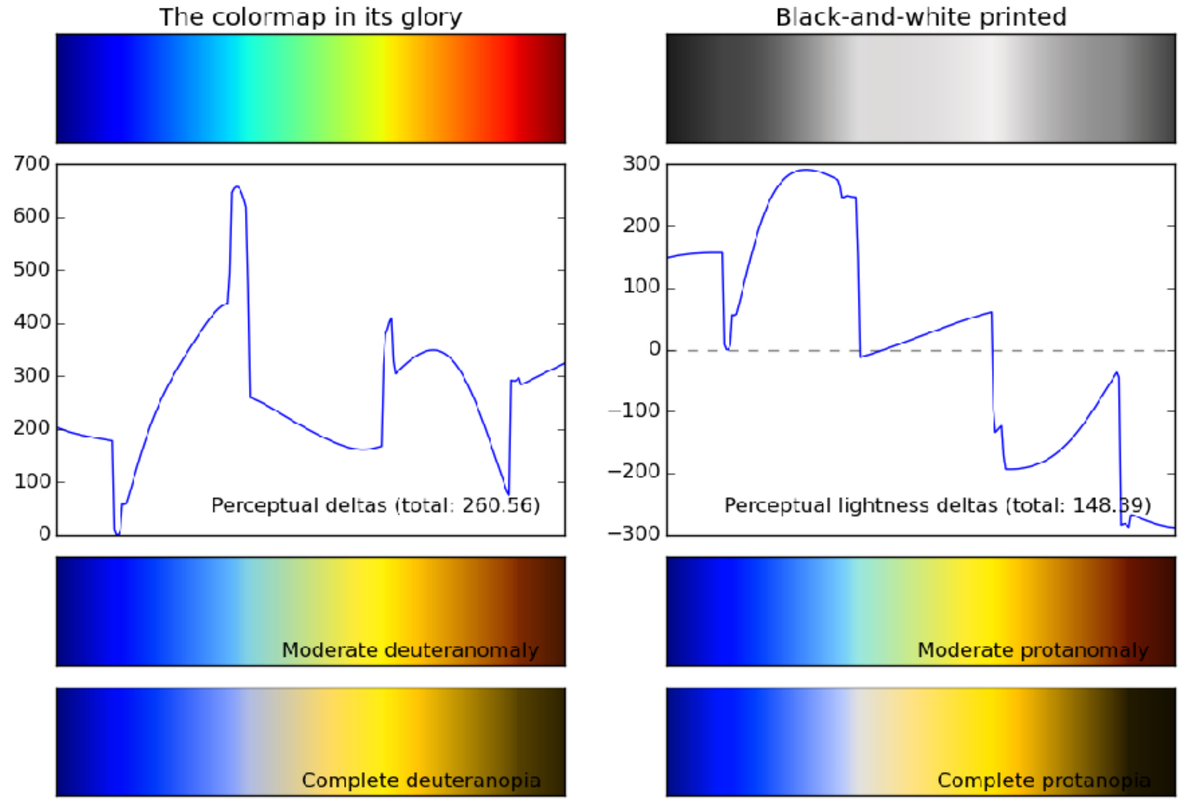
https://bids.github.io/colormap/, https://gramaz.io/d3-cam02/#ciecam02

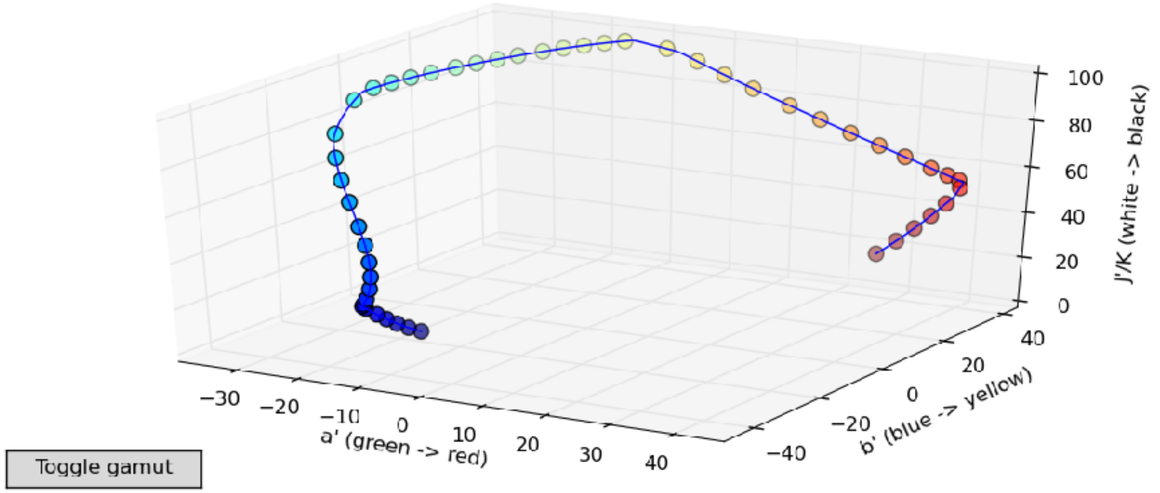
Perceptual banding

Color is easily misused.

Use maps that are perceptually uniform.

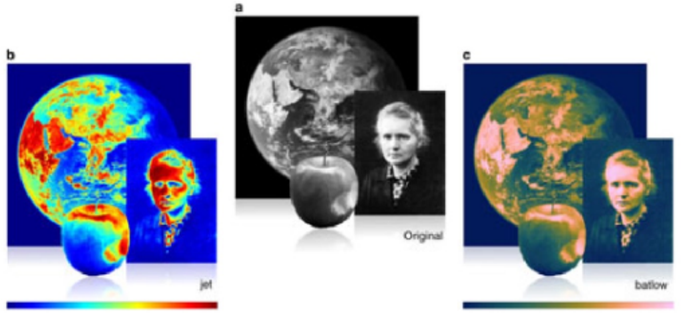
Rainbow (jet) is not.





https://bids.github.io/colormap/, https://gramaz.io/d3-cam02/#ciecam02

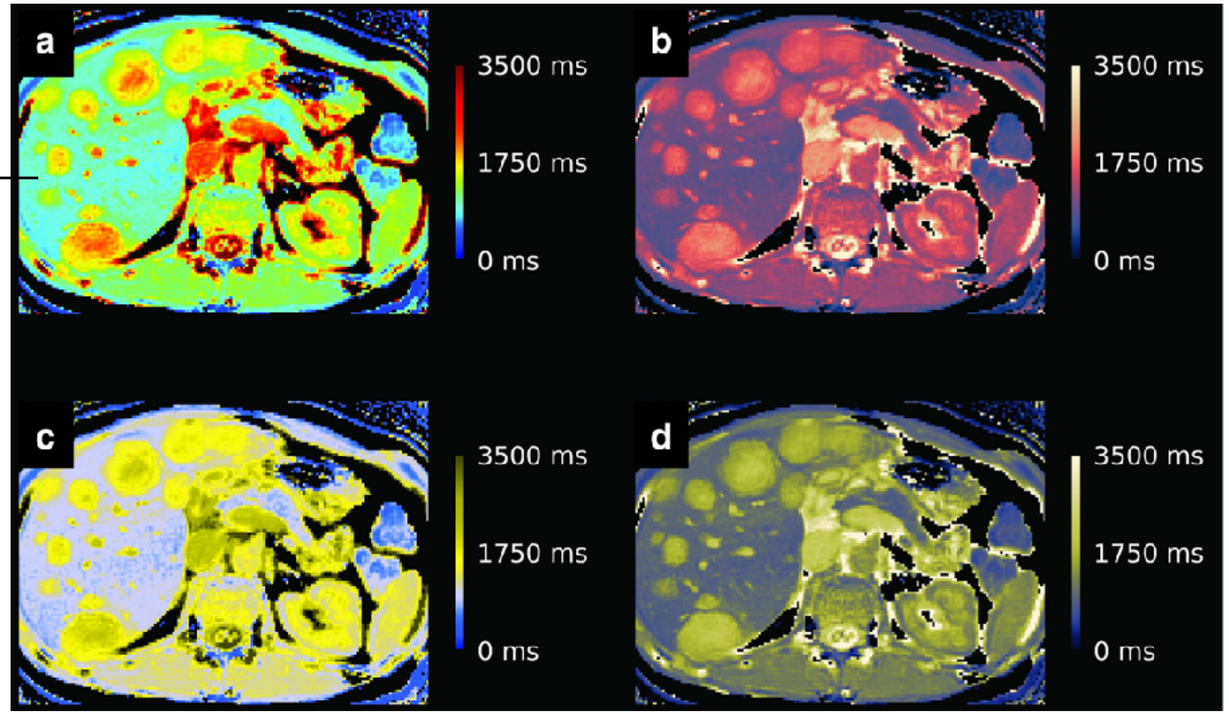
The bad e.g., perceiving features



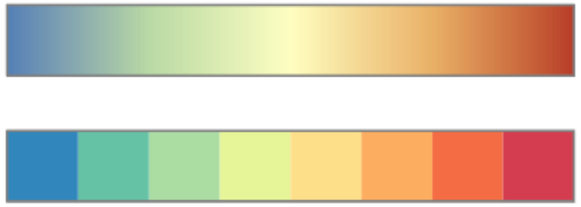
Rainbow maps are not all bad

Ware, Stone, Szafir (2023)

- *The good:* Accurate for reading a value off a map using a key
- **Improve** rainbow maps e.g., make them **perceptually uniform**



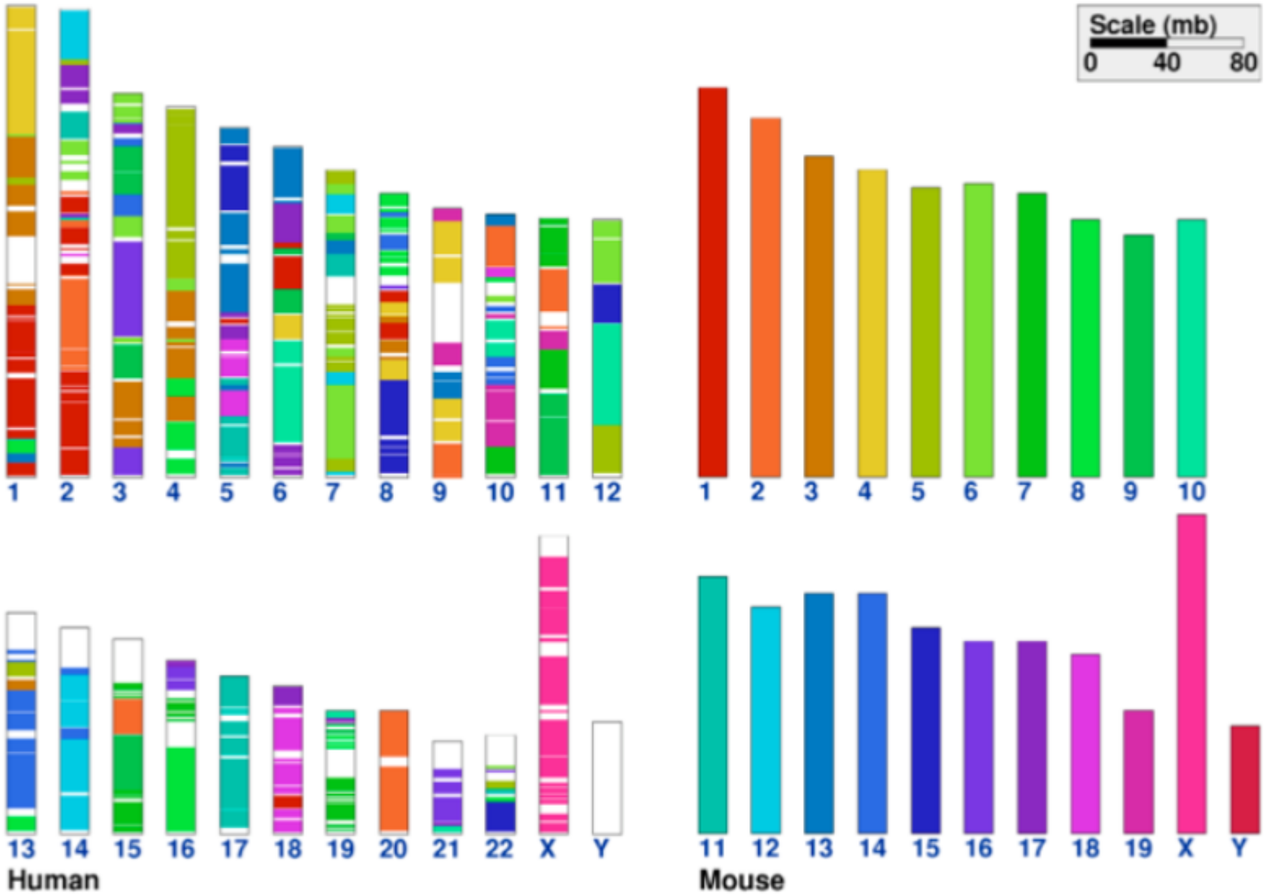
Sollmann et al. JMRI 2024



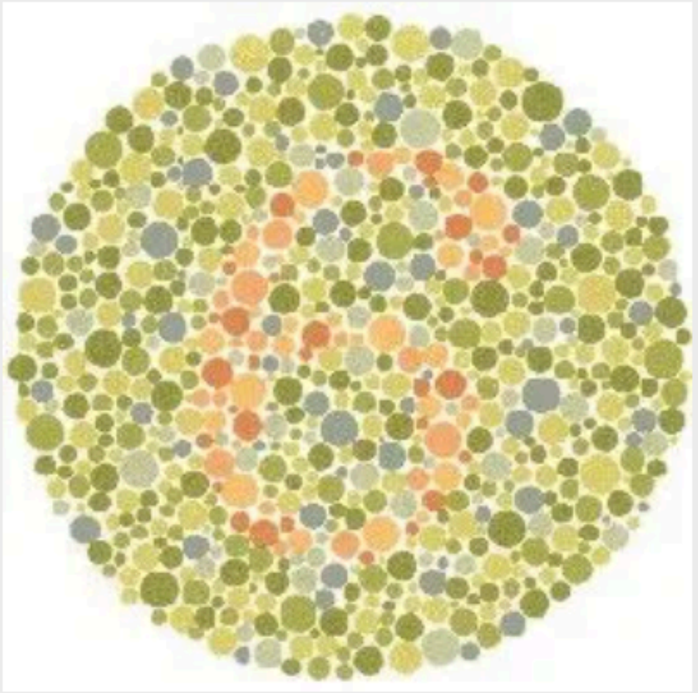
spectral colormap, D3 color schemes

How many colors?

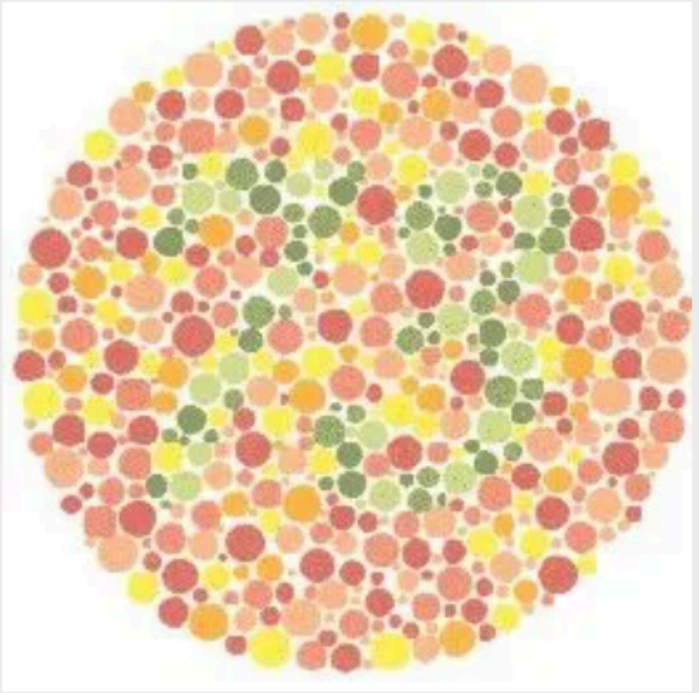
Easily descriminable — i.e., usually 3-9 colors



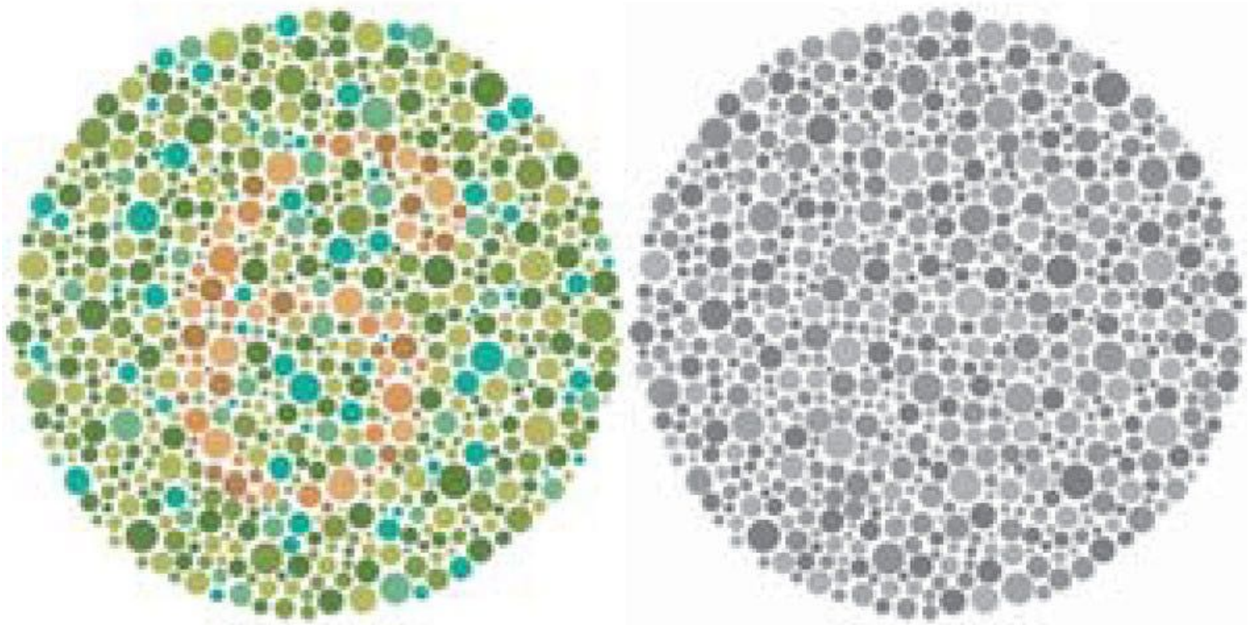
[Cinteny: flexible analysis and visualization of synteny and genome rearrangements in multiple organisms. Sinha and Meller. BMC Bioinformatics, 8:82, 2007.]



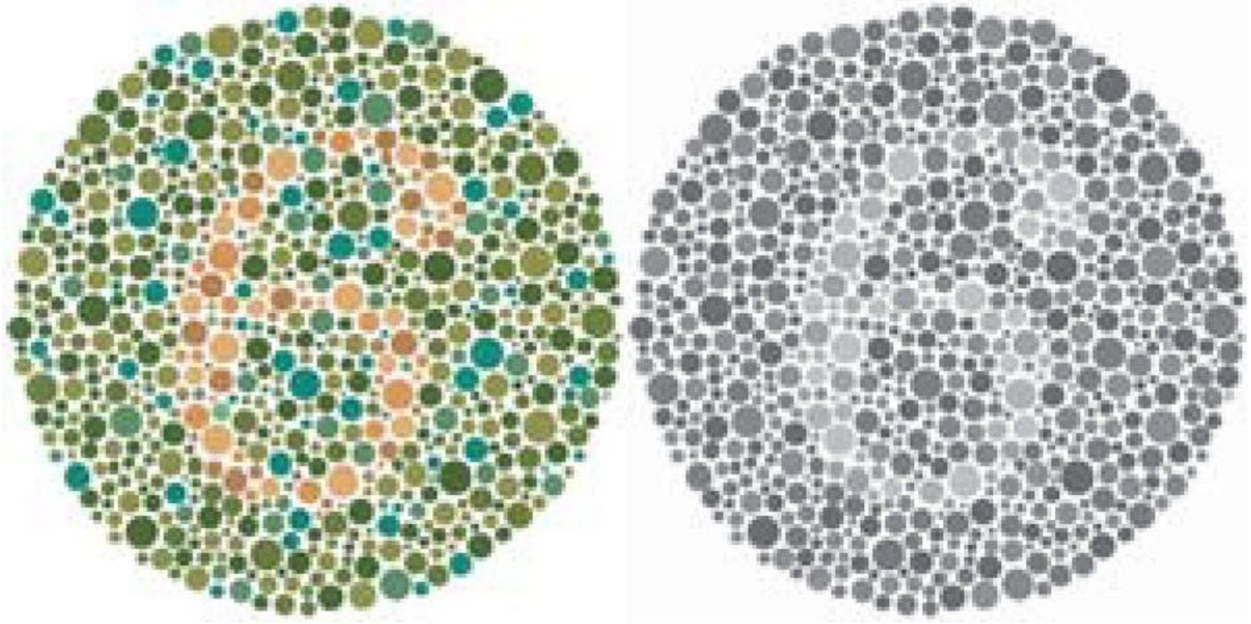
Enter The Number



Enter The Number



Impossible to see in grayscale 🫠

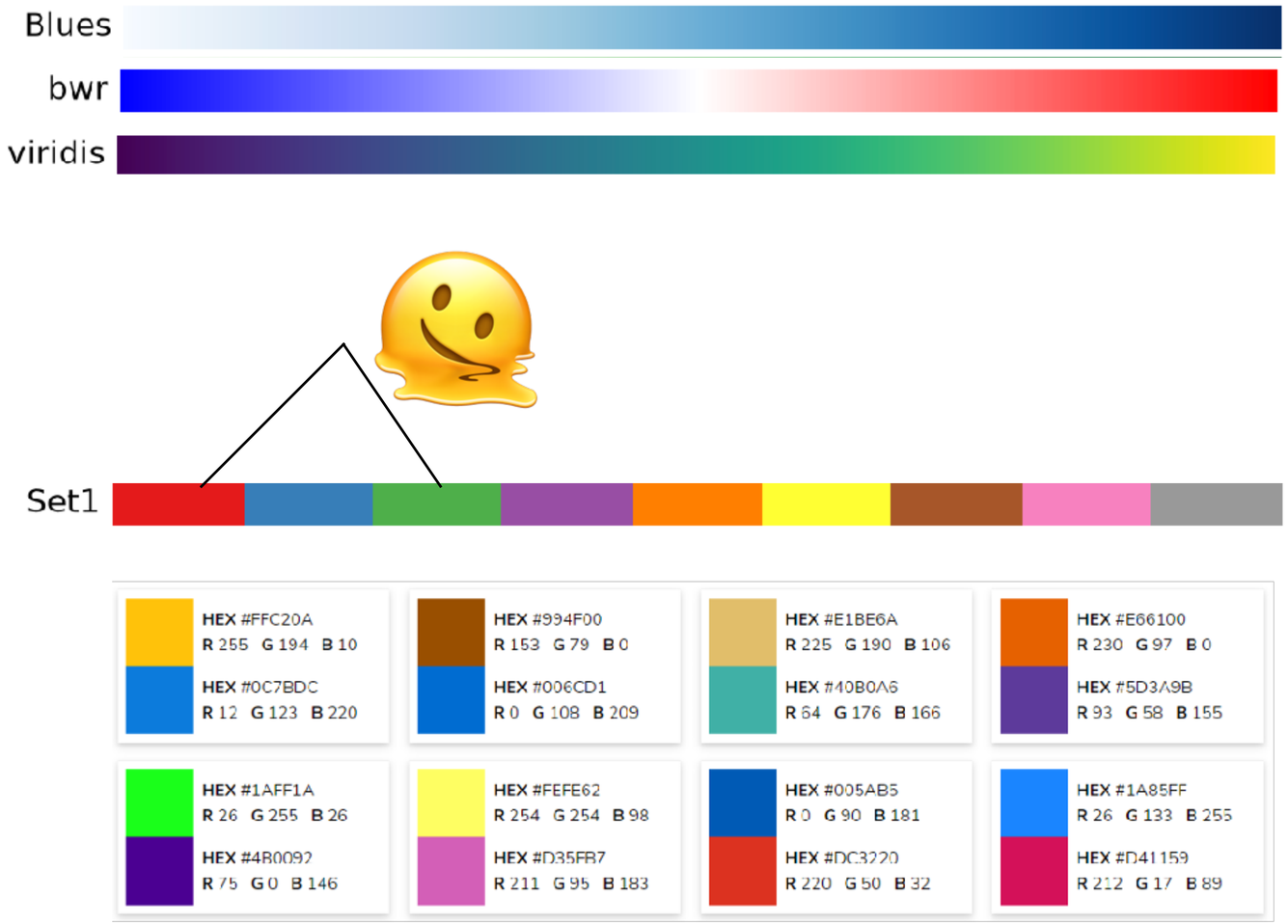


More **contrast** can help perceive differences

Tip: Flip your vis in black and white — ***does it survive***?

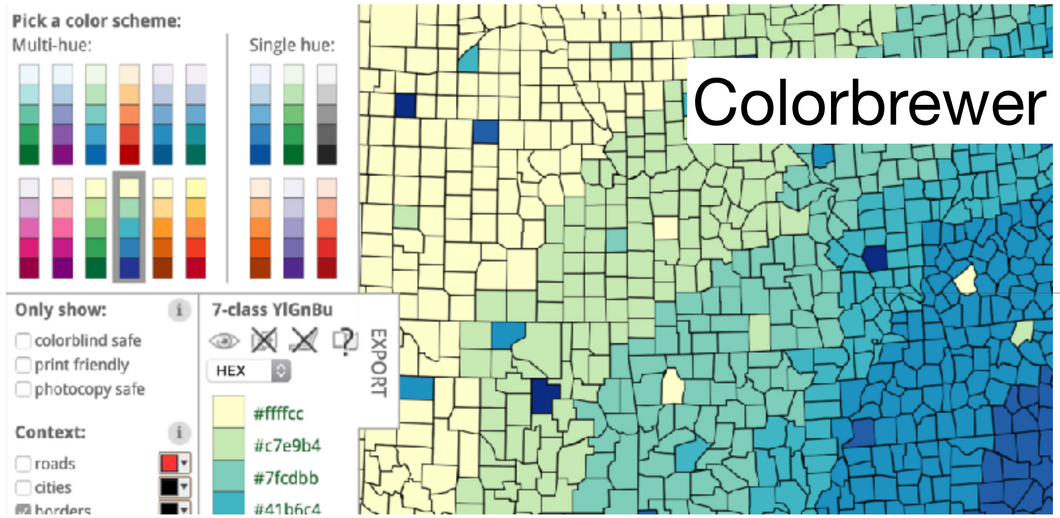
Use color schemes appropriate to your data.

- **Gradients** for continuous data (sequential, multicolor, diverging)
- **Binned colors** for categories
- Leverage data **semantics**
- The default color scheme may not always be the best (matplotlib Set1 not accessible)
- Be **accessible**

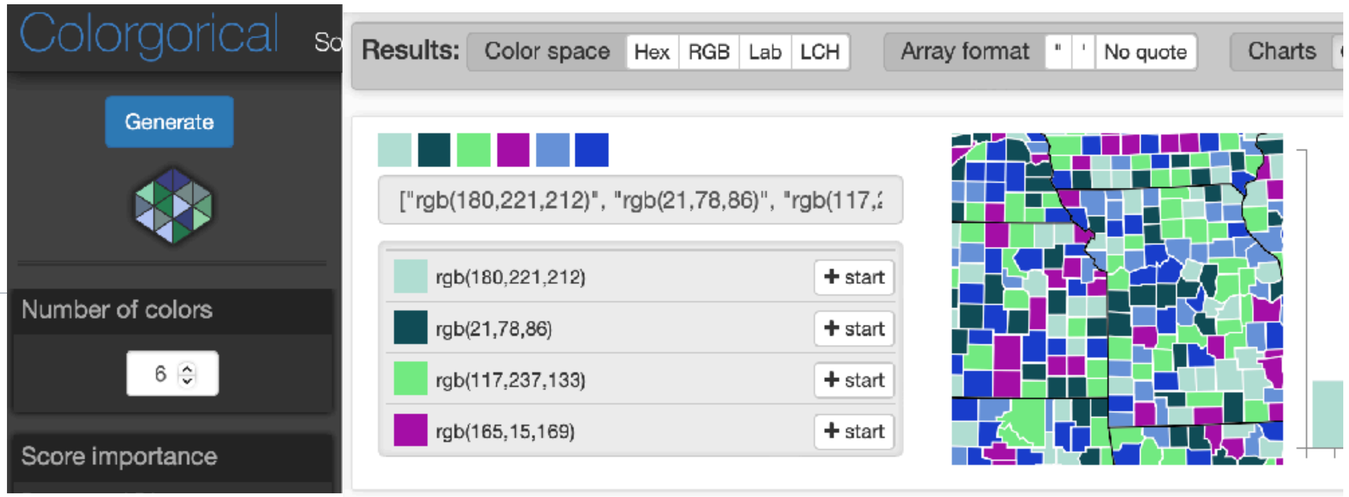


⚒ (Some) color tools

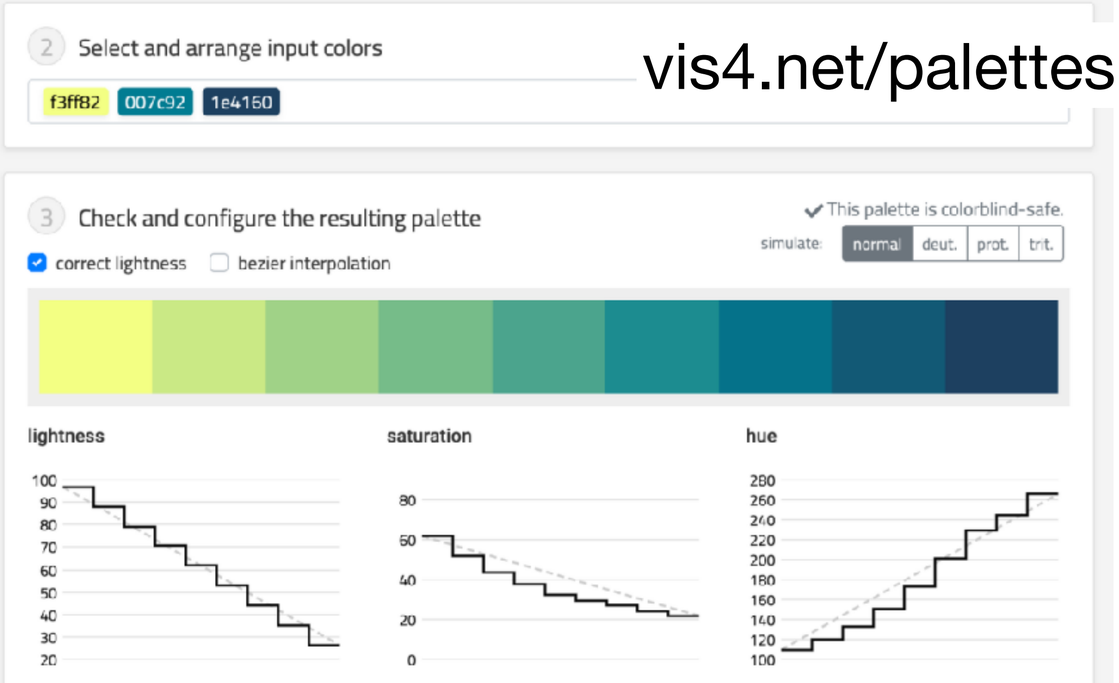
Colorbrewer



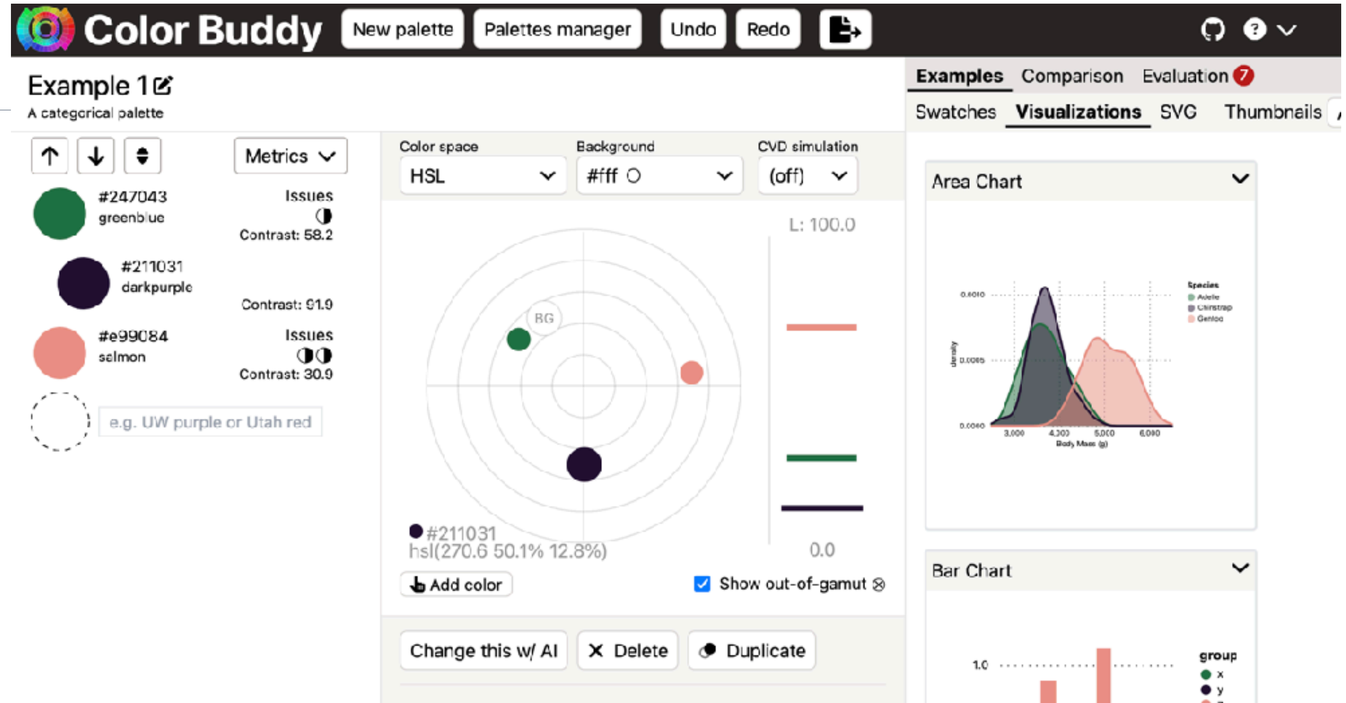
Colorgorical



vis4.net/palettes

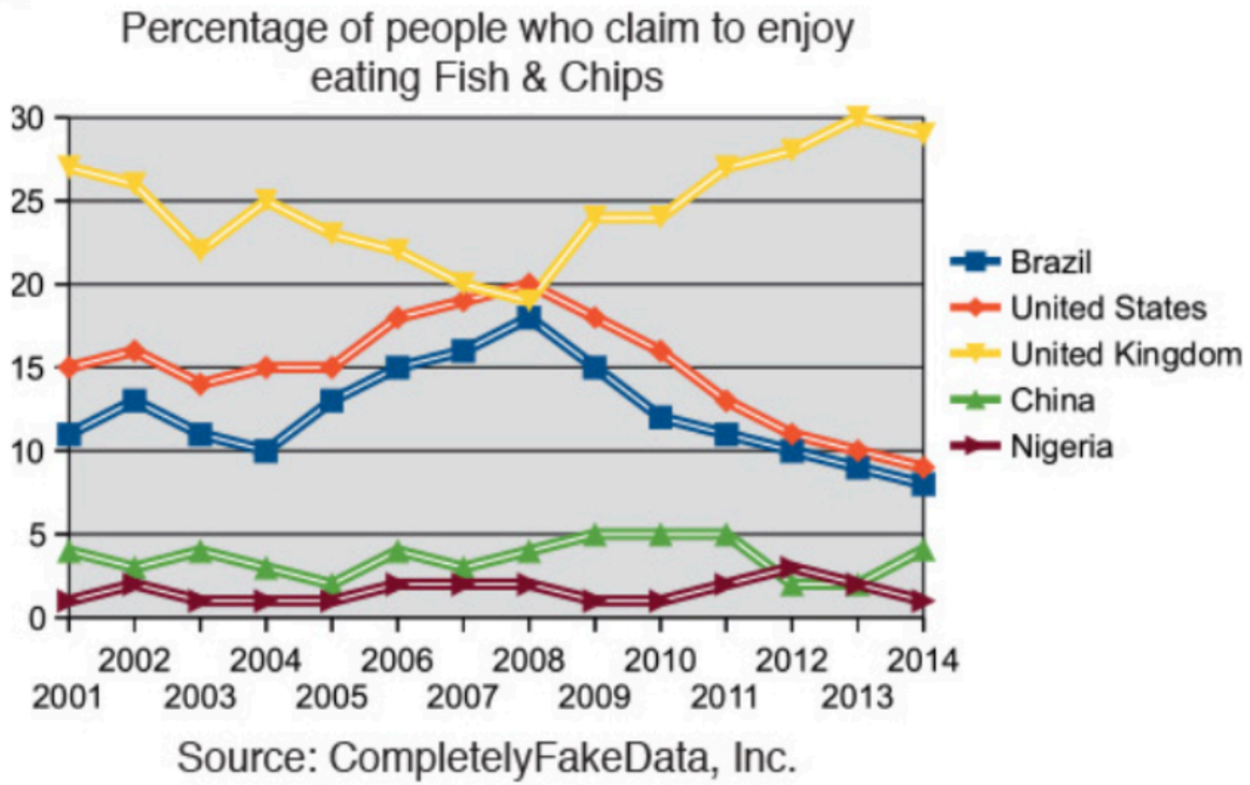


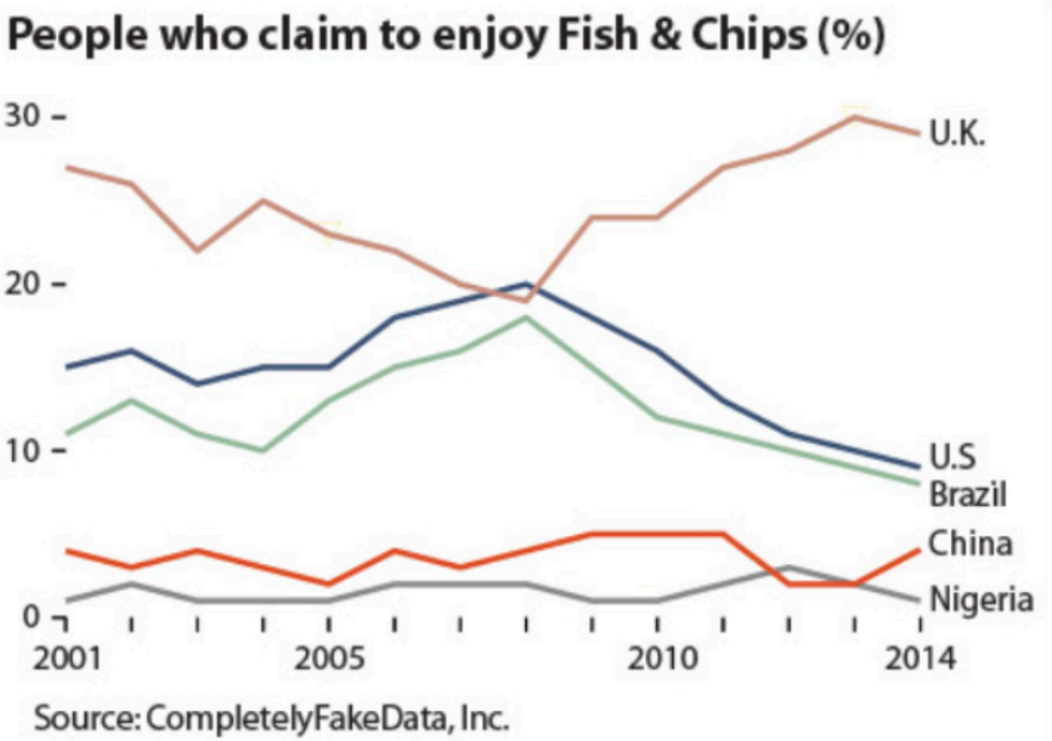
Color Buddy



Aesthetics

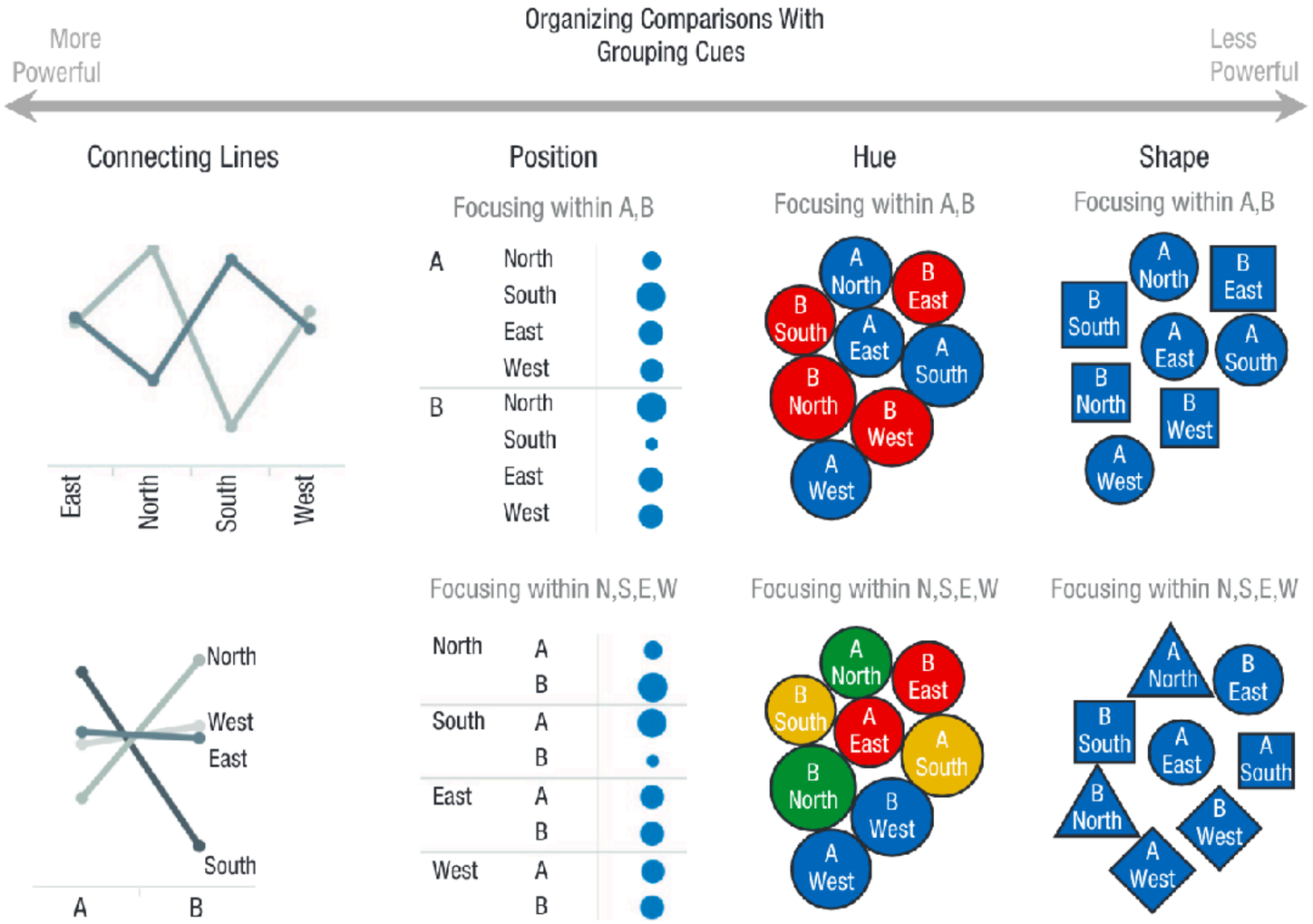
Thoughtful design choices go a long way for your audience. Just don’t overdo it.


Percentage of people who claim to enjoy eating Fish & Chips
30
25
20
15
10
5
0
2001
2002
2003
2004
2005
2006
2007
2008
2009
2010
2011
2012
2013
2014
Brazil
United States
United Kingdom
China
Nigeria
Source: CompletelyFakeData, Inc.


People who claim to enjoy Fish & Chips (%)
30
20
10
0
2001
2005
2010
2014
U.K.
U.S
Brazil
China
Nigeria
Source: CompletelyFakeData, Inc.

Design a hierarchy of information

Group for comparison

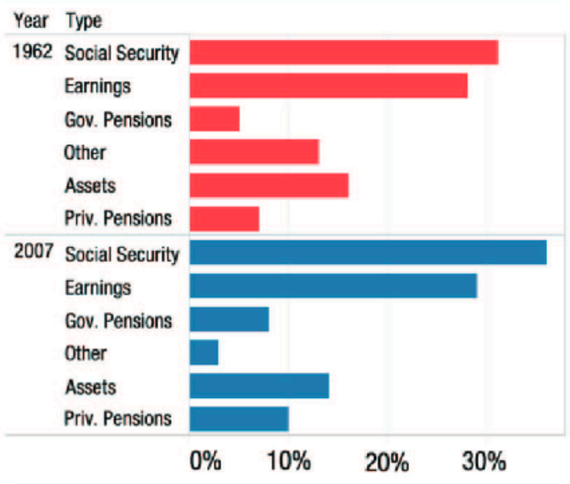


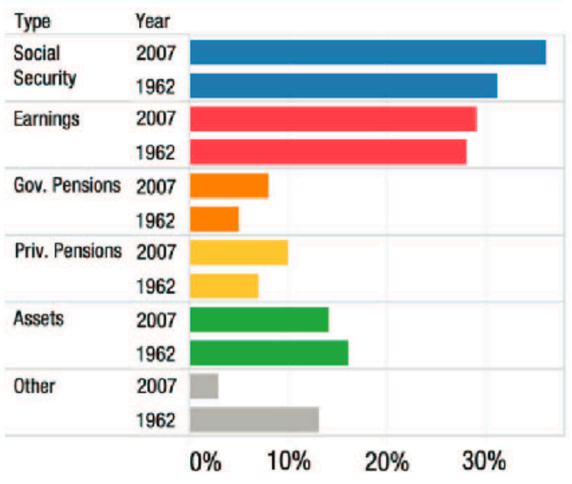
The science of visual data communication: What works, Franconeri et al. 2021

Design a hierarchy of information

Group for comparison

More likely to first compare **categories within same year** (*less of an eye jump*)





More likely to first compare **different years for a category** (*less of an eye jump*)

Weak color grouping makes theme comparison hard in each word group

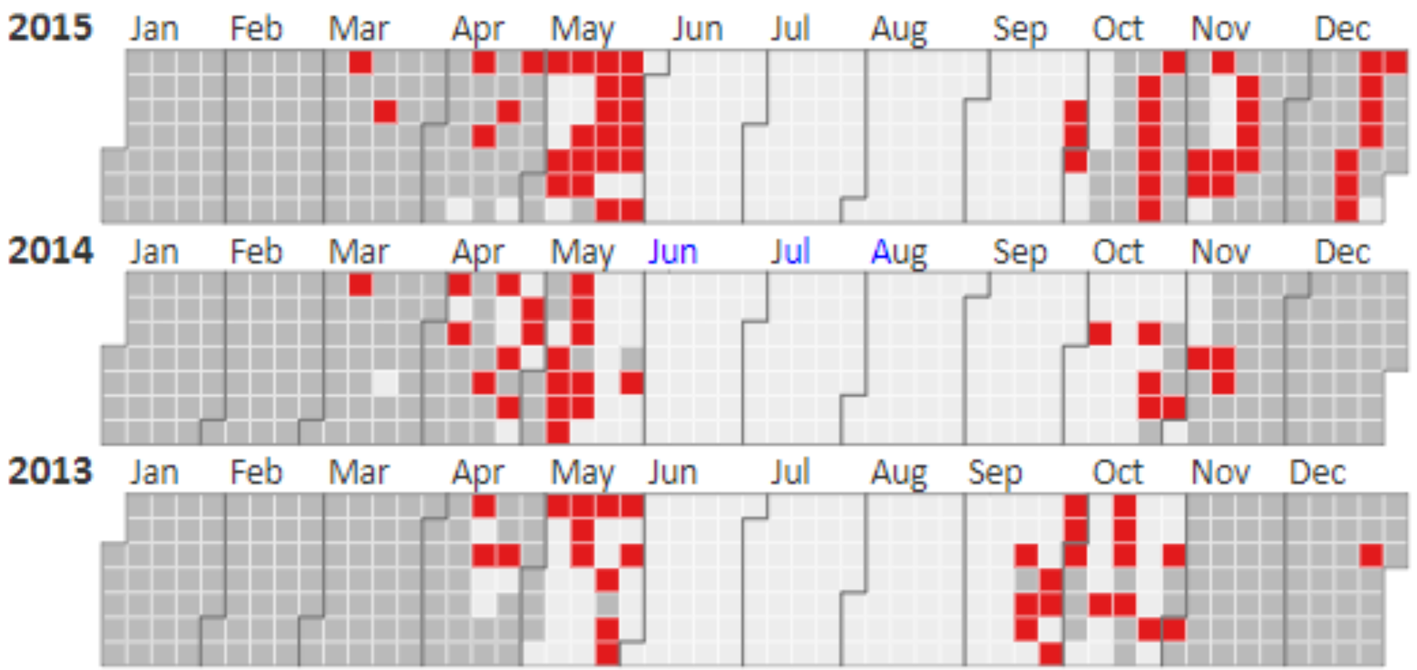
Strong position/ proximity grouping strategy makes theme comparison easier

The science of visual data communication: What works, Franconeri et al. 2021

Manage complexity

Small multiples

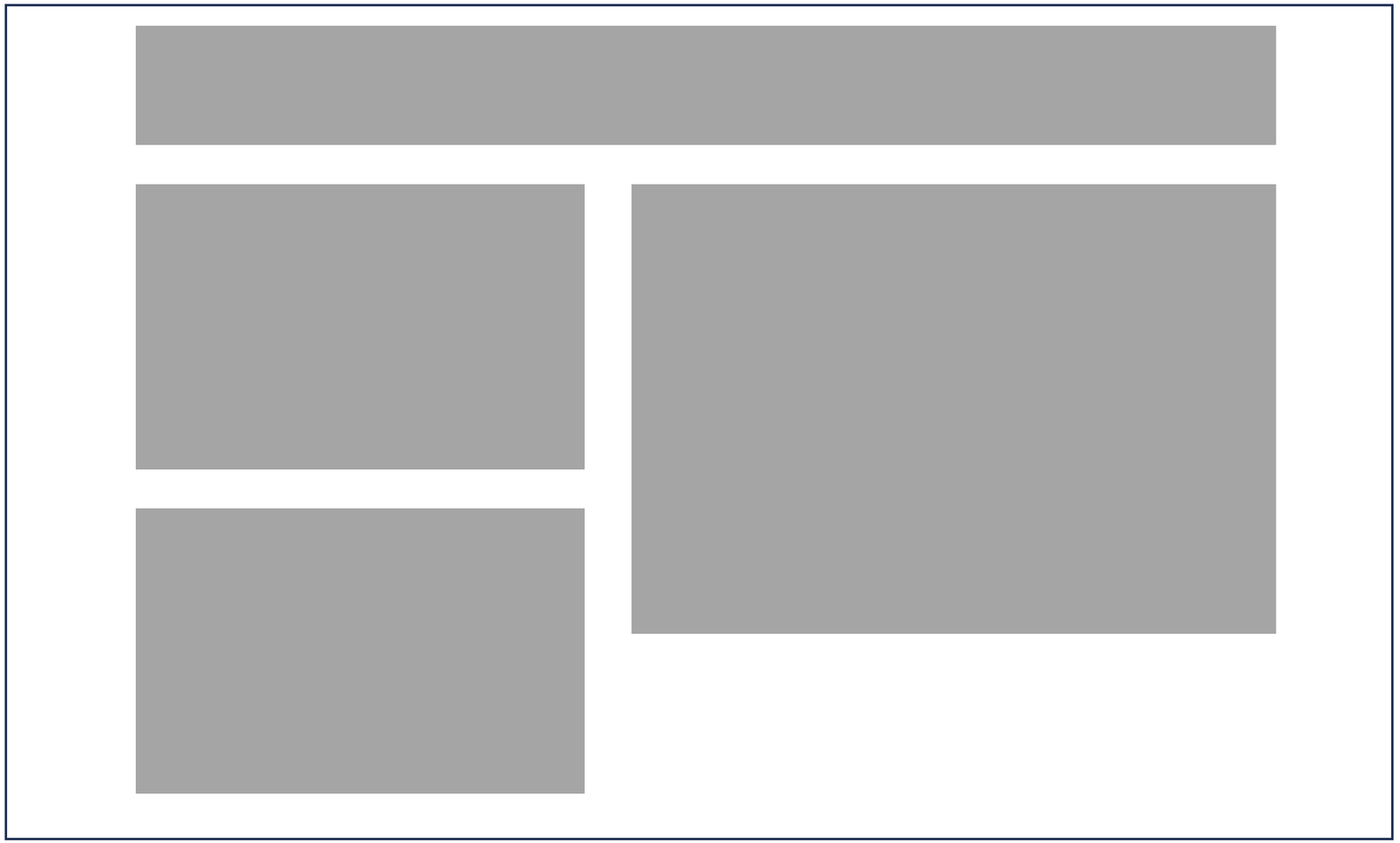
- Manage information complexity by breaking into “chunks”
- Once you can read one, you can read all = easier to find patterns
- Grouping also at play here! (how are the subcharts grouped?)



Tominski & Schumann, *Interactive Visual Data Analysis*, CRC Press 2020

Design a hierarchy of information

Arrange and flow on a grid



Interactive Hierarchical Data Exploration through Dimensional Bundling

Laura Garrison[1,2], Juliane Müller[3], Stefanie Schreiber[3,4], Steffen Oeltze-Jafra[3,4], Helwig Hauser[1,2], & Stefan Bruckner[1,2]

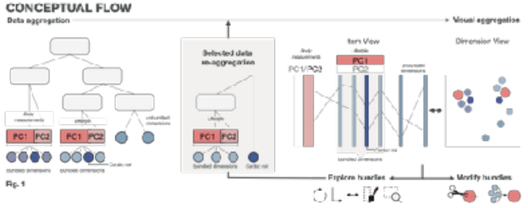
[1]Dept. of Informatics, Univ. of Bergen, Bergen, Norway
[2]Mohn Medical Imaging & Visualization Centre, Bergen, Norway
[3]Center for Behavioral Brain Sciences, Magdeburg, Germany
[4]Dept. of Neurology, Otto-von-Guericke Univ., Magdeburg, Germany

This research is supported by the University of Bergen and the Trond Mohn Foundation in Bergen (BFS2018, Visualizing Data Science for Large Scale Hypothesis Management in Imaging Biomarker Discovery (VID)), and by the Federal State of Saxony-Anhalt, Germany (FKZ: I 68).

MOTIVATION

The identification of interesting patterns and relationships is essential to exploratory data analysis. This becomes increasingly difficult in high dimensional datasets, and even more difficult when these data contain mixed numerical and categorical data types. To the first high-dimensionality problem, while dimensionality reduction techniques are frequently applied to reduce the analysis space, these approaches may unintentionally bury key dimensions within a larger grouping and obfuscate meaningful patterns. To the second problem of mixed data types within a dataset, previous approaches have either treated these different data types separately, or used one type to filter the other. This presents a missed opportunity to identify relationships between these different data types.

CONCEPTUAL FLOW



Our integrated Dual Analysis + DimLift conceptual workflow (Fig. 1) and interface (Fig. 2) combines statistical analysis with user interactions.

INTEGRATED DUAL ANALYSIS

The first phase of this project [1] extends the Dual Analysis framework [2], a technology that enables the simultaneous joint exploration of both items and dimensions, to allow for joint analysis of numerical and categorical data. This joint analysis is made possible through our computation of common statistical measures for all dimensions.

INTERFACE OVERVIEW

DIMLIFT

Algorithm 1: Dimensional bundle creation for two or more dimensions

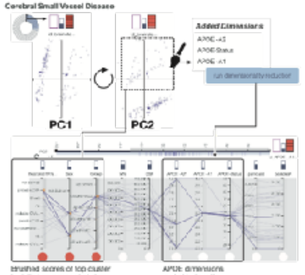
```
initialize pool ← all dimensions in dataset
do
    mark all dims in pool as possibly contributing
    initialize new bundle
    perform PAMO on pool
    for all dimensions in pool
        if PC1 loading ≥ contribution threshold
            move dimension from pool to new bundle
        else
            mark dimension as non-contributing
while pool contains dimensions marked as
    non-contributing
for all bundles
    perform PAMO on bundle
    store PC1 and PC2 for bundle
```

The second phase proposes a novel approach to dimensionality reduction via the creation of *dimensional bundles* [3]. Generated through an iterative dimensionality reduction or user-driven approach, dimensional bundles are expressive groups of dimensions that contribute similarly to the variance of a dataset.

Dimensions and bundles are plotted against interrelatable statistical measures:

- coefficient of unalikeability
- standard deviation/stDev
- variance/VA
- modality
- percentage missingness

CASE STUDY



We exemplify the power of our integrated dual analysis and dimensional bundling techniques for data exploration and hypothesis generation in an expert case study on clinical cohort data (Fig. 3). Interactive exploration and reconstruction methods via a layered parallel coordinates plot allow users to list interesting and subtle relationships to the surface, e.g., Boston STRIVE criteria alongside diagnostic group and key test/imaging dimensions. Users may modify bundles as they form new hypotheses, e.g., addition of APOE to an existing bundle.

Explore and identify interesting patterns in high dimensional mixed-type datasets.

REFERENCES [1] J. Müller, L. Garrison, S. Schreiber, S. Bruckner, H. Hauser, and S. Oeltze-Jafra, "Integrated Dual Analysis of Quantitative and Qualitative High-Dimensional Data," in IEEE Transactions on Visualization and Computer Graphics, vol. 27, no. 5, pp. 2863-2876, 1 June 2021. [2] C. Turkay, P. Filzmoser, and H. Hauser, "Brushing Dimensions - A Dual Visual Analysis Model for High-Dimensional Data," IEEE Transactions on Visualization and Computer Graphics, vol. 17, no. 12, pp. 2591-2600, 30-1. [3] L. Garrison, J. Müller, S. Schreiber, S. Oeltze-Jafra, H. Hauser and S. Bruckner, "DimLift: Interactive Hierarchical Data Exploration Through Dimensional Bundling," in IEEE Transactions on Visualization and Computer Graphics, vol. 27, no. 5, pp. 2908-2922, 1 June 2021.

Semantic Snapping for Guided Multi-View Visualization Design

Yngve S. Kristiansen, Laura Garrison, and Stefan Bruckner, *Member, IEEE Computer Society*

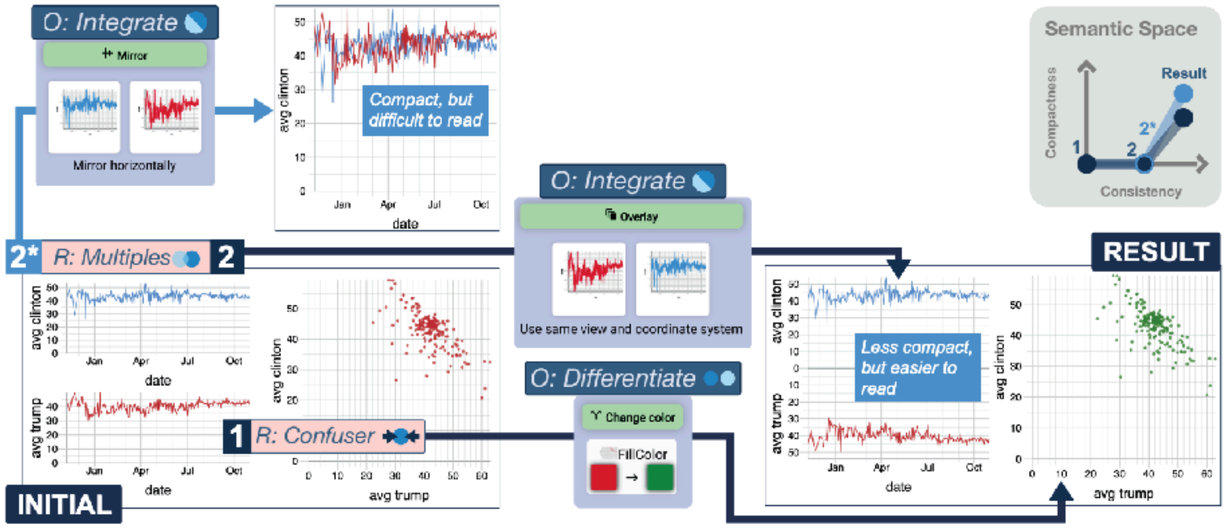
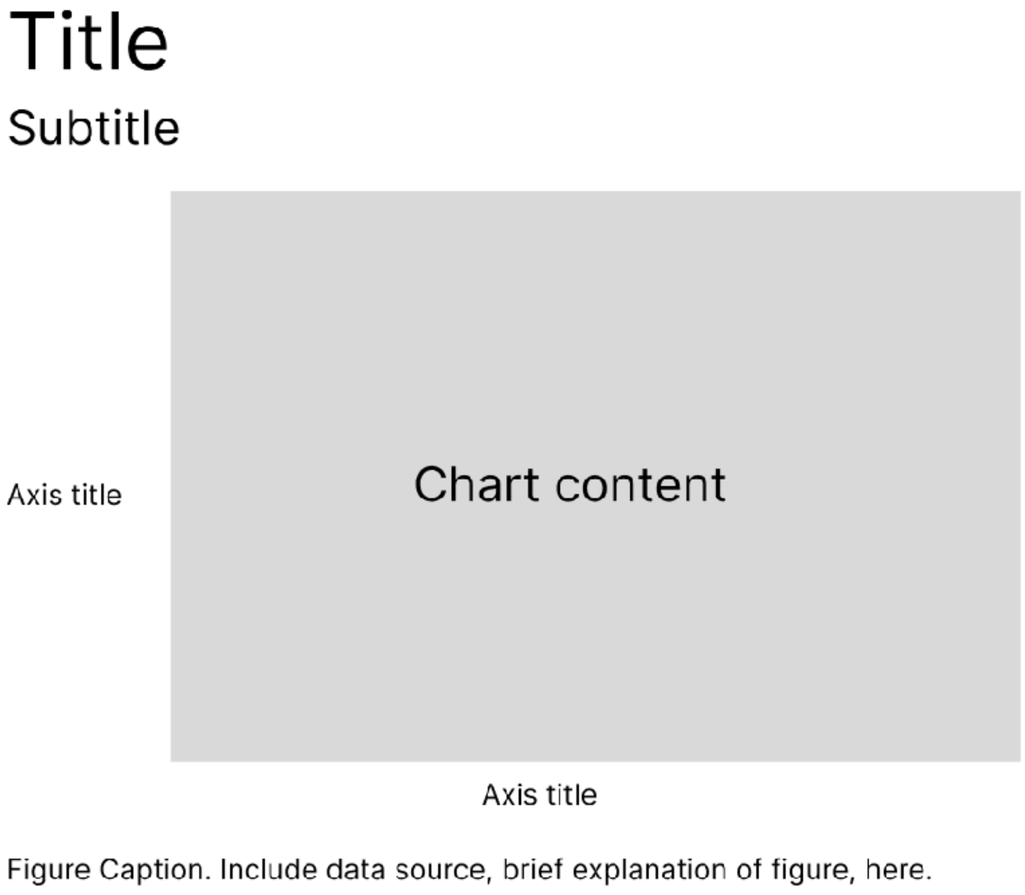


Fig. 1. Semantic snapping allows the user to perform iterative operations to improve the compactness and consistency of a multi-view visualization. Underlying algebraic rules called *relations* define the available *operations* for each iteration. In this example showing the 2016 US Election poll percentages and pollsters, from our initial composition we (1) identify a *confuser* relation between the bottom left and rightmost views showing the same color (red). We *differentiate* these views by selecting green as the fill color for the scatter plot. We next identify a *multiples* relation in the two left views. We resolve this through one of two *integration* operations. (2*) *Overlay* produces an unsatisfactory result, so we revert and (2) perform a *mirroring* operation to arrive at our resulting composition. The semantic map to the right illustrates our path through semantic space.

Design a hierarchy of information

Text is an equal partner

- Basic anatomy of a chart
  - Title
  - Axis labels
  - Caption
- Golden ratio (~1.618) or similar typescale to ladder text
  - https://typescale.com/ different scale options



Remember that tables are also a visualization

Declutter!

Ottley & Schwabish 2026

Table 1: The interview participants, their professional roles, sectors, experience, and domains in which they authored or maintained style guides.

Participant	Locale	Role	Sector	# Guides	Select Domain Coverage	Experience	Maintenance
Alpha	US	Graphics Director	Journalism	3	Journalism, Education, Policy Institute	20+ yrs	–
Beta	US	Senior Economist	U.S. Government	1	Government (Public Health)	10+ yrs	–
Gamma	UK	Data Visualisation Lead	U.K. Local Government	1	Government (Urban Services)	7+ yrs	In progress
Delta	US	Data Visualization Designer	Freelance / Consulting	7	Foundations, Business Intelligence	12+ yrs	–
Epsilon	US	Information Designer	Freelance / Consulting	5	Accounting, Pharmaceutical, FinTech	8+ yrs	–
Zeta	US	Product Design Lead	Technology	1	Technology (Software)	15+ yrs	In progress (6 updates)
Eta	DE	Product Specialist	Visualization Software	1	Policy Institute	7+ yrs	–
Theta	DE	Data Visualization Designer	Freelance / Consulting	5	Transportation, Intergovernmental	8+ yrs	1 year
Iota	PT	Data and Visual Journalist	Finance	2	Journalism, Finance	4+ yrs	In Progress

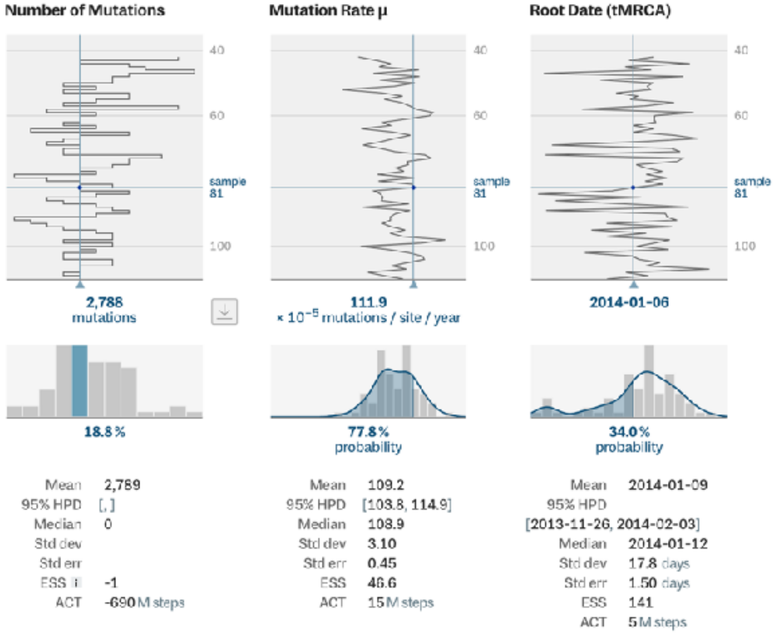
3. **Generating Initial Themes:** Style guides were often described as responses to organizational PROBLEMS, while their development and implementation took the form of corresponding INTERVENTION strategies. Additional themes captured variations in GOVERNANCE structures (i.e., who enforces guidance) and mechanisms for FLEXIBILITY or exception handling, foreshadowing the four-dimensional structure developed in later phases.

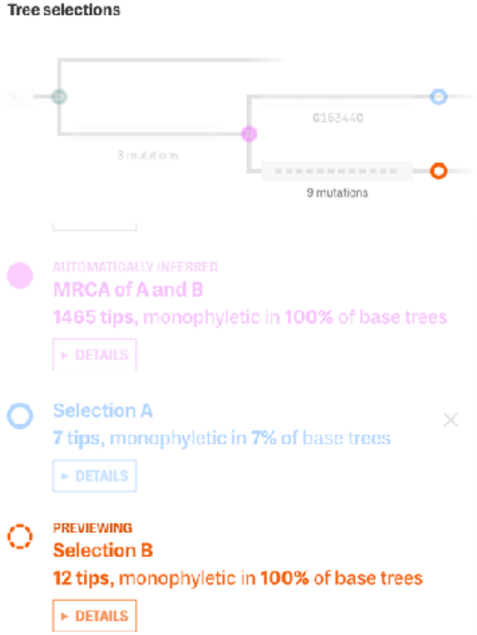
- RULES & MECHANISMS: The mechanisms through which guidance is expressed and operationalized.
- INSTITUTIONAL ENFORCERS: The actors and structures responsible for maintaining and applying guidance.
- SITUATED FLEXIBILITY: The conditions under which guidance is enforced, adapted, or overridden.

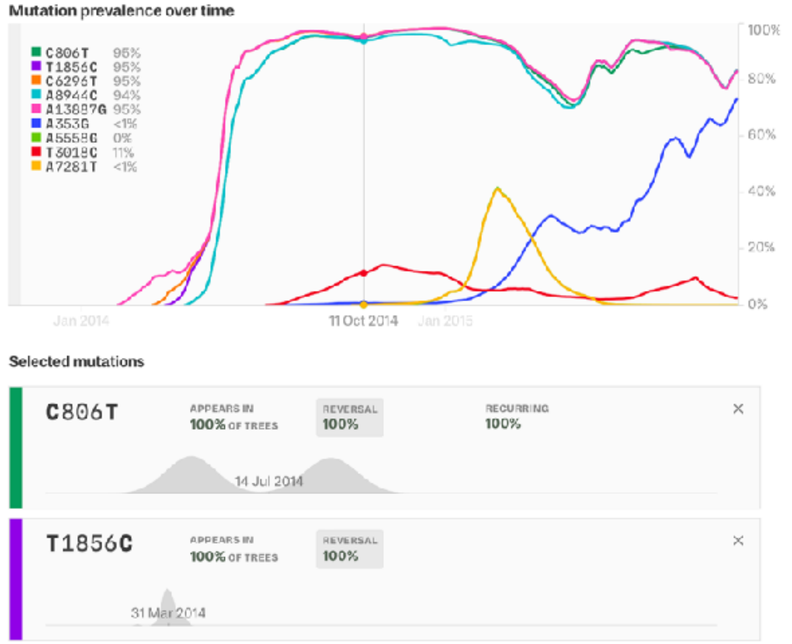
Be mindful of “overdesigning”

People are distrustful of something that is too pretty

Amy







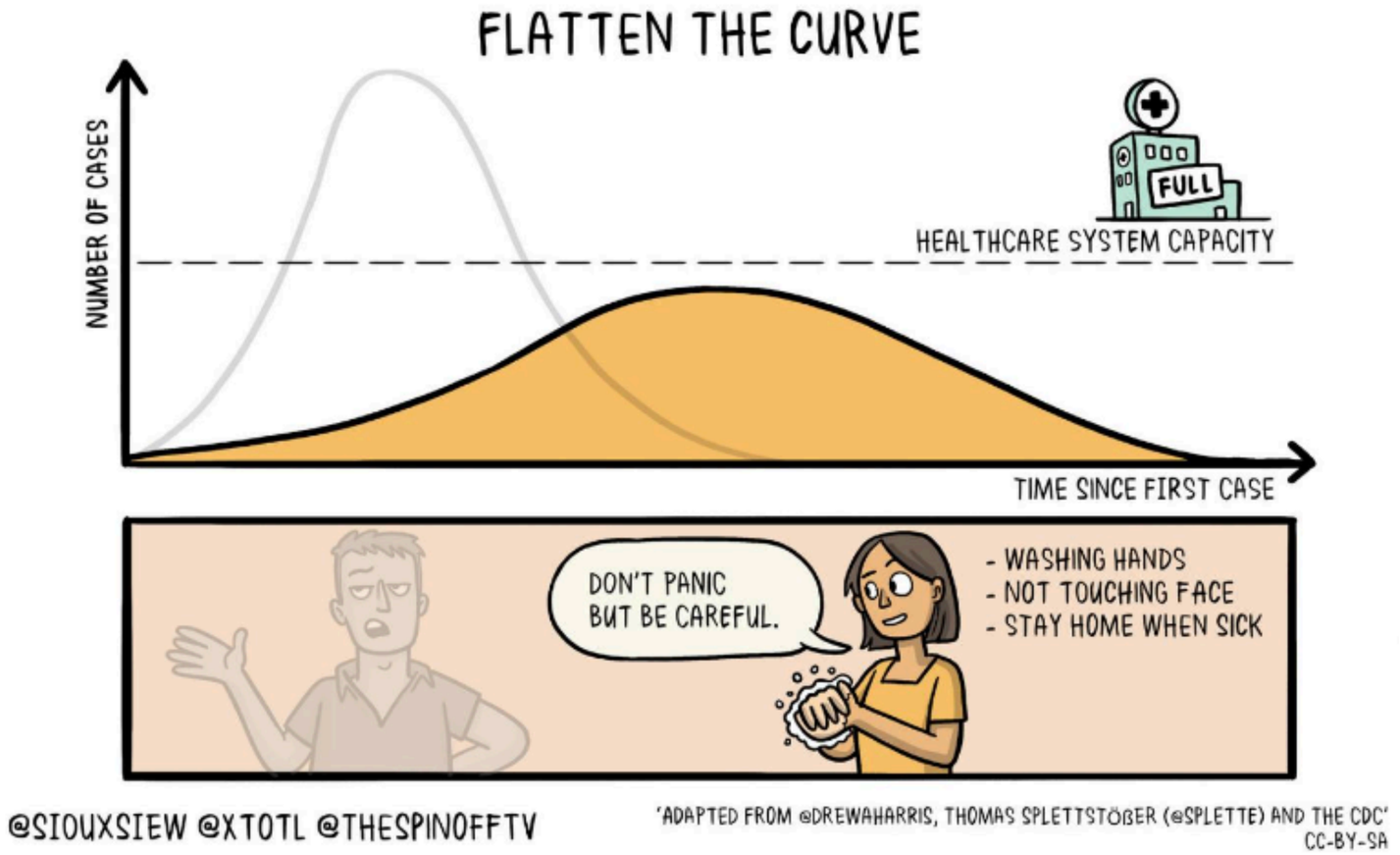
Delphy, Fathom Information Design

Story

Create a visualization with a clear, interesting, and relevant message.

Telling Stories

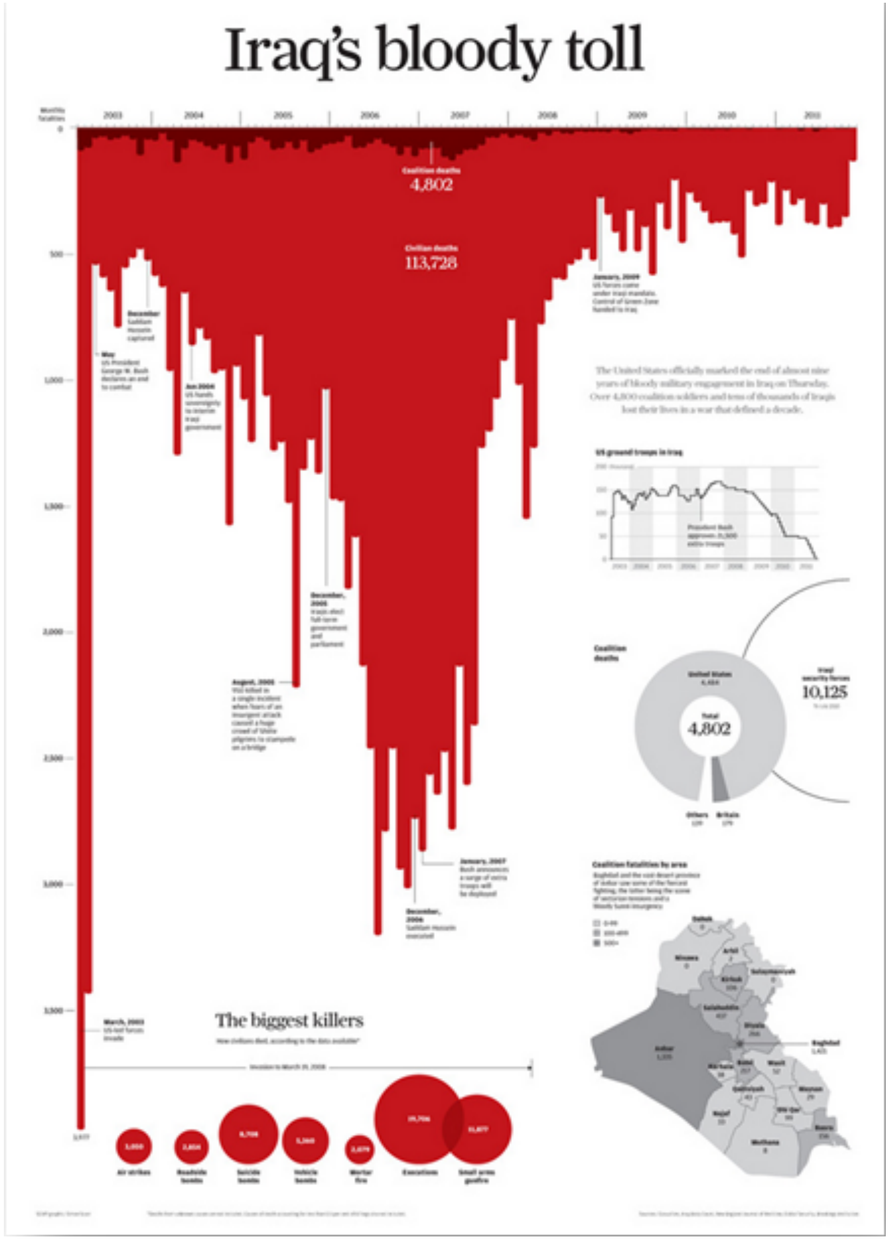
- **Reduce** uncertainty, **resolve** ambiguity, **attribute** consequences, **assign** blame [1]
- Engage audience →behavioral change more likely [2]
- More memorable content [3]



[1] L. Neeley et al., "Linking Scholarship and Practice: Narrative and Identity in Science," Frontiers in Communication Vol. 5. 2020.
[2] M. C. Green and T. C. Brock, "The Role of Transportation in the Persuasiveness of Public Narratives," Journal of Personality and Social Psychology, Vol. 79, 2000.
[3] A. C. Graesser et al., "Advanced Outlines, Familiarity, and Text Genre on Retention of Prose," The Journal of Experimental Education, Vol. 48, 1980

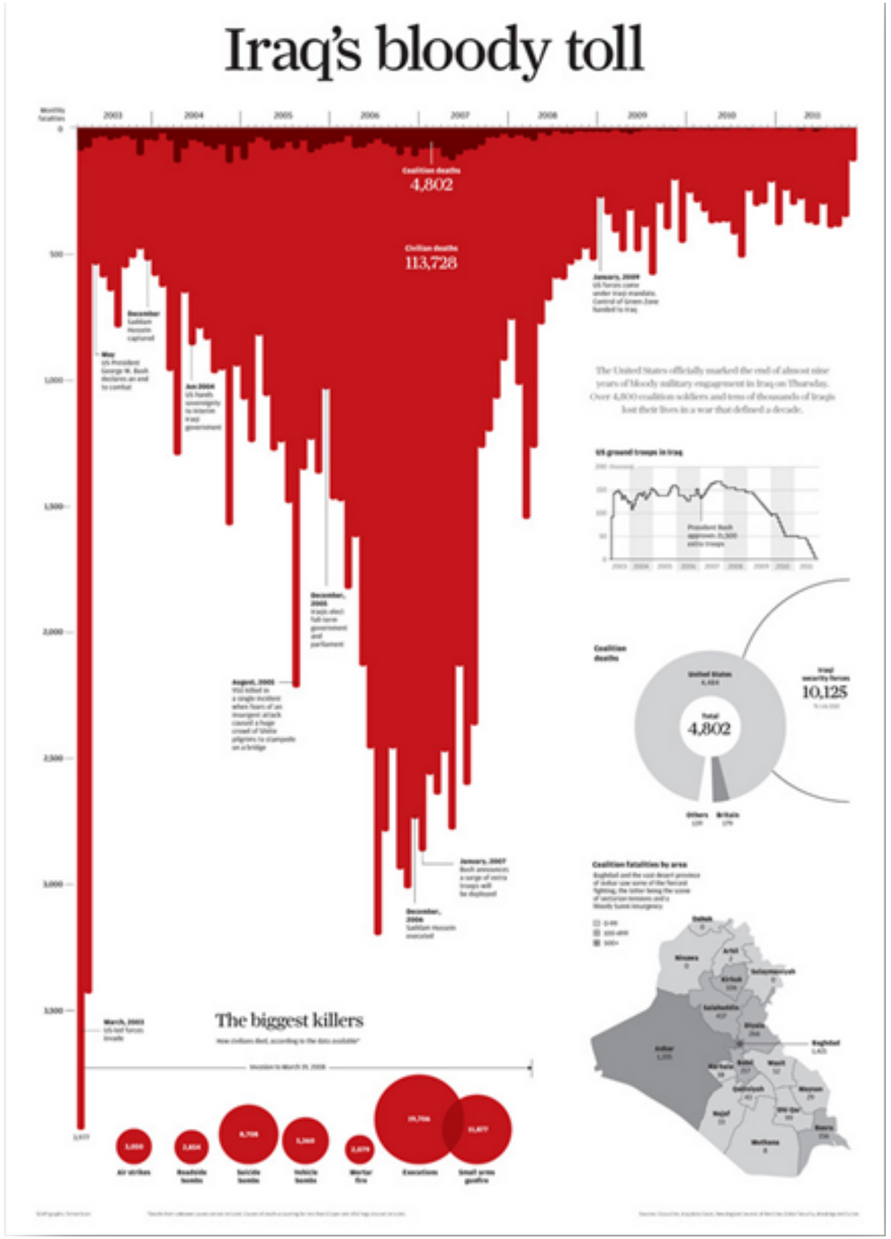
The Pudding

24 hours in an
invisible epidemic
by alvin chang
scroll down

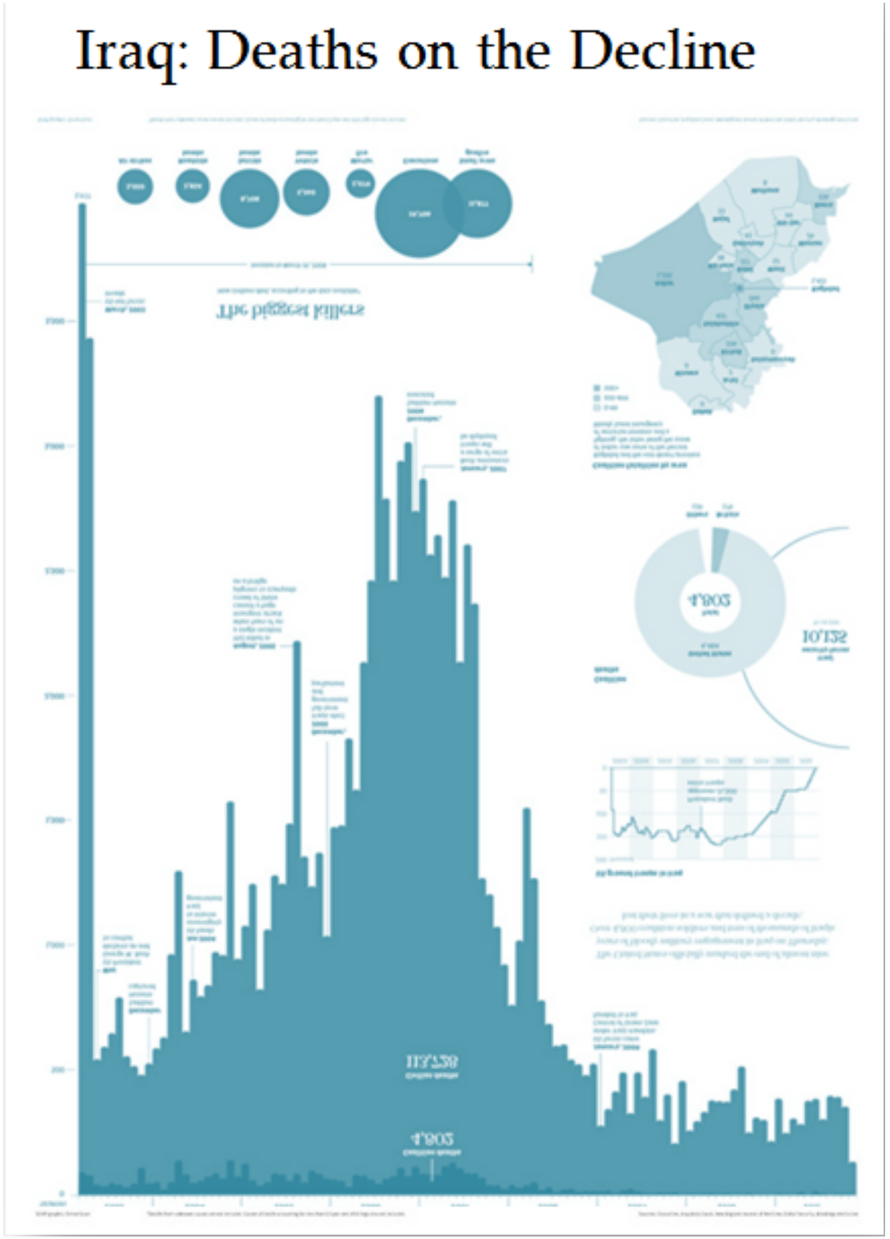


Simon Scarr - Iraq's bloody toll

Consider impactful metaphors.



Simon Scarr - Iraq's bloody toll



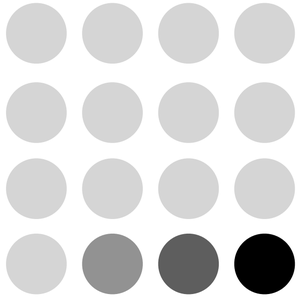
Andy Cotgreave: Should you Trust a Data Visualization?

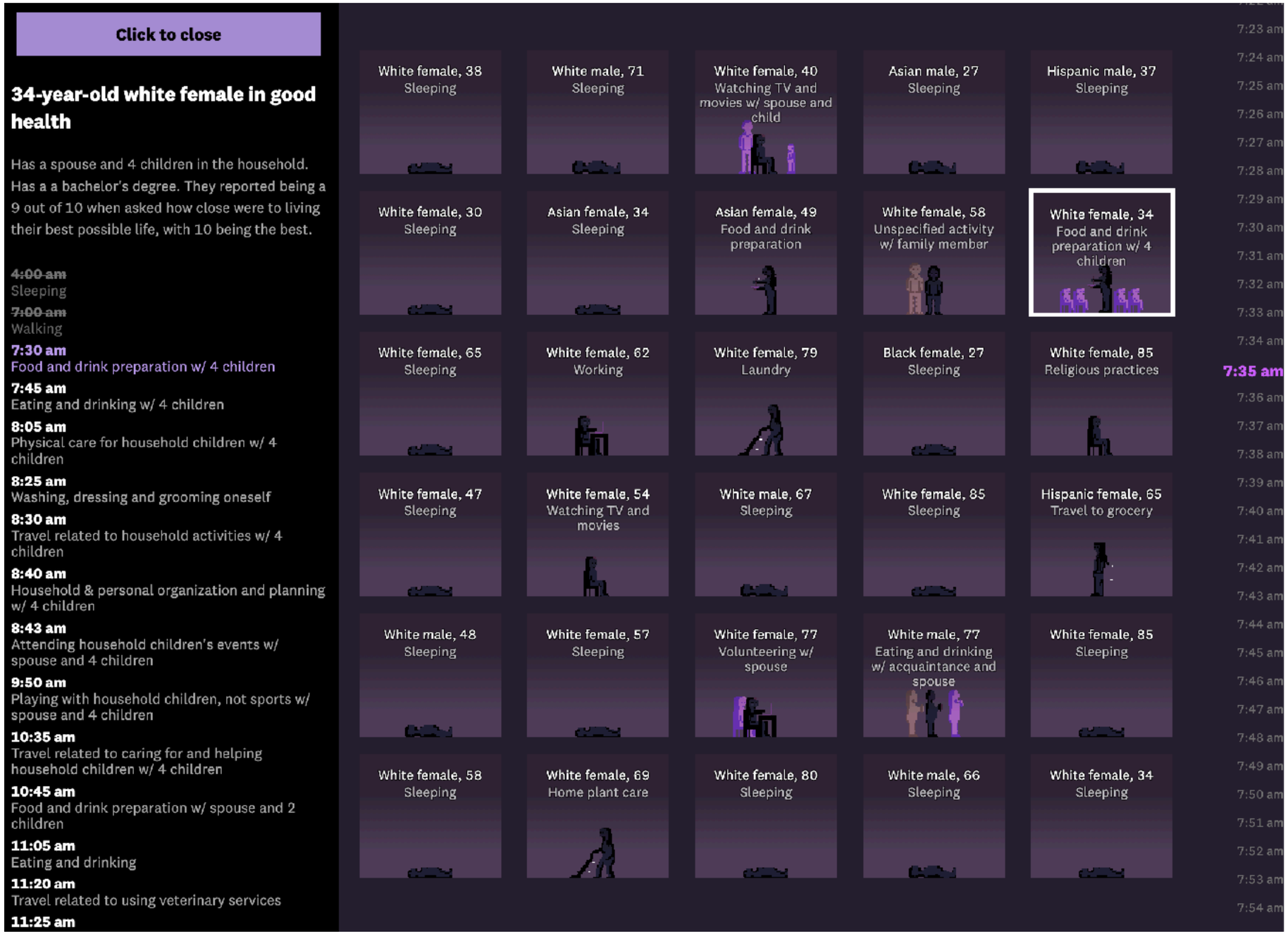
Same data, different message.

Information design is rhetorical (and emotional).

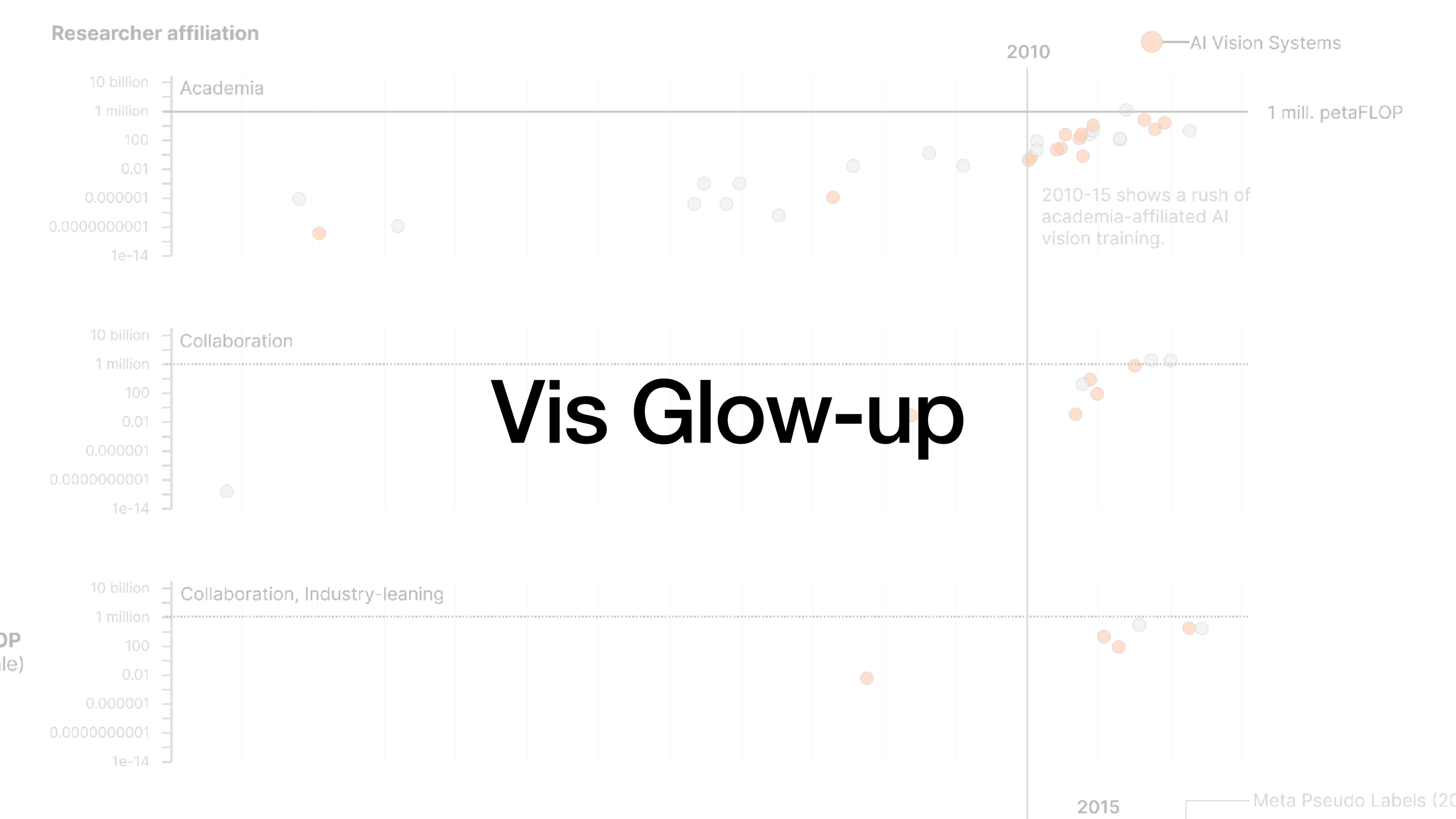
Roxanne

Plurality



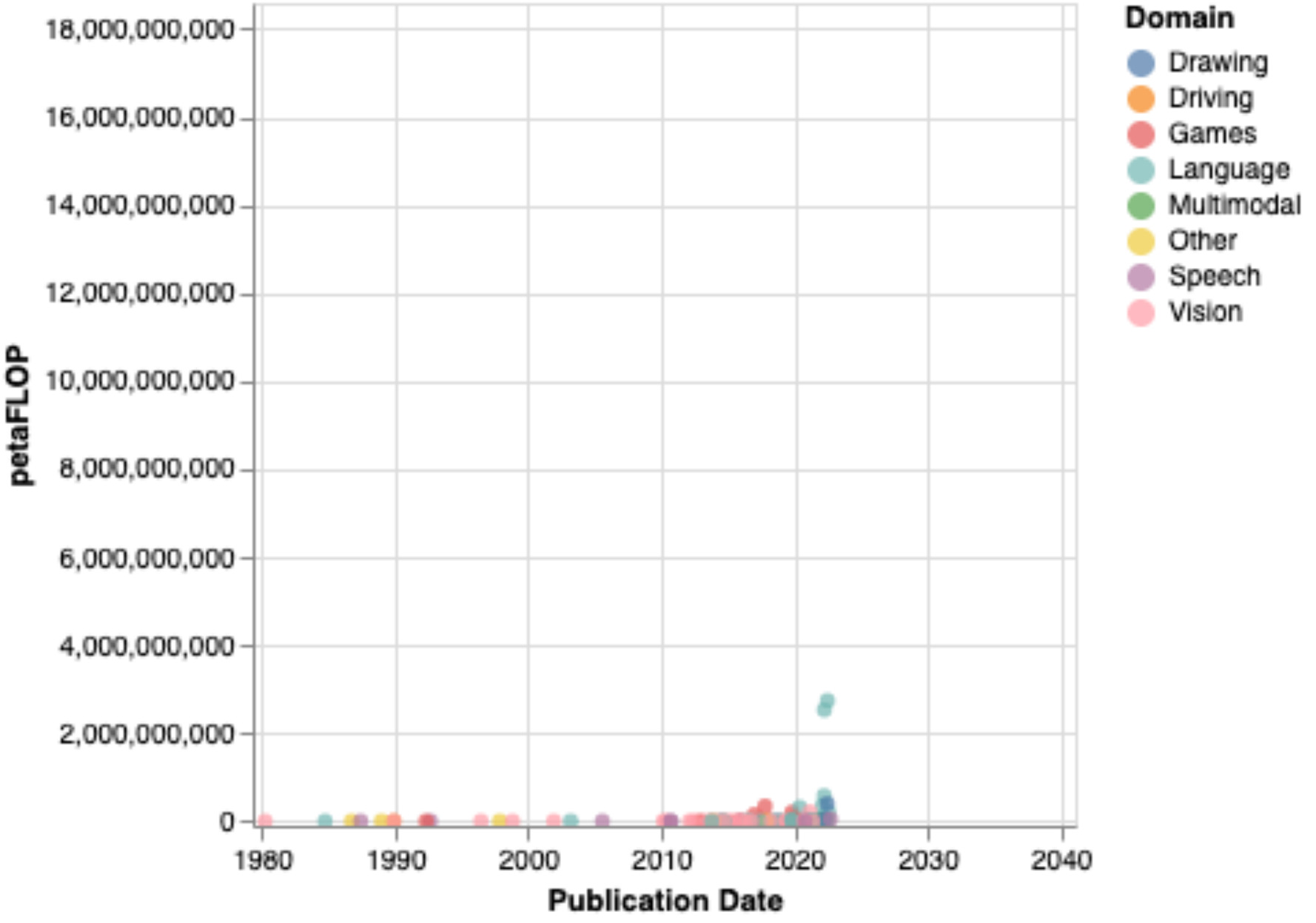


The Pudding: https://pudding.cool/2023/09/invisible-epidemic/


Researcher affiliation
2010
AI Vision Systems
10 billion
1 million
100
0.01
0.000001
0.0000000001
1e-14
Academia
1 mill. petaFLOP
2010-15 shows a rush of academia-affiliated AI vision training.
Collaboration
Collaboration, Industry-leaning
2015
Meta Pseudo Labels (20

Vis Glow-up

Development of AI



Define your goals

Can you describe your goal(s) in < 3 sentences?

I want to share a visual about the total computation needed to train popular AI systems, targeting a lay audience. I want to emphasize advancements in the vision domain, broken down by researcher affiliation (industry, academia, etc.).

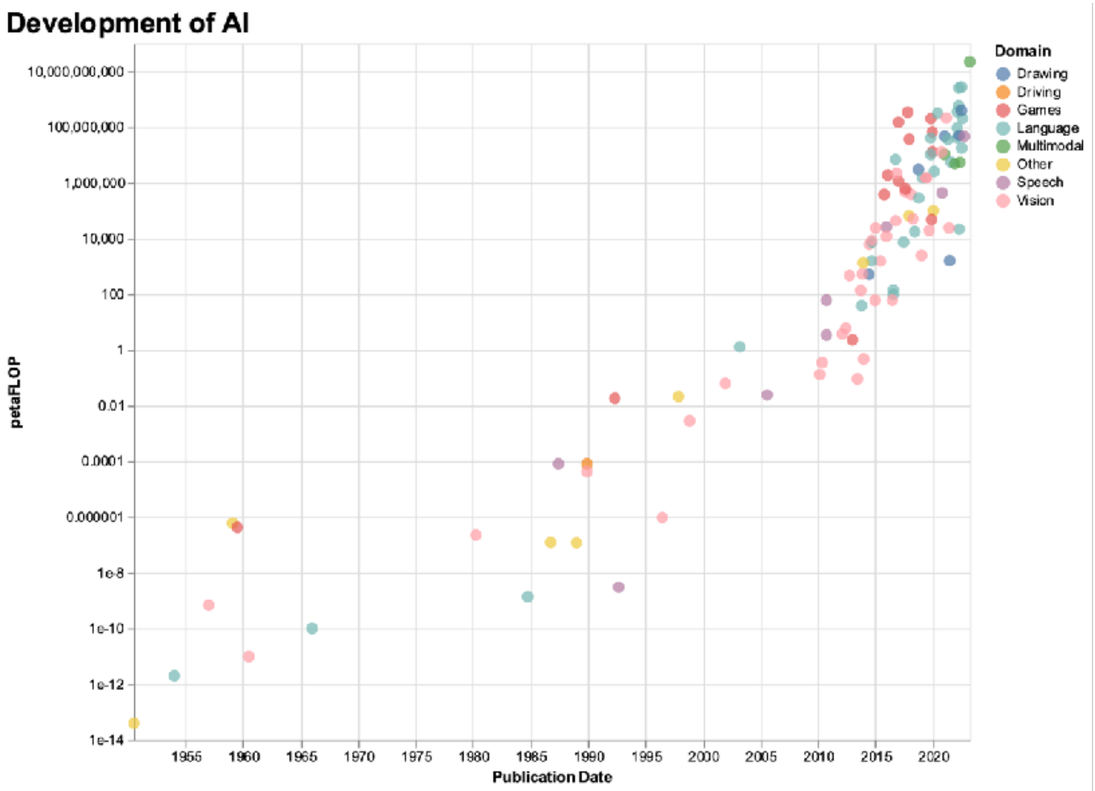
Information

Show what is necessary, with some context

Information

Choose appropriate axis scaling

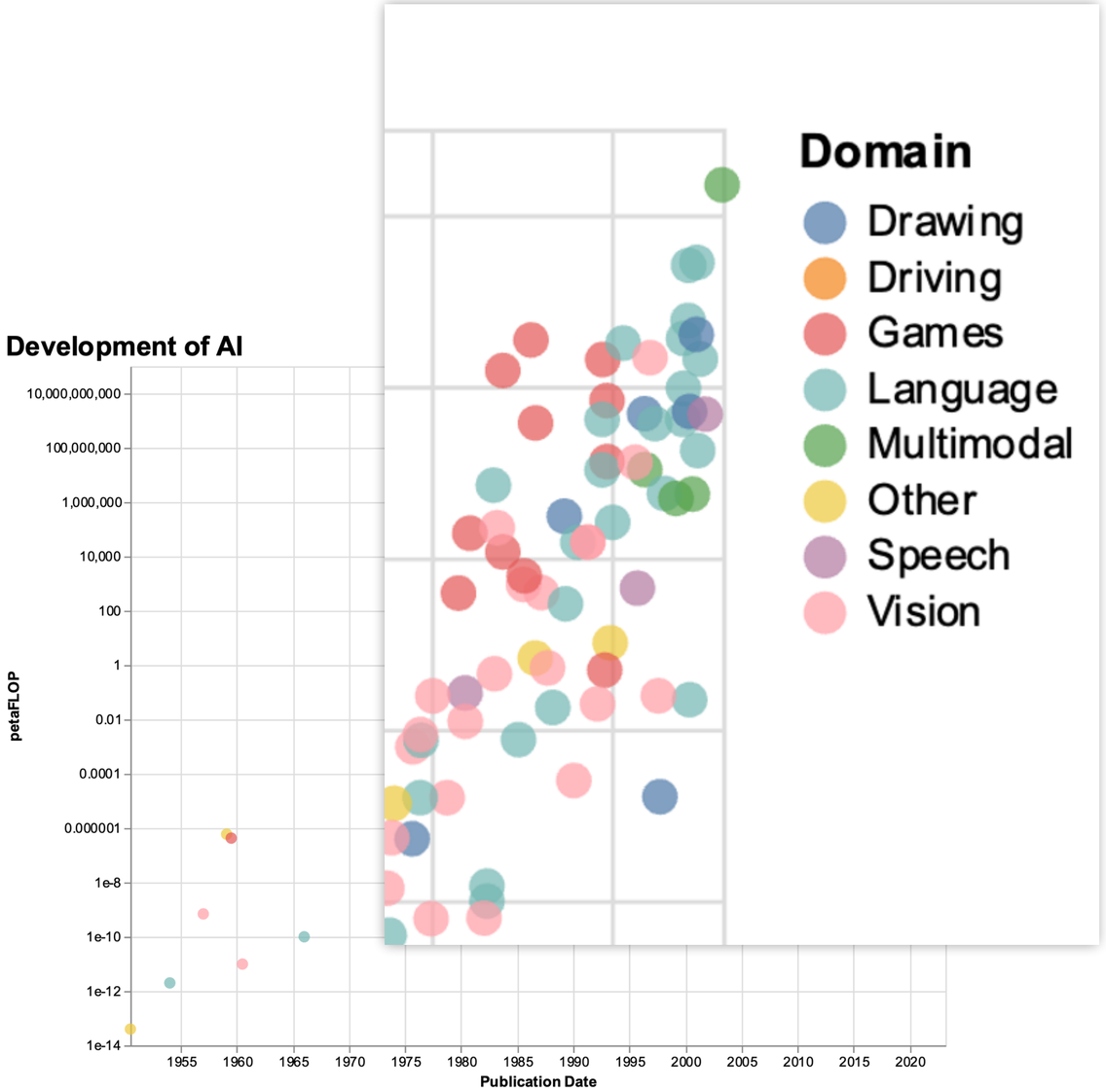
Log axis scaling better than **linear** to see spread of data



Information

Use **color schemes** appropriate to your data

Distinct colors for unordered categorical attributes



The science of visual data communication: what works, Franconeri et al. 2021

Acknowledge uncertainties, unknowns

Data biases?
Missingness?

Source: Sevilla et al. (2023), Our World in Data [https://ourworldindata.org/grapher/artificial-intelligence-training-computation]

Notes:

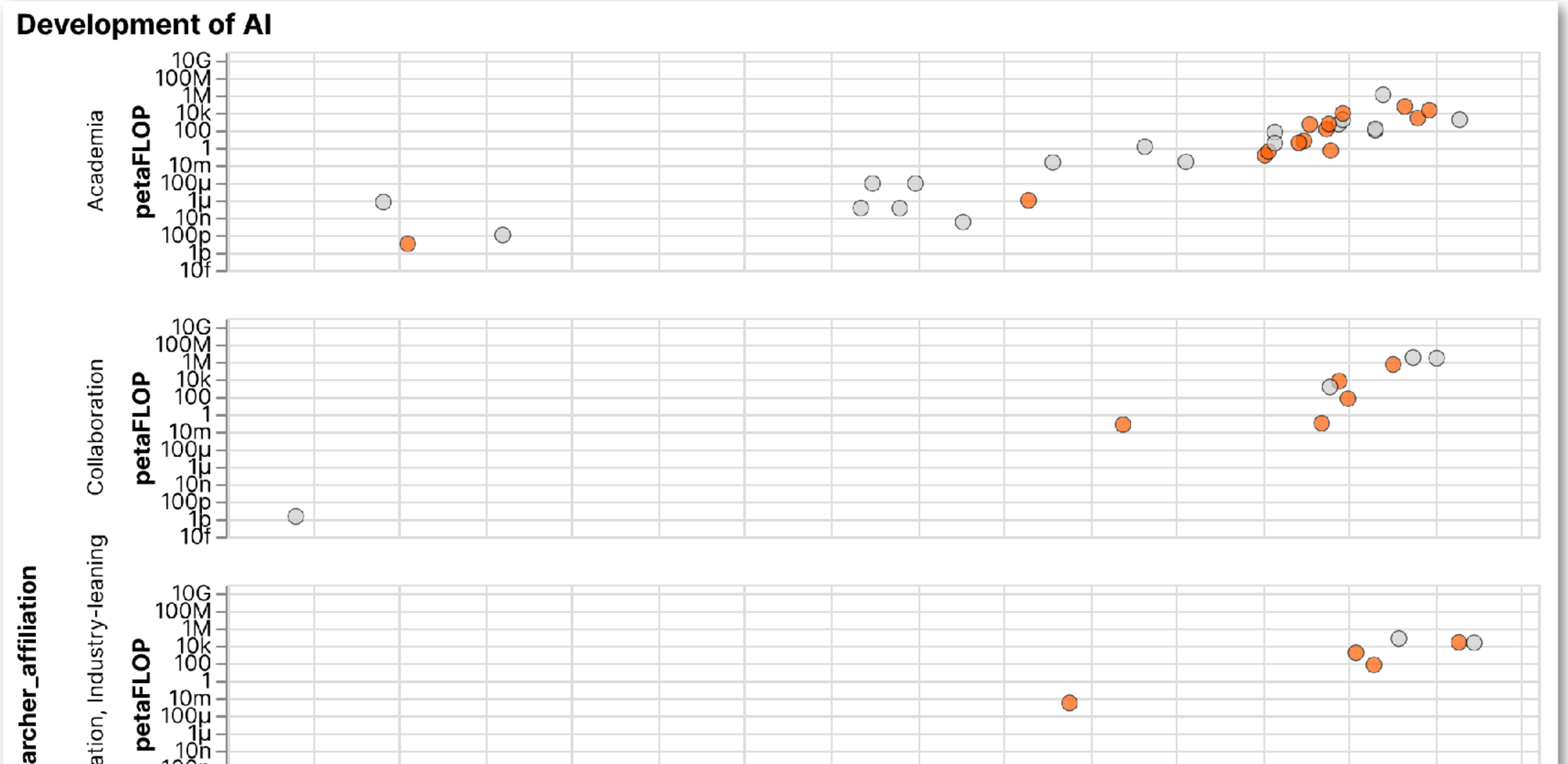
1. *Computation is estimated based on published results in the AI literature and comes with some uncertainty. The authors expect the estimates to be correct within a factor of 2.*
2. *Computation training data listed as NaN for **58%** of AI systems included in source dataset. As such, our picture of this story is only partially complete.*

Aesthetics

Visually subset

Aesthetics: Design with grey

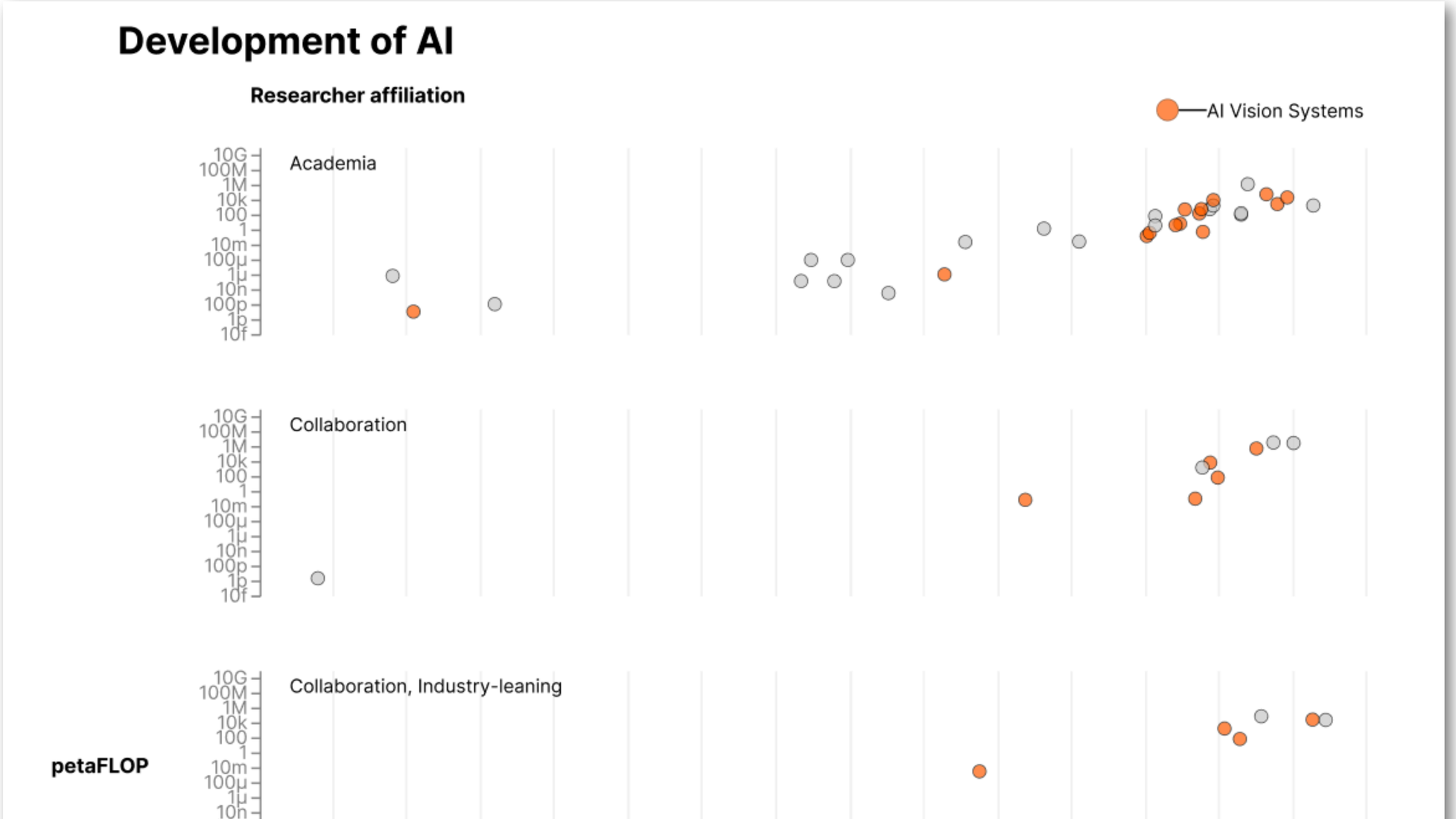
use color strategically to highlight elements of interest



Thoughtful aesthetics can help avoid taxing working memory.

Declutter your visual!

Integrate legends as direct annotations on data

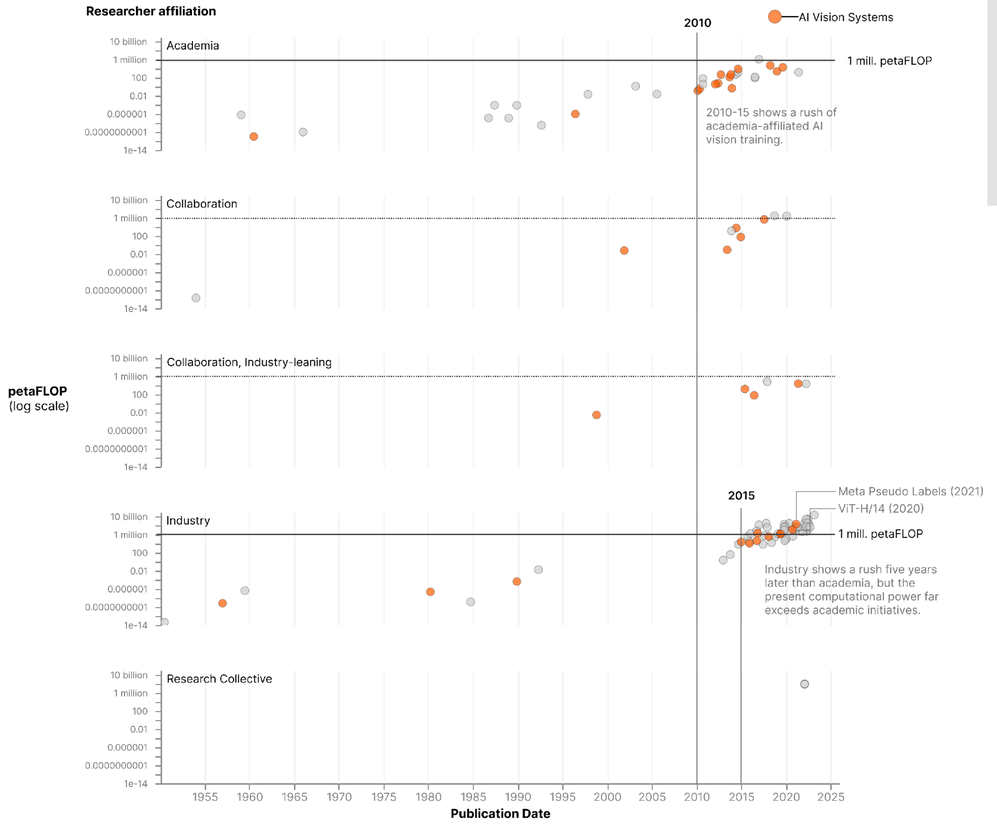


Story

Use figure title to explain trends, highlight takeaway(s)

Recent years show industry leading in computation used to train AI vision systems, although academia led the first push.

Computation is measured in total petaFLOP, which is 10^{15} floating-point operations (FLOP).



What is a FLOP?

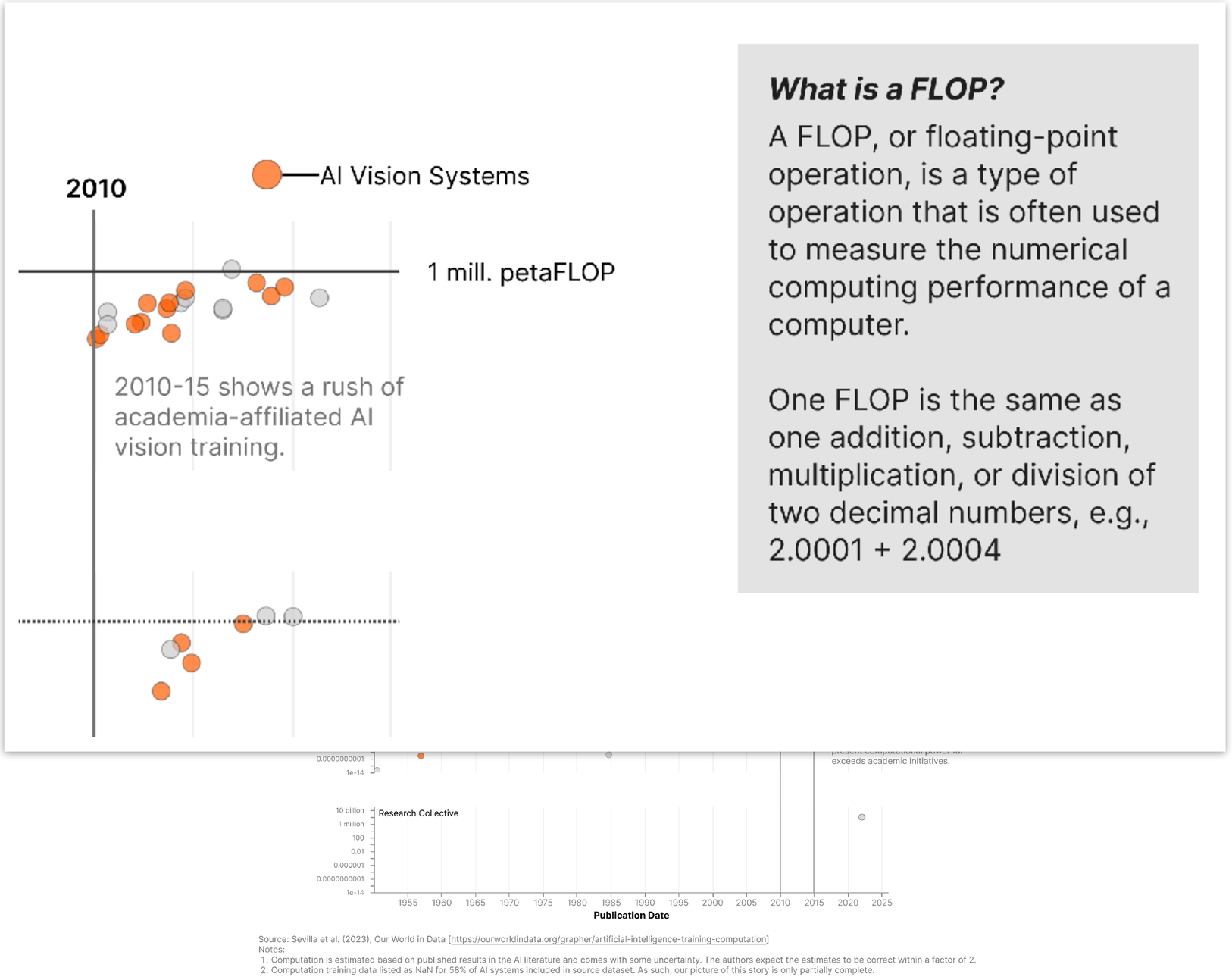
A FLOP, or floating-point operation, is a type of operation that is often used to measure the numerical computing performance of a computer.

One FLOP is the same as one addition, subtraction, multiplication, or division of two decimal numbers, e.g., 2.0001 + 2.0004

Source: Sevilla et al. (2023), Our World in Data [https://ourworldindata.org/grapher/artificial-intelligence-training-computation]
Notes:
1. Computation is estimated based on published results in the AI literature and comes with some uncertainty. The authors expect the estimates to be correct within a factor of 2.
2. Computation training data listed as NaN for 58% of AI systems included in source dataset. As such, our picture of this story is only partially complete.

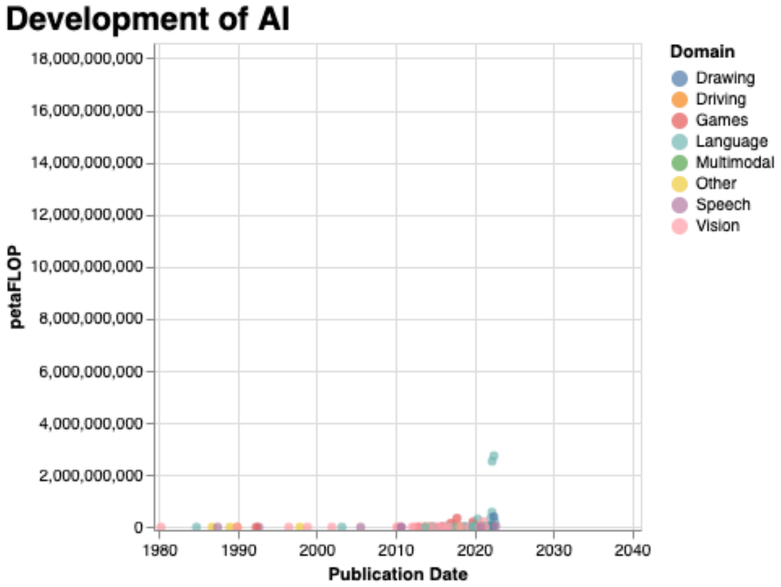
Story

Add explainers



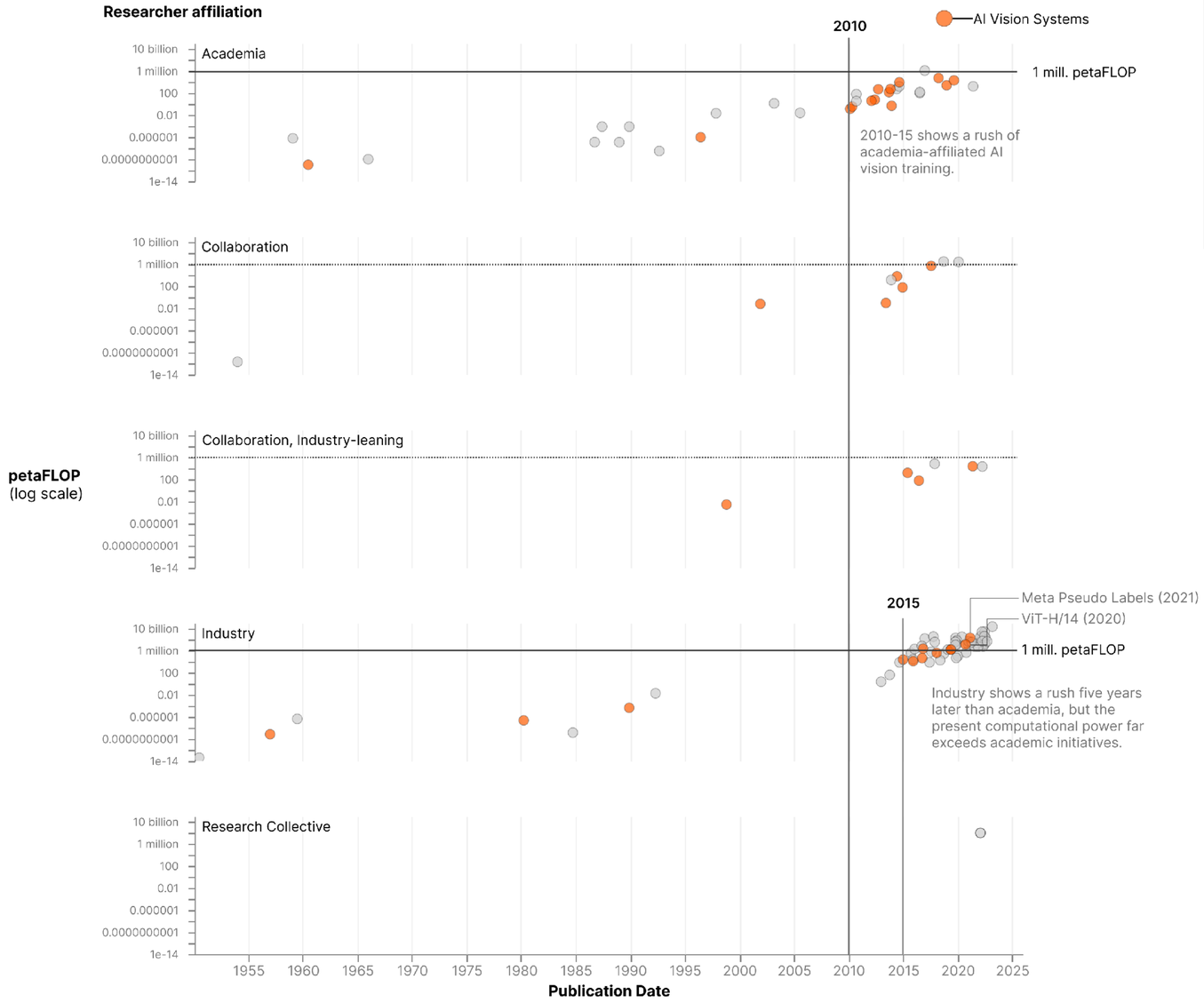
Source: Sevilla et al. (2023), Our World in Data [https://ourworldindata.org/grapher/artificial-intelligence-training-computation]
Notes:
1. Computation is estimated based on published results in the AI literature and comes with some uncertainty. The authors expect the estimates to be correct within a factor of 2.
2. Computation training data listed as NaN for 58% of AI systems included in source dataset. As such, our picture of this story is only partially complete.

Vis Glow-up



Recent years show industry leading in computation used to train AI vision systems, although academia led the first push.

Computation is measured in total petaFLOP, which is 10^{15} floating-point operations (FLOP).



What is a FLOP?

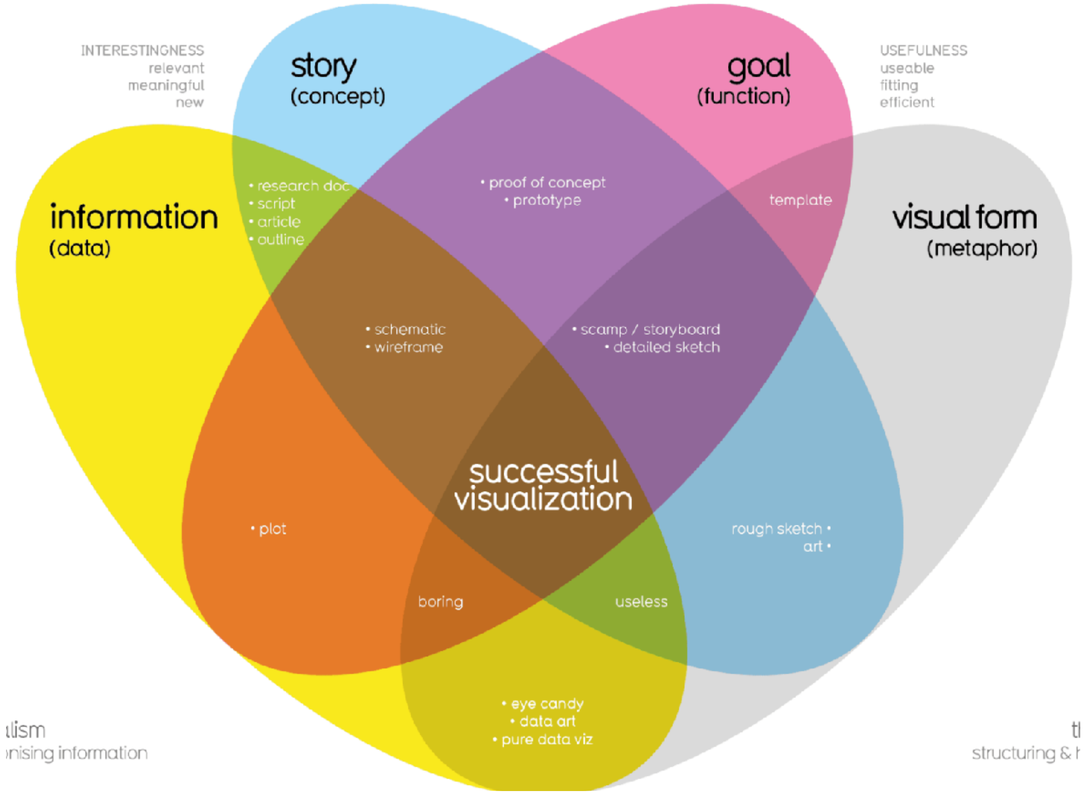
A FLOP, or floating-point operation, is a type of operation that is often used to measure the numerical computing performance of a computer.

One FLOP is the same as one addition, subtraction, multiplication, or division of two decimal numbers, e.g., 2.0001 + 2.0004

Source: Sevilla et al. (2023), Our World in Data [https://ourworldindata.org/grapher/artificial-intelligence-training-computation]
Notes:
1. Computation is estimated based on published results in the AI literature and comes with some uncertainty. The authors expect the estimates to be correct within a factor of 2.
2. Computation training data listed as NaN for 58% of AI systems included in source dataset. As such, our picture of this story is only partially complete.

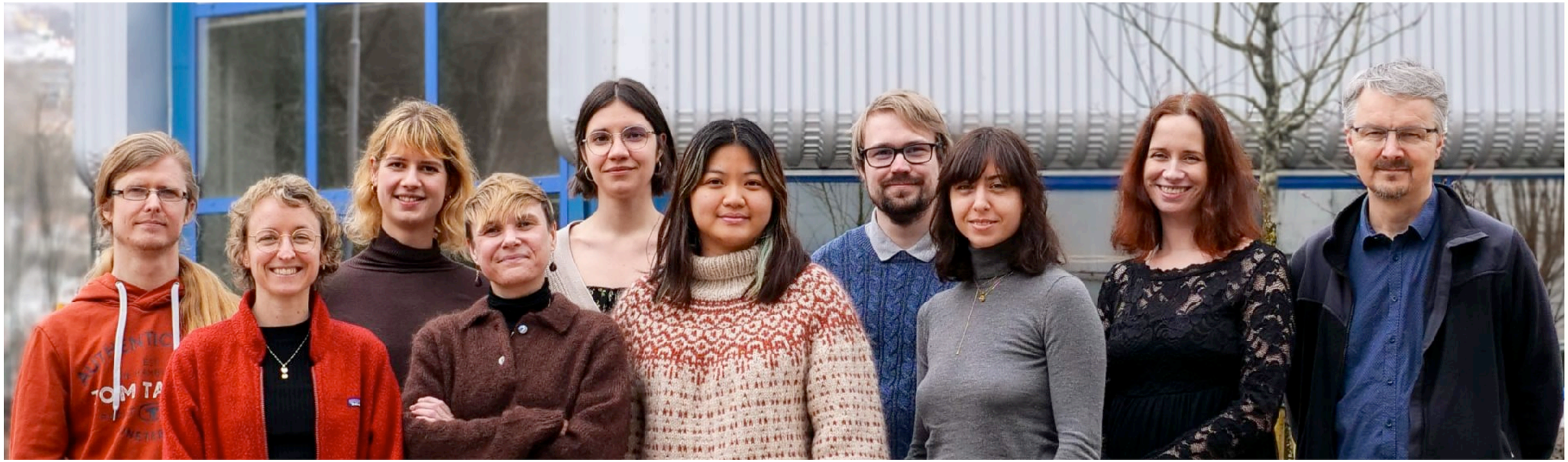
⚒ Fundamentals of a “good” visualization

1. Show **information** integrity
2. Main **goal** to create something useful and fitting to your audience
3. Make an interesting **story**
4. Don’t skip (or overdo) **aesthetics**



Want more?

- Cairo, A., 2016. The truthful art: Data, charts, and maps for communication. New Riders.
- Christiansen, J., 2022. Building Science Graphics: An Illustrated Guide to Communicating Science Through Diagrams and Visualizations. CRC Press.
- Franconeri, S.L., Padilla, L.M., Shah, P., Zacks, J.M. and Hullman, J., 2021. The science of visual data communication: What works. *Psychological Science in the public interest*, *22*(3), pp.110-161.
- Gleicher M. Considerations for visualizing comparison. IEEE TVCG. 2017 Aug 29;24(1):413-23.
- Nature Points of View: Data visualization article collection (https://blogs.nature.com/methagora/2013/07/data-visualization-points-of-view.html)
- Rogowitz, B.E., Treinish, L.A. and Bryson, S., 1996. How not to lie with visualization. *Computers in physics*, *10*(3), pp.268-273.
- Schwabish J. Better data visualizations: A guide for scholars, researchers, and wonks. Columbia University Press; 2021 Dec 31.



Laura Garrison
Associate Professor in Visualization
University of Bergen, Norway

vis.uib.no
laura-garrison.com
laura.garrison@uib.no
@lauragarrison.bsky.social

Thank you!
Questions?

Acknowledgements

- Alberto Cairo, Cédric Scherer, John Schwabish, Jen Christiansen, Michael Correll, Amy Zhang, Roxanne Ziman